A PRACTICAL MOBILE HANDBOOK

Website Design with AI

Start with one click.
Build something real.



Start learning →

Your learning space

Lessons
Follow the course

Code Lab
Copy and practise

Resources
Files and quick help

Paul Victor · Beloved Technology

Welcome 1/2

Your phone is the guide

Read on your phone. Build on your computer.

1 · Look

See what you are about to make.

2 · Do

Follow one step and try it.

3 · Check

Compare your result. Make one change.

Welcome 2/2

Your phone is the guide

Tap Lessons when you want the menu. Code opens the code index. Resources holds downloads and quick references.

Select code to copy it. If long lines copy poorly in your PDF reader, use Copy code online for the exact text.

The lessons work offline. Online previews and downloads need internet.

Welcome 1/1

Welcome to your first website

Never used a computer? Start with the mouse and keyboard lesson. You do not need an AI account.

Take one small step at a time. You are learning when you can try it yourself, even if you make a mistake.

Welcome 1/1

Meet your course author

Paul Victor

Web developer, ICT trainer and digital media creator. Paul teaches practical skills through small projects and visible results.

Beloved Technology

From your first click to your first website. Learn by making something that works.

Welcome 1/2

Download your course files

Open the online Code Lab

Download the practice ZIP

1 Download the ZIP. Press Ctrl + J in your browser, then choose Show in folder.

2 Right-click the ZIP → Extract All. Choose Documents → Website Class → Extract.

Welcome 2/2

Download your course files

3 Open Website_Design_Practice_Files. Use START_HERE.html to find an exercise.

4 Copy that exercise into a separate My Practice folder. Edit your copy.

Welcome 1/1

Find your way around the course

Home returns to the dashboard. Lessons opens the course menu. Code finds examples. Resources holds your quick references.

Use the arrow buttons to move through pages. At the end of a lesson, Next takes you to the following lesson.

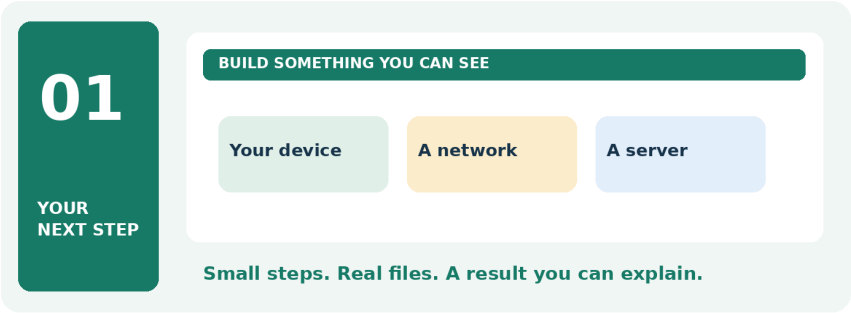
A page number shows your place in the book. It does not track completed work.

Computer & internet 1/2

Part 01 Computer and internet

Find your files and understand how a webpage reaches your screen.

VISUAL EXPLANATION



Your finish line

Open Notepad and save a file. Follow a request through the browser, server and network.

Computer & internet 2/2

Part 01 Computer and internet

TRY

Take this part one practical at a time. Keep your previous work and change only the file named in the current step.

CHECK

When you finish, write one sentence explaining what this subject does. Then show its result on your computer.

Computer & internet 1/5

Meet the mouse and keyboard

YOU NEED

your computer, mouse and keyboard.

Let us practise

1. Find the power button marked with a circle and a short line. Press it once and wait. Sign in with your own Windows account. On a shared computer, use the account provided by its owner.

Computer & internet 2/5

Meet the mouse and keyboard

2 Move the mouse slowly. Watch the arrow move on the screen. That arrow is called the pointer.

3 Click an empty part of the desktop. The desktop is the main screen you see after signing in.

4 Find the Windows Start button on the taskbar, usually along the bottom. Click it once.

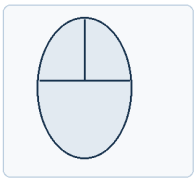
Computer & internet 3/5

Meet the mouse and keyboard

5 Type Notepad. Click the Notepad result. A writing window opens.

VISUAL EXPLANATION

Find the controls before writing code



Left button: click

Right button: show choices

Ctrl + S Hold Ctrl, tap S

Type this

My name is Ada.

Computer & internet 4/5

Meet the mouse and keyboard

Press Enter. Type: I am learning to build websites.

Backspace removes the character just before the blinking line. That blinking line shows where your next letter will appear. Hold Shift while pressing a letter to make one capital letter.

Try it: change Ada to your name. Use the mouse to click where you want to type.

Computer & internet 5/5

Meet the mouse and keyboard

You are ready when: you can open Notepad and type two sentences without looking back at the steps.

Computer & internet 1/5

Select copy paste and undo

YOU NEED

Notepad. Keep the sentences from Computer lesson 1 open.

Follow these steps

1. Place the pointer just before your name.

2. Hold the left mouse button. Move across your name. Release the button. The highlighted letters are selected.

Computer & internet 2/5

Select copy paste and undo

3 Hold Ctrl and tap C. This copies the selected letters.

4 Click at the end of your second sentence. Press Enter, then Ctrl + V. Your copied name appears.

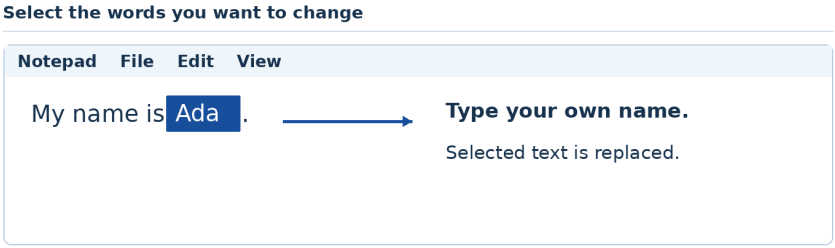
5 Press Ctrl + Z. Your last change is undone.

Computer & internet 3/5

Select copy paste and undo

6 Press Ctrl + A. All the text is selected. Click once inside the text to remove the selection without deleting it.

VISUAL EXPLANATION



Keep these shortcuts nearby

Ctrl + C

Copy the selected text

Computer & internet

Select copy paste and undo

Ctrl + V

Paste the copied text

Ctrl + Z

Undo the last change

Ctrl + A

Select everything in the current writing area

Ctrl + S

Save your work

Computer & internet 5/5

Select copy paste and undo

Be careful: typing while text is highlighted replaces that text. If you do it accidentally, press Ctrl + Z.

TRY

copy one sentence and paste it on a new line. Then undo the paste.

Computer & internet 1/4

Make a folder and save your work

YOU NEED
File Explorer and Notepad.

A file holds one piece of work. A folder holds files. We will keep our website files together so we can find them again.

Make your course folder

1. Hold the Windows key and press E. This opens File Explorer.

2. Click Documents on the left.

Computer & internet 2/4

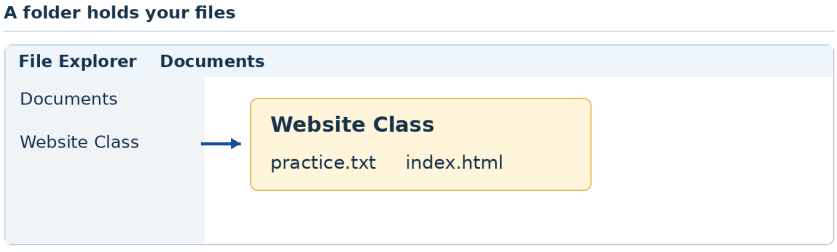
Make a folder and save your work

3 Right-click an empty space. Choose New, then Folder.

4 Type Website Class and press Enter.

5 Double-click Website Class to open it.

VISUAL EXPLANATION



Computer & internet 3/4

Make a folder and save your work

Save your sentences

1. Return to Notepad by clicking its icon on the taskbar.

2. Click File, then Save as. On versions that show Save first, use that option for this new file.

3. Open Documents, then Website Class in the save window.

Computer & internet 4/4

Make a folder and save your work

4 In File name, type practice.txt. Click Save.

5 Close Notepad using the X at the top right. In File Explorer, double-click practice.txt.

You should see: the same sentences you saved.

TRY

add a third sentence. Press Ctrl + S. Close and reopen the file to check that the new sentence stayed.

Computer & internet

What the internet actually is

1 **YOUR DEVICE**
A phone or computer

↓

2 **CONNECTION**
Wi-Fi, cable or mobile data

↓

3 **INTERNET**
Many connected networks

↓

4 **OTHER COMPUTERS**
They send information back

YOU NEED

your classroom computer and a simple drawing on paper.

Computer & internet 2/5

What the internet actually is

The internet is a worldwide connection of networks. A network is a group of connected devices that can exchange information. Your phone, a school computer and a company's server can communicate through these connections.

Find the connection around you

1. Find the router that supplies your connection, or the phone whose hotspot you use. You do not need to change its settings.

Computer & internet 3/5

What the internet actually is

2 On Windows, click the network symbol near the clock. Look at the connected network's name. Do not change its settings yet.

3 Draw your computer, the router or hotspot, and the internet provider. Join them with lines.

4 Say the route aloud or write it in your notes: “My computer connects here. This connection reaches our provider.”

Computer & internet 4/5

What the internet actually is

Wi-Fi is a wireless connection to a nearby network. It does not guarantee internet access. A router can still be switched on when the provider's connection is down.

An ISP is an internet service provider: the company carrying your connection to other networks. Data can travel through fibre cables, other cables, radio links and mobile networks. The internet is not simply “inside the Wi-Fi box.”

Computer & internet 5/5

What the internet actually is

Try it: identify whether your classroom connection uses a cable, Wi-Fi or a phone hotspot. If you know the network owner, ask which company provides the connection.

Check yourself: Wi-Fi can work while the router has no onward internet connection. A Wi-Fi symbol alone does not prove that websites are reachable.

Reference: N1.

Computer & internet 1/5

Open a browser and visit a website

YOU NEED

Microsoft Edge or Google Chrome and a working internet connection.

The web is one service that uses the internet. Email and other services use it too. A browser is the program we use to open webpages. A search engine is a service that helps us find them.

Computer & internet 2/5

Open a browser and visit a website

Visit an address directly

1. Click Start. Type Edge and open Microsoft Edge. Chrome is also suitable if already installed.

2. Press Ctrl + L. This selects the address bar near the top.

3. Type https://example.com and press Enter. Wait for the page to load.

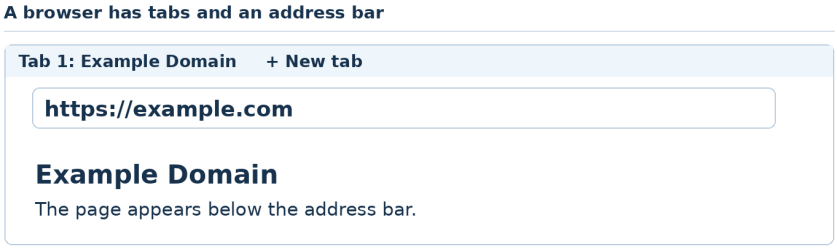
Computer & internet

Open a browser and visit a website

4 Press Ctrl + T to open another tab. Type a search such as what is a webpage and press Enter.

5 Compare the results page with the page you visited directly. Click the first tab to return.

VISUAL EXPLANATION



Computer & internet 4/5

Open a browser and visit a website

A tab keeps one page open while you visit another. The Back button returns to the previous page in that tab. Refresh asks the browser to load the current page again.

A website contains related pages. One page may be its Home page, another its Contact page. A link is an item you can click to move to a page or a place on a page.

Computer & internet 5/5

Open a browser and visit a website

TRY

visit example.com without using a search result. Write the browser name and the search service name in notes.txt. They are different tools.

Reference: N2.

Computer & internet 1/5

Names addresses and places to store a site

YOU NEED

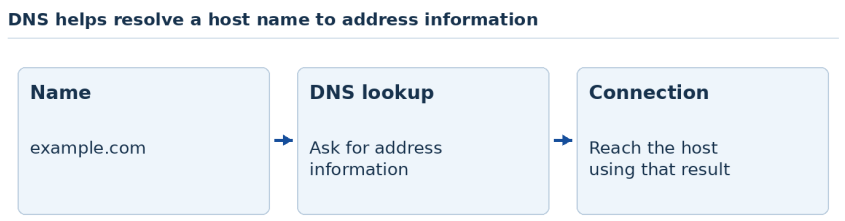
the browser and paper. No purchase is needed.

People remember a name such as example.com more easily than a numeric network address. DNS helps look up the network information needed to reach a name. An IP address helps networks deliver data to the right destination.

Computer & internet

Names addresses and places to store a site

VISUAL EXPLANATION



A domain is a registered name. Hosting is the service running or storing the website. You can buy a name without yet having a working website. You can also test a website on your own computer without buying either.

Computer & internet 3/5

Names addresses and places to store a site

Read this address together

https://example.com/cakes?size=small#prices

https

Use HTTP through an encrypted connection

example.com

The host name to reach

/cakes

The path requested from that host

Computer & internet 4/5

Names addresses and places to store a site

?size=small

Extra query information sent with the request

#prices

A place within the page; not sent as part of the HTTP request

This is an address example, not a promise that a cakes page exists there.

Computer & internet 5/5

Names addresses and places to store a site

TRY

in your notebook, invent a website name and two page paths. Circle the domain name and underline each path. Do not register or buy the name.

One name can use several IP addresses, and several websites can share infrastructure. Think of DNS as a lookup system, not a permanent one-name-one-computer rule.

References: N2, N3.

Computer & internet

Follow a webpage from request to screen

1 **BROWSER**
You enter a web address

↓

2 **DNS**
Finds the server address

↓

3 **REQUEST**
Browser asks for the page

↓

4 **RESPONSE**
Server sends files back

↓

5 **SCREEN**
Browser displays the page

Computer & internet 2/6

Follow a webpage from request to screen

YOU NEED

paper cards labelled Browser, DNS and Server.

A server is a computer or program that responds to requests. Your browser is the client asking for the page.

Computer & internet

Follow a webpage from request to screen

Act it out

1. Place three cards on a table: Browser, DNS and Server. Begin at Browser and ask DNS for the host’s address. A remembered result may already be available.

2. DNS provides address information. The browser connects to the destination.

Computer & internet

Follow a webpage from request to screen

3 The browser sends an HTTP request for a path, such as /.

4 The server sends an HTTP response: status information and usually some content.

5 The browser reads the HTML. It may request CSS, JavaScript, pictures and fonts separately. Then it builds the page you see.

Computer & internet 5/6

Follow a webpage from request to screen

HTTP is a set of rules for web requests and responses. HTTPS protects the connection using encryption and certificate checks. It does not prove that a seller is honest or that a webpage's claims are true.

A response code is a short result label. 200 usually means success. 404 means the requested resource was not found. 500 indicates a server-side failure. A redirect tells the browser to try another address.

Computer & internet 6/6

Follow a webpage from request to screen

TRY

explain why a heading might appear before a large photograph finishes loading. The browser often needs more than one response to finish a page.

References: N2, N4.

Computer & internet 1/5

Watch real requests in the browser

YOU NEED

Edge or Chrome. The following steps open a panel built into the browser. You can close it without changing your page.

You do not need to understand the whole panel. Today we will find the page request and its response.

1 Open https://example.com.

2 Press F12, or Ctrl + Shift + I, to open developer tools. Some laptops also need Fn with F12.

Computer & internet 2/5

Watch real requests in the browser

3 Click Network. If it is hidden, use the extra-tab menu marked with arrows.

4 Press Ctrl + R to reload the page. Rows should appear in the panel.

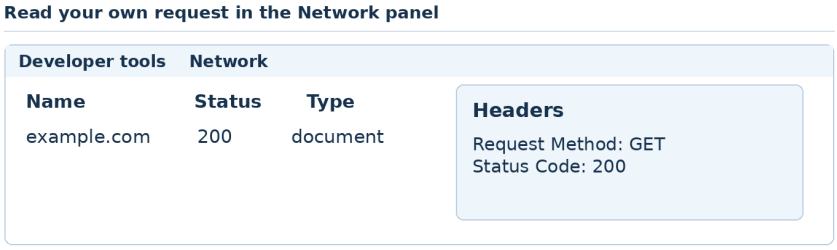
5 Click the row for the document. Look under Headers for Request URL, Request Method and Status Code.

Computer & internet 3/5

Watch real requests in the browser

6 Read the actual method and status you see. Open Response or Preview to inspect the returned content.

VISUAL EXPLANATION



Computer & internet 4/5

Watch real requests in the browser

Do not worry if a different website has many rows. One page can request many files. Some results may come from a cache, which is a saved copy used to reduce repeated downloads.

TRY

write down the requested address and status. A 200 status usually means the request succeeded. Close the developer tools using its X when you finish.

Computer & internet 5/5

Watch real requests in the browser

If the panel is empty, open Network before refreshing. If a website fails, the panel may show a network error rather than an HTTP status because no response arrived.

Reference: N5. The illustration shows an example result; read your own screen.

Computer & internet

Data travels in pieces

YOU NEED

six small pieces of paper and a pencil.

Information travels across networks in small units called packets. Routers forward packets towards their destination. The receiving systems handle assembling the delivered data into something useful.

Send a paper message

1. Write WE, ARE, LEARNING, TO, BUILD, WEBSITES on six cards.

Computer & internet 2/5

Data travels in pieces

2 Number them 1–6 and write the same destination name on each card.

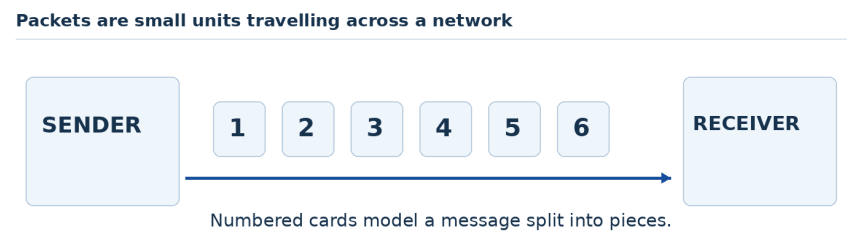
3 Let classmates pass them along two different routes across the room.

4 At the destination, arrange the cards in order. Notice whether one arrives late or is missing.

Computer & internet

Data travels in pieces

VISUAL EXPLANATION



This is a simplified model. Real networking adds addressing, checks and transport rules. For example, TCP can detect missing data and arrange retransmission. Different web connections may use different transport protocols.

Computer & internet

Data travels in pieces

Bandwidth describes how much data a connection can carry in a given time. Latency is the delay before information gets where it needs to go. A connection can have good capacity but still have noticeable delay.

A large image contains more data than a short sentence. That is one reason a page full of huge photos can feel slow and use more of a mobile data allowance.

Computer & internet 5/5

Data travels in pieces

TRY

which should usually need more download data, one short paragraph or a high-resolution video? Explain your answer without using a speed-test number.

References: N1, N2.

Computer & internet 1/5

Online local upload and download

YOU NEED

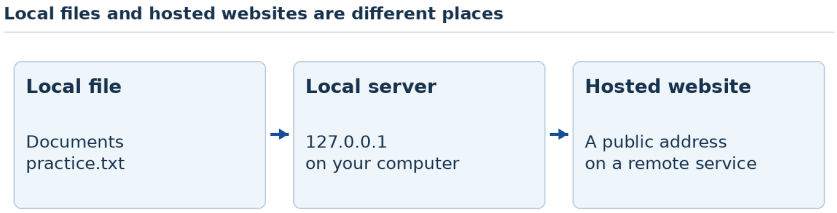
your browser, File Explorer and practice.txt from the earlier folder lesson.

Download means receiving a copy from a remote service. Upload means sending a copy to a remote service. Saving a file in Documents is not automatically publishing it on the web.

Computer & internet

Online local upload and download

VISUAL EXPLANATION



Compare two places

1. Open practice.txt in Notepad. It is stored on your computer.

2. Open example.com in your browser. You are visiting a remote website.

Computer & internet 3/5

Online local upload and download

3 On your own practice computer, briefly disconnect your own Wi-Fi connection. Do not disconnect a shared classroom router.

4 Edit and save practice.txt. It should still work.

5 Try reloading the website. It may fail, or show stored content. A cached page does not prove that the internet is connected.

Computer & internet 4/5

Online local upload and download

6 Reconnect Wi-Fi using the same network and the approved password arrangements.

Later, you will open a page using file:/// and run a server using http://127.0.0.1. Both can refer to work on your computer. A public https:// address usually reaches a hosted service.

Computer & internet 5/5

Online local upload and download

If every site fails, check the connection. If just one path returns 404, check that address. If words load but a picture does not, check that picture's request.

TRY

explain why sending someone C:\Users\Ada\Documents\index.html will not give them your website.

Reference: N2.

Computer & internet

Meet the five subjects we will practise

1. **HTML**
   Content and structure
2. **CSS**
   Colour, spacing and layout
3. **JAVASCRIPT**
   Actions in the browser
4. **PHP**
   Work on the server
5. **SQL**
   Read and change stored data

Computer & internet 2/5

Meet the five subjects we will practise

YOU NEED

this picture and a discussion. You will write each kind of code yourself.

HTML

Describe the content and structure of the page

CSS

Control colours, spacing, type and layout

JavaScript

Respond to actions in the visitor's browser

Computer & internet 3/5

Meet the five subjects we will practise

PHP

Run on the server, check submitted information and prepare a response

HTML is a markup language. CSS is a style-sheet language. JavaScript and PHP are programming languages. They work together, but they do not do the same job.

Computer & internet 4/5

Meet the five subjects we will practise

Imagine a cake order page. HTML provides the labelled quantity field. CSS makes it readable. JavaScript can show a quick estimate as the visitor types. PHP checks the submitted quantity and calculates again on the server.

The browser can display HTML, apply CSS and run JavaScript. It does not execute PHP source code. A PHP server runs that code and sends the result back.

Computer & internet 5/5

Meet the five subjects we will practise

TRY

name the tool you would use to change a blue heading, add a paragraph, respond to a button click and check a submitted form on the server.

Answers after you try: CSS, HTML, JavaScript, PHP.

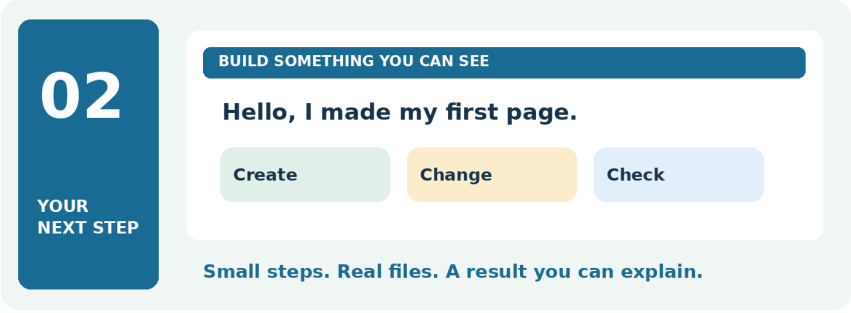
We will practise these foundations before moving into AI-assisted building and frameworks. You are not expected to remember all the syntax today.

References: N2, N6, N7, N8.

Part 02 Your first webpage

Start with Notepad, then make VS Code your everyday work space.

VISUAL EXPLANATION



Your finish line

Save index.html, open it in a browser and change the visible words.

Notepad & VS Code 1/5

Make your first webpage in Notepad

Notepad menu bar

REAL TOOL INTERFACE



Click File, then Save as. Name the file index.html. Choose All files so Notepad does not add .txt.

YOU NEED

Notepad and your Website Class folder.

Make your first webpage in Notepad

A webpage is a document that a browser can show. Today you will make one. Do not worry about remembering the symbols yet.

Type this carefully

Open a new Notepad document with File → New. In versions with tabs, New tab is also suitable. Type the whole example below.

Notepad & VS Code 3/5

Make your first webpage in Notepad

CODE EXAMPLE Copy online ↗

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport"
        content="width=device-width, initial-scale=1">
  <title>My first website</title>
</head>
<body>
  <h1>Hello, I am Ada</h1>
  <p>This is my first webpage.</p>
</body>
</html>
```

On many English keyboards, hold Shift and press the comma key for <. Hold Shift and press the full-stop key for >. If these keys differ, press Windows + Ctrl + O to open the on-screen keyboard and look for the symbols.

Make your first webpage in Notepad

Save it as a webpage

1. Click File → Save as. Open Documents → Website Class.

2. In File name, type index.html.

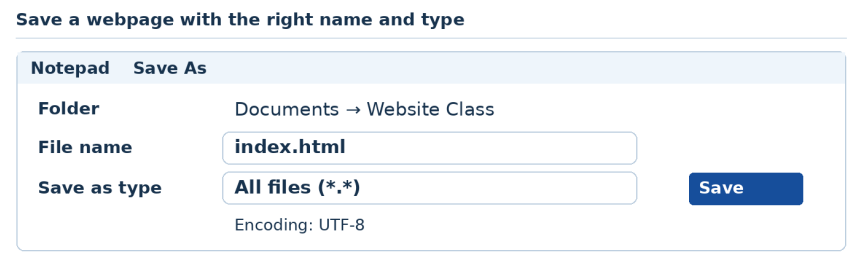
3. Change Save as type to All files. Choose UTF-8 if Encoding is shown.

4. Click Save. Keep this folder open for the next lesson.

Notepad & VS Code 5/5

Make your first webpage in Notepad

VISUAL EXPLANATION



Important: index.html is different from index.html.txt. The ending .html is what we need.

Notepad & VS Code 1/4

Open your page and change it

YOU NEED

File Explorer, Notepad and Edge or Chrome.

See your page

1. In Website Class, right-click index.html.
2. Choose Open with, then Microsoft Edge or Google Chrome.

Notepad & VS Code 2/4

Open your page and change it

3 You should see a large greeting and a smaller sentence.

VISUAL EXPLANATION



Change the greeting

1 Return to Notepad. If the file is closed, use File → Open. Select All files in the file-type choice if index.html is hidden.

Notepad & VS Code 3/4

Open your page and change it

2 Change Ada to your own name. Leave <h1> and </h1> in place.

3 Press Ctrl + S to save.

4 Return to the browser. Press Ctrl + R to refresh the page.

If you still see Ada, check that you saved. Then check that the browser opened the index.html inside Website Class.

Notepad & VS Code 4/4

Open your page and change it

If you see the code itself, check the filename. In Windows 11, File Explorer → View → Show → File name extensions reveals file endings. In Windows 10, use the View tab → File name extensions. If the file ends in .html.txt, right-click it, choose Rename, remove only the final .txt, press Enter and confirm the extension change.

You have succeeded when: your own name appears in the browser.

Source: S1.

Notepad & VS Code 1/4

Understand the few tags you used

YOU NEED

your index.html in Notepad.

HTML gives the browser instructions about the page. A tag is an instruction inside angle brackets, such as <p>. For now, learn these four parts.

VISUAL EXPLANATION



Notepad & VS Code 2/4

Understand the few tags you used

Make two changes

1. Find <title>My first website</title>.

2. Replace only My first website with Ada School Project. Save and refresh. Look at the browser tab, not the page heading.

3. Find the sentence inside and .

Notepad & VS Code 3/4

Understand the few tags you used

4 Change it to I can create and edit a webpage. Save and refresh again.

Add another paragraph

Click just before </body>. Press Enter to make a new line. Type:

CODE EXAMPLE Copy online ↗

```
<p>I want to build a website for a small business.</p>
```

Save and refresh. You should now see two paragraphs beneath the greeting.

Notepad & VS Code 4/4

Understand the few tags you used

Say it back: “The title names the browser tab. The h1 is the main heading on the page. A p is a paragraph.”

TRY

add one more paragraph about a skill you want to learn. Change the words, not the surrounding tags.

Make a safe copy before changing more

YOU NEED
File Explorer.

A backup is a spare copy. It helps you return to a working page if an experiment goes wrong.

Copy the folder

1. Save your page and close Notepad.

2. Open Documents in File Explorer.

Notepad & VS Code 2/5

Make a safe copy before changing more

3 Click Website Class once. Press Ctrl + C, then Ctrl + V.

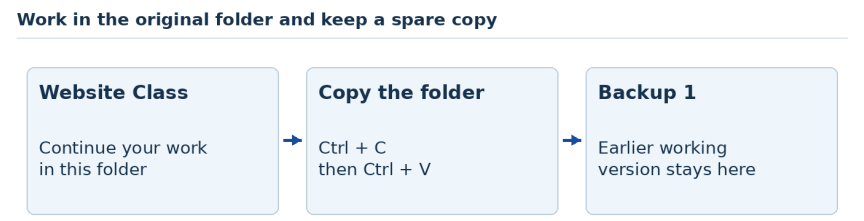
4 Windows creates a copy. Right-click that copied folder and choose Rename.

5 Call it Website Class Backup 1. Press Enter.

Notepad & VS Code 3/5

Make a safe copy before changing more

VISUAL EXPLANATION



Continue working in Website Class. Leave the backup alone.

A small confidence test

Open the original index.html in Notepad. Change the heading to I made this page. Save and open it in the browser.

Notepad & VS Code 4/5

Make a safe copy before changing more

Now open index.html inside Website Class Backup 1. It should still contain the earlier version. You have two separate copies.

If you lose your way: close the extra browser tabs and return to the original folder. Having five copies open at once makes it easy to edit the wrong file.

Notepad & VS Code 5/5

Make a safe copy before changing more

Ready to move on: open the correct file, change a sentence, save it and refresh without help. If this still feels difficult, repeat Computer lesson 3 and Setup lessons 1–2 before installing another tool.

Install Visual Studio Code

YOU NEED

Edge or Chrome. You need permission from the owner before installing on a shared computer.

VS Code is an editor, like Notepad, with extra help for writing code. It is not the same product as Visual Studio.

Download and install

1. Open your browser. Click the address bar at the top.

Notepad & VS Code 2/5

Install Visual Studio Code

2 Type https://code.visualstudio.com and press Enter.

3 Choose Download for Windows. Use the official website, not an advertisement from a search result.

4 Open the downloaded installer from the browser's Downloads list or your Downloads folder.

Install Visual Studio Code

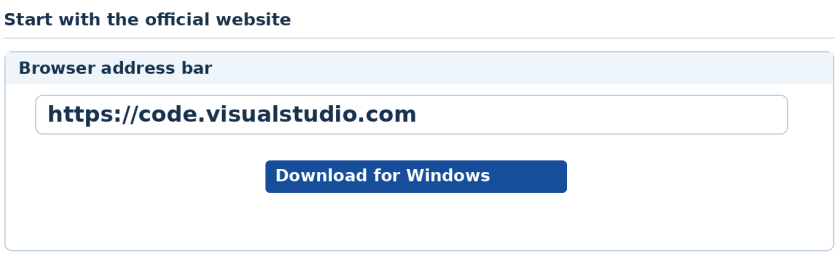
5 Read the setup screens. Accept the licence only if you agree, then follow Next and Install. Keep the default location unless you have a reason to change it. Choose the User Installer when it is offered for your personal Windows account.

6 Choose Launch Visual Studio Code when installation finishes, or open it from Start.

Notepad & VS Code 4/5

Install Visual Studio Code

VISUAL EXPLANATION



If setup says the computer is unsupported, open Windows Settings → System → About and compare your system with the linked requirements. Continue the Notepad lessons on that computer; use a supported system for later tools. Do not use unofficial old installers.

Notepad & VS Code 5/5

Install Visual Studio Code

You should see: the VS Code welcome screen.

There is no need to sign in or install an AI extension just to edit HTML files.

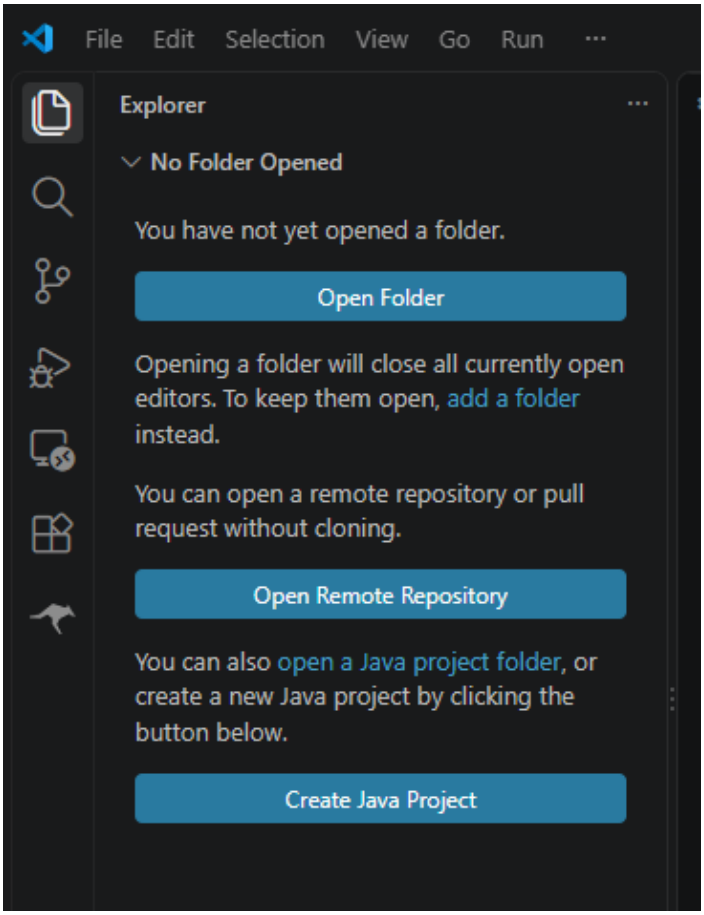
Sources: S2, S3.

Notepad & VS Code 1/6

Open your folder in VS Code

VS Code Explorer

REAL TOOL INTERFACE



Notepad & VS Code 2/6

Open your folder in VS Code

Click Open Folder. Choose Website Class, then choose the folder for this practical.

YOU NEED

VS Code and the original Website Class folder.

Find your page

1. In VS Code, click File → Open Folder.

2. Select Documents → Website Class. Click Select Folder.

Open your folder in VS Code

3 If asked about trust, trust it only because this is your own classroom folder.

4 On the left, click index.html. Its code opens in the large middle area.

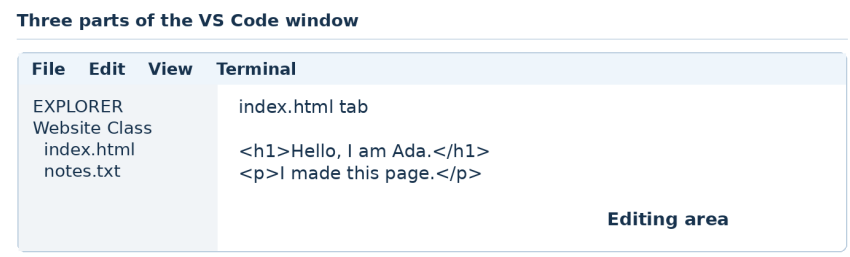
5 Change one sentence. Press Ctrl + S.

6 Open the page in your browser and refresh.

Notepad & VS Code 4/6

Open your folder in VS Code

VISUAL EXPLANATION



The left file list is called Explorer. A file's tab appears above the writing area. Clicking a tab switches between open files.

Notepad & VS Code 5/6

Open your folder in VS Code

Create a new file

At the top of Explorer, click the New File icon. If you cannot find it, right-click the folder name and choose New File. Type notes.txt and press Enter.

Type: This is my VS Code practice folder. Save.

You should see: index.html and notes.txt in the left file list.

Notepad & VS Code 6/6

Open your folder in VS Code

TRY

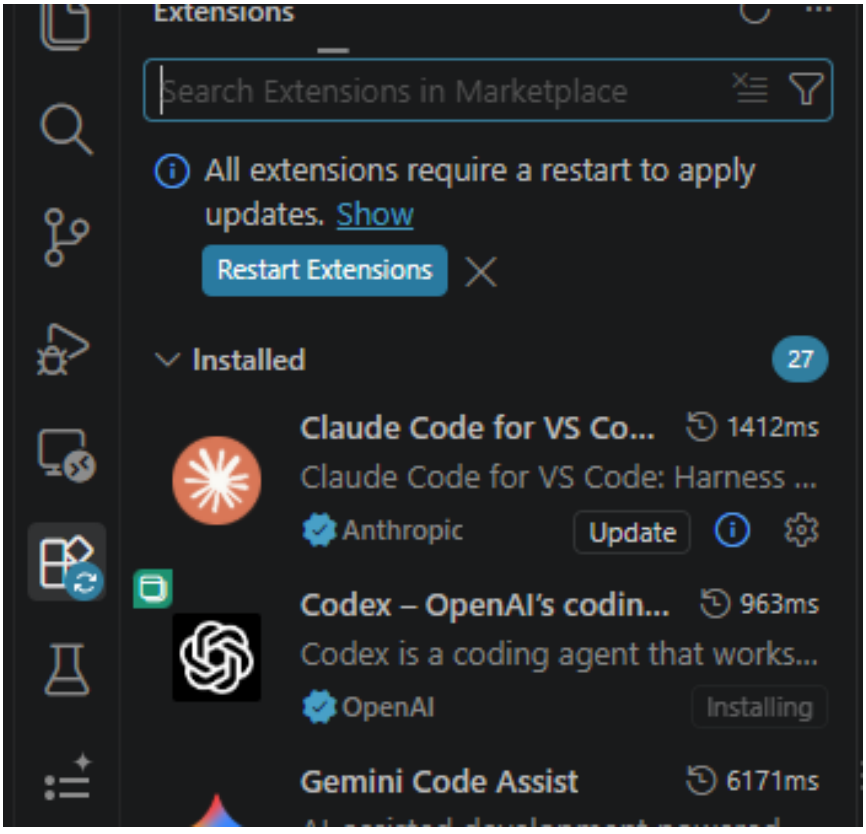
close notes.txt using the X on its tab. Open it again by clicking its name in Explorer.

Source: S4.

Add Live Server to VS Code

VS Code Extensions

REAL TOOL INTERFACE



Notepad & VS Code 2/6

Add Live Server to VS Code

Click the Extensions icon on the left. Use the search box at the top. For this lesson, search Live Server and check the publisher before installing.

YOU NEED

VS Code with Website Class open. Internet is needed for installation.

An extension adds a tool to VS Code. Live Server opens your HTML page and refreshes it when you save. It is separate from the AI tools we will add later.

Add Live Server to VS Code

Install the correct extension

1. Click the Extensions icon on the left. It looks like a group of small squares. Or press Ctrl + Shift + X.

2. In the search box, type Live Server.

3. Select Live Server by Ritwick Dey. Check the publisher before installing.

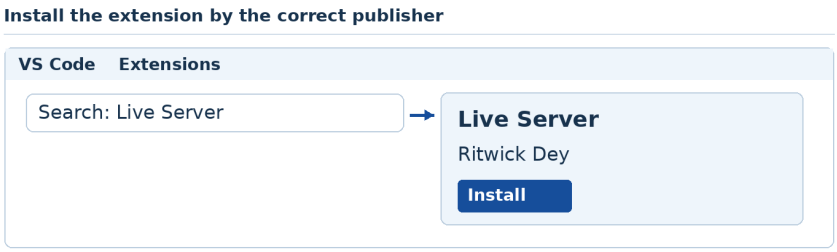
Notepad & VS Code 4/6

Add Live Server to VS Code

4 Click Install. Return to Explorer using the files icon.

5 Right-click index.html and choose Open with Live Server.

VISUAL EXPLANATION



Add Live Server to VS Code

Your browser should open the page at a local address. It may begin with 127.0.0.1 and include a number such as 5500. Use the address that actually opens on your computer.

Change one sentence in VS Code and press Ctrl + S. Watch the browser update.

Add Live Server to VS Code

If the menu option is missing: check that installation finished and you opened a .html file. Close and reopen VS Code if needed. You can still use the manual browser method from Setup lesson 2.

Microsoft Live Preview is another useful extension. We use Live Server here so the class follows one set of instructions. Neither tool by itself runs Laravel or Django.

Source: S5.

Notepad & VS Code 1/5

Get the practice files ready

Use File Explorer and VS Code. The ZIP is a companion to this handout, not a replacement for typing and understanding the examples.

1. Open Download your course files from Contents of this page and choose the practice ZIP.

2. In File Explorer, right-click the ZIP → Extract All. Choose a folder inside Documents → Website Class.

Notepad & VS Code 2/5

Get the practice files ready

3 Open the extracted Website_Design_Practice_Files folder. You should see html-1 through html-5, css-1 through css-5, javascript-1 through javascript-5, php-1 through php-5, sql and website-1 through website-5.

4 Make a separate folder called My Practice. Create your own exercise folders inside it using those same names.

Get the practice files ready

5 In VS Code, use File → Open Folder to open the folder for the current practical.

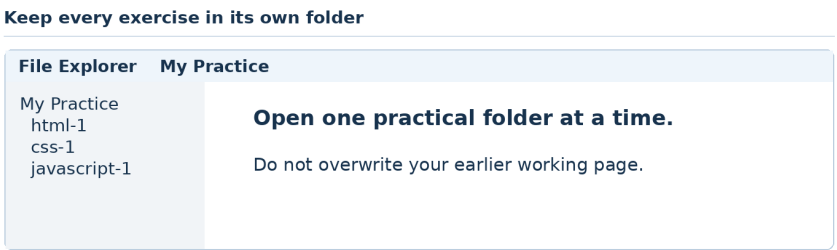
For learning, type the displayed code into your own files. Use the supplied answer folder when comparing or recovering a mistake. Every supplied HTML, CSS, JavaScript and PHP page is complete.

Notepad & VS Code 4/5

Get the practice files ready

The HTML snippets labelled “Inside body” replace only the content between <body> and </body> in the starter below. Keep its surrounding document structure. We show that repeated shell once so you can focus on each new lesson.

VISUAL EXPLANATION



Notepad & VS Code 5/5

Get the practice files ready

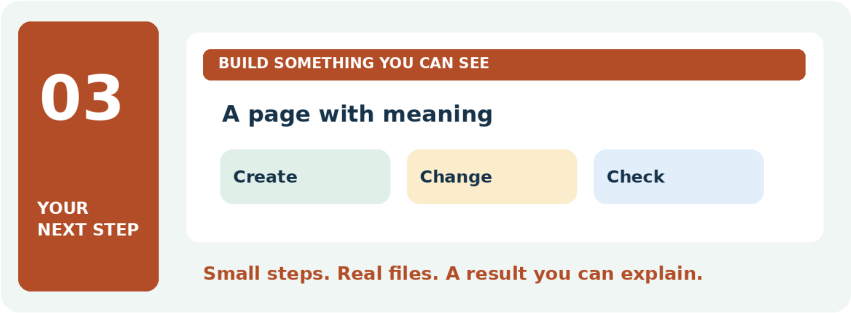
The books.svg image for HTML 3 is supplied. Copy it into your HTML 3 folder. It is an original classroom illustration, so no image search is required.

HTML 1/1

Part 03 HTML gives content a structure

Build five useful pages with headings, lists, images, links and forms.

VISUAL EXPLANATION



Your finish line

Finish a profile, menu, event page, registration form and club homepage.

HTML 1/4

Save and reopen the right file

1. In File Explorer, turn on file extensions: View → Show → File name extensions on Windows 11, or View → File name extensions on Windows 10.

2. Confirm your file is index.html, not index.html.txt. In Notepad, Save As → All files → index.html → UTF-8.

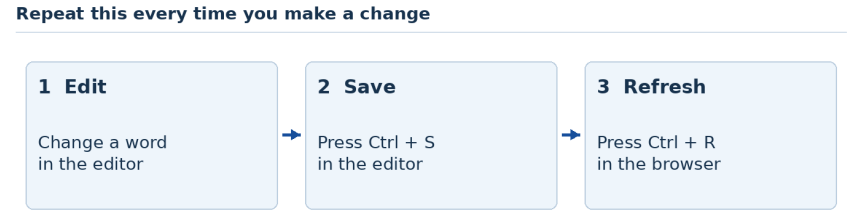
HTML 2/4

Save and reopen the right file

3 To edit: right-click index.html → Open with → Notepad. To view: right-click → Open with → Edge or Chrome.

4 After every edit, Ctrl + S in Notepad and Ctrl + R in the browser.

VISUAL EXPLANATION



HTML 3/4

Save and reopen the right file

Code appears as text

Check the .html ending and open it in a browser.

Old words remain

Save the editor file, then refresh the browser. Check the folder path.

Picture is missing

Match src to the exact filename, including its extension.

Link does nothing useful

Check href and confirm the destination file or id exists.

HTML 4/4

Save and reopen the right file

CHECK

Create a heading with your name. Close the page. Find the file again and reopen it. You have made a webpage you can keep.

HTML 1/4

Build the document shell

HTML describes content and its meaning. It is a markup language. Tags are not decoration commands.

Create index.html in your first practical folder. Use Notepad with Save as type set to All files, or VS Code. Start every HTML practical with this complete shell:

HTML 2/4

Build the document shell

CODE EXAMPLE Copy online ↗

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport"
        content="width=device-width, initial-scale=1">
  <title>My practice page</title>
</head>
<body>
  <!-- Put the practical's body content here. -->
</body>
</html>
```

Save and open it in the browser. It will be blank until you add body content. Change title to match each exercise.

HTML 3/4

Build the document shell

doctype and html

Use modern HTML and en for the document language.

head and title

Hold document information and the browser-tab title.

meta charset

Use UTF-8 character encoding.

meta viewport

Use the device width for the page viewport.

HTML 4/4

Build the document shell

body

Contains the visible page content.

<!-- -->

A code comment; it is visible in source, so never put secrets there.

HTML 1/5

Read tags and attributes

An element usually has an opening tag, content and a closing tag. Close inner elements before outer ones.

VISUAL EXPLANATION

The tag names the element and the attribute adds detail

```
<a href="about.html">About</a>
```

a: link element href: destination About: text

HTML 2/5

Read tags and attributes

In About, a is the tag name. href is an attribute. Its value gives the destination. About is the visible text.

Use h1 for the page's main topic, h2 for sections and h3 for a smaller topic inside a section. These levels describe structure. Use CSS later to change visual sizes.

HTML 3/5

Read tags and attributes

Some elements, including img, input, meta and link, do not have closing tags. Do not add </img> or </input>.

Your five HTML levels

1. Text and headings: a personal introduction.

2. Lists and table structure: a restaurant menu.

Read tags and attributes

3 Links, images and meaningful sections: an event page.

4 Form controls and labels: workshop registration.

5 Navigation and interactive native elements: a club page.

HTML 5/5

Read tags and attributes

After each practical, save and refresh. Right-click the page → View page source to compare the markup. A browser may recover from invalid HTML, so you can also visit https://validator.w3.org/nu/, choose File upload, select your fictional practice file and click Check.

References: N6 and the W3C page-structure source S36.

HTML 1/5

A personal introduction page

YOUR GOAL

Make a simple profile page for a student or craft worker.

Preview live ↗ Copy files ↗

Open html-1 in VS Code. Put the HTML below inside the body of index.html.

Save your files. Right-click index.html → Open with Live Server. After each edit, save and observe the browser.

HTML 2/5

A personal introduction page

In index.html: inside <body>

CODE EXAMPLE Copy online ↗

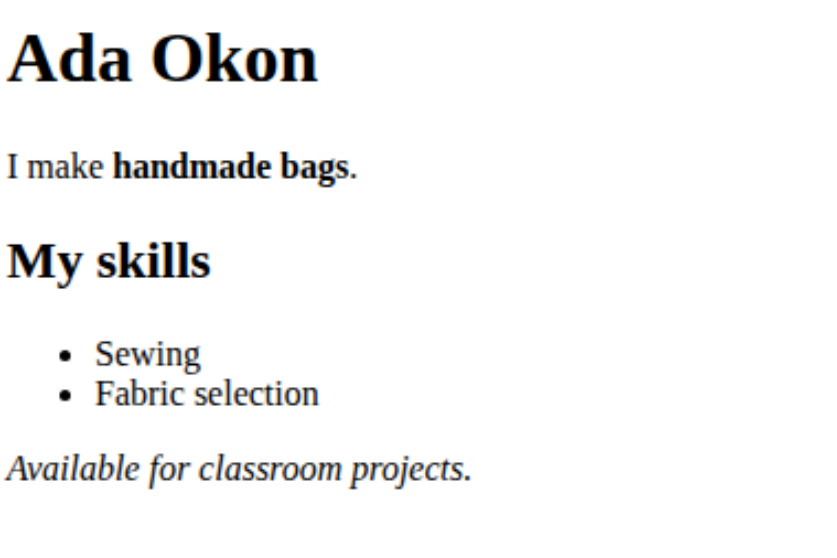
```
<main>
  <h1>Ada Okon</h1>
  <p>I make <strong>handmade bags</strong>.</p>
  <h2>My skills</h2>
  <ul>
    <li>Sewing</li>
    <li>Fabric selection</li>
  </ul>
  <p><em>Available for classroom projects.</em></p>
</main>
```

HTML 3/5

A personal introduction page

Starting screen

REAL WEBSITE OUTPUT



main

Contains the main content.

HTML 4/5

A personal introduction page

h1 and h2

Name the page and introduce a section.

p

Groups a paragraph of text.

strong and em

Mark importance and emphasis; usually bold and italic.

ul and li

Create an unordered list and its individual items.

HTML 5/5

A personal introduction page

TRY

Replace the name and skills with your own. Add a third skill.

CHECK

Name is visible. Two original skills appear as separate bullet points.

FIX IT

Match opening and closing tags. Check filenames in src and href. Save, then refresh.

HTML 1/5

A restaurant menu page

YOUR GOAL

Present real-world menu information with categories and prices.

Preview live ↗ Copy files ↗

Open html-2 in VS Code. Put the HTML below inside the body of index.html.

HTML 2/5

A restaurant menu page

In index.html: inside <body>

CODE EXAMPLE Copy online ↗

```
<header><h1>Market Kitchen</h1></header>
<main>
  <h2>Lunch menu</h2>
  <table>
    <caption>Sample prices in NGN</caption>
    <tr><th scope="col">Meal</th><th scope="col">Price</
th></tr>
    <tr><td>Jollof rice</td><td>2500</td></tr>
    <tr><td>Vegetable soup</td><td>3000</td></tr>
  </table>
  <h2>How to order</h2>
  <ol><li>Choose a meal.</li><li>Ask the attendant.</li>
</ol>
</main>
<footer><p>Classroom menu only.</p></footer>
```

HTML 3/5

A restaurant menu page

Starting screen

REAL WEBSITE OUTPUT



Market Kitchen

Lunch menu

Sample prices in NGN

Meal	Price
Jollof rice	2500
Vegetable soup	3000

How to order

1. Choose a meal.
2. Ask the attendant.

Classroom menu only.

HTML 4/5

A restaurant menu page

header, main and footer

Separate the introduction, main content and closing note.

h1, h2 and p

Give the page and sections names, then normal text.

table and caption

Create and name a data table; not a page-layout tool.

tr, th and td

Make a row, heading cell and ordinary data cell.

HTML 5/5

A restaurant menu page

scope="col"

Associates a heading with its column.

ol and li

Make ordered steps and each step.

TRY

Add a third meal and its price. Keep each price in the same row as its meal.

CHECK

Two meals and two prices appear. The ordering instructions are numbered.

HTML 1/6

An event information page

YOUR GOAL

Make an event page with a working internal link and meaningful image text.

Preview live ↗ Copy files ↗

Open html-3 in VS Code. Put the HTML below inside the body of index.html.

HTML 2/6

An event information page

In index.html: inside <body>

CODE EXAMPLE Copy online ↗

```
<header>
  <h1>Community Reading Day</h1>
  <a href="#details">See event details</a>
</header>
<main>
  <figure>
    <img src="books.svg" alt="Three books standing toget
her"
         width="300" height="150">
    <figcaption>Bring a book to share.</figcaption>
  </figure>
  <section id="details">
    <h2>Event details</h2>
    <p><time datetime="2027-03-20">20 March 2027</time><
/p>
    <p>Practice venue: Community Hall.</p>
  </section>
</main>
```

HTML 3/6

An event information page

Starting screen

REAL WEBSITE OUTPUT

Community Reading Day

See event details



Bring a book to share.

Event details

20 March 2027

Practice venue: Community Hall.

HTML 4/6

An event information page

header, main and section

Group introductory, main and related section content.

h1, h2 and p

Supply the visible heading hierarchy and paragraph.

a and href="#details"

Link to the element whose id is details; a short page may barely scroll.

figure and figcaption

Keep an illustration and its caption together.

HTML 5/6

An event information page

img, src and alt

Load an image file and provide equivalent useful text.

width and height

Reserve the image dimensions; CSS can later make it responsive.

time and datetime

Show a human-readable date with a machine-readable value.

id

Identifies this section uniquely on the page.

HTML 6/6

An event information page

TRY

Change the event and date in both visible text and datetime. Check that they agree.

CHECK

The image appears. The link changes the URL fragment to #details.

HTML 1/6

A workshop registration form

YOUR GOAL

Build correctly labelled fields and observe browser validation.

Preview live ↗ Copy files ↗

Open html-4 in VS Code. Put the HTML below inside the body of index.html.

HTML 2/6

A workshop registration form

In index.html: inside <body>

CODE EXAMPLE Copy online ↗

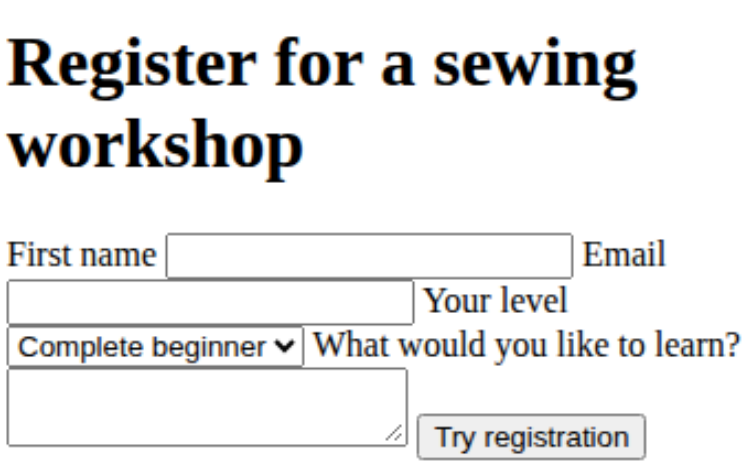
```
<main>
  <h1>Register for a sewing workshop</h1>
  <form action="index.html" method="get">
    <label for="name">First name</label>
    <input id="name" name="name" required>
    <label for="email">Email</label>
    <input id="email" name="email" type="email" required
>
    <label for="level">Your level</label>
    <select id="level" name="level">
      <option value="new">Complete beginner</option>
      <option value="some">Some experience</option>
    </select>
    <label for="note">What would you like to learn?</lab
el>
    <textarea id="note" name="note"></textarea>
    <button type="submit">Try registration</button>
  </form>
</main>
```

HTML 3/6

A workshop registration form

Starting screen

REAL WEBSITE OUTPUT



main and h1

Identify the main content and form purpose.

HTML 4/6

A workshop registration form

form, action and method

Submit to index.html using GET; this demo only reloads with query values.

label and for

Connect a visible label to an input id.

input, id and name

Create a field, identify it and name the submitted value.

type="email" and required

Apply browser email-format and missing-value checks.

HTML 5/6

A workshop registration form

select and option

Offer a menu; value is the submitted choice.

textarea

Provides a multiline writing field.

button type="submit"

Submits the enclosing form.

TRY

Add a labelled phone field using type="tel". Do not collect real personal information.

HTML 6/6

A workshop registration form

CHECK

Empty required fields are blocked. A valid fictional entry appears in the URL, but is not stored.

HTML 1/6

A school club home page

YOUR GOAL

Build navigation, a self-contained article and an expandable answer.

Preview live ↗ Copy files ↗

Open html-5 in VS Code. Put the HTML below inside the body of index.html.

HTML 2/6

A school club home page

In index.html: inside <body>

CODE EXAMPLE Copy online ↗

```
<header>
  <h1>Bright Readers Club</h1>
  <nav aria-label="Main navigation">
    <a href="index.html">Home</a>
    <a href="about.html">About</a>
  </nav>
</header>
<main>
  <article>
    <h2>This week</h2>
    <p>Bring one short story for discussion.</p>
  </article>
  <details>
    <summary>Do I need to buy a book?</summary>
    <p>No. Bring a book you already have.</p>
  </details>
</main>
```

Continue the same practical. Keep the earlier lines; add the following lines in the named file.

HTML 3/6

A school club home page

about html

CODE EXAMPLE Copy online ↗

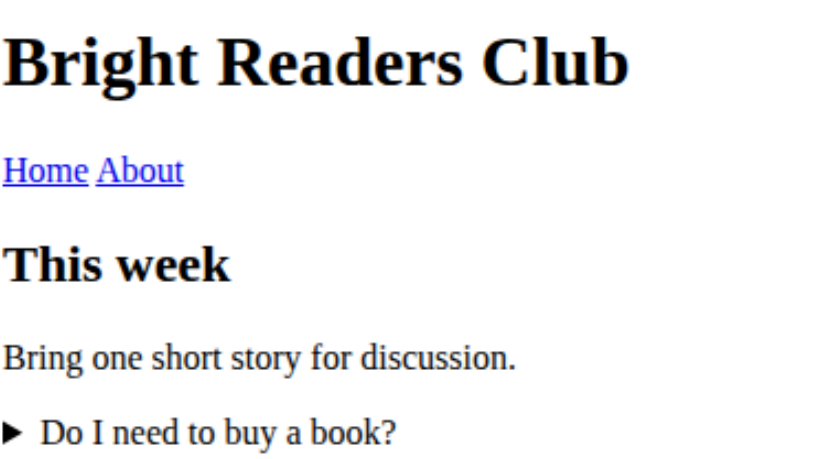
```
<!doctype html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport"
        content="width=device-width, initial-scale=1">
  <title>About our club</title>
</head>
<body>
<main><h1>About our club</h1>
<p>We practise reading together.</p>
<a href="index.html">Home</a></main>
</body>
</html>
```

HTML 4/6

A school club home page

Starting screen

REAL WEBSITE OUTPUT



header, main, h1, h2 and p

Organise the page and its readable content.

HTML 5/6

A school club home page

nav and aria-label

Identify and name a navigation region.

a and relative href values

Link to files in the same folder.

article

Groups a self-contained club update.

details and summary

Provide an expandable answer with a clickable summary.

HTML 6/6

A school club home page

TRY

Add a Contact link and contact.html. Test the return link to Home.

CHECK

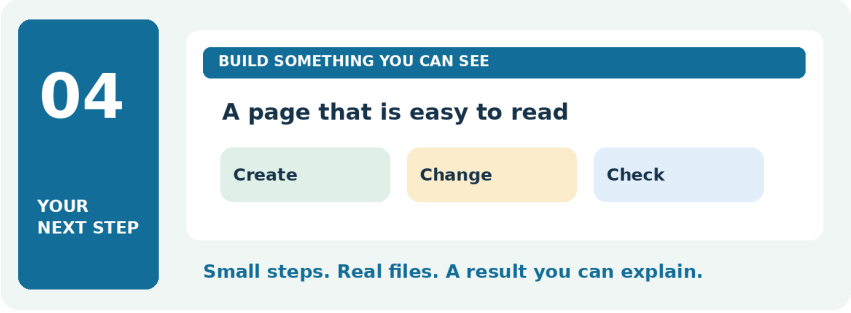
About opens and Home returns. The question expands without JavaScript.

CSS 1/1

Part 04 CSS gives the page its appearance

Use colour, spacing and layout to make the same content easier to read.

VISUAL EXPLANATION



Your finish line

Build a business introduction, product card, navigation, course grid and enquiry form.

CSS 1/4

See which rule changes the page

1. Open the practical’s folder in VS Code. index.html and style.css should appear together in Explorer.

2. Confirm the head contains <link rel="stylesheet" href="style.css">. This course uses style.css in the CSS practicals and styles.css in the later AI starter. Use the name shown for that project.

CSS 2/4

See which rule changes the page

3 Change the body colour to #126b59. Save and refresh. If nothing changes, check the filename before changing more code.

4 Right-click a visible heading in the browser → Inspect. The Styles panel shows rules that apply to it. Untick one rule, then tick it again.

CSS 3/4

See which rule changes the page

VISUAL EXPLANATION

Read a CSS rule as a simple instruction

```
.card { padding: 24px; }
```

Selector: .card Property: padding Value: 24px

FIX IT

Edits made in Inspect are temporary. Make the same change in your saved CSS file to keep it. A rule with a line through it is being overridden or cannot apply.

CSS 4/4

See which rule changes the page

Your small experiment: try padding: 8px and then padding: 32px. Watch the space inside the box change. Restore the value from the practical before continuing.

CSS

Select and style the page

CSS controls appearance. It does not replace HTML's meaning. Use a separate style.css file connected by a link element in head.

VISUAL EXPLANATION

Read a CSS rule as a simple instruction

```
.card { padding: 24px; }
```

Selector: .card | Property: padding | Value: 24px

CSS

Select and style the page

A selector chooses the elements. A declaration gives a property and value. Braces surround the declarations. Colons separate properties from values; semicolons separate declarations.

h1 chooses heading elements. .card chooses elements with class="card". #notice chooses the element whose id is notice. Reuse a class; keep an id unique in a page.

CSS 3/5

Select and style the page

A later rule can win when competing rules have equal priority and specificity. A class selector is more specific than an element selector in an ordinary style sheet. Inspect an element in the browser to see which rule won. Do not add !important whenever you are confused.

Your five CSS levels

1. Typography and colour.

CSS 4/5

Select and style the page

2 Box model, borders and spacing.

3 Flexbox, navigation and interaction states.

4 Grid and responsive breakpoints.

5 Form styling and reusable custom properties.

CSS 5/5

Select and style the page

Only CSS is the new subject here. Each practical supplies a small HTML fixture to give the styles something to act on. Source: N11.

CSS 1/6

A clear business introduction

YOUR GOAL

Style a service heading and short description.

Preview live ↗ Copy files ↗

Open css-1 in VS Code. Put the HTML below inside the body of index.html.

In head, add <link rel="stylesheet" href="style.css">. Create style.css beside index.html.

CSS 2/6

A clear business introduction

In index.html: inside <body>

CODE EXAMPLE Copy online ↗

```
<main class="intro">
  <h1>Clean Home Services</h1>
  <p>Careful cleaning for small offices.</p>
  <a href="mailto:hello@example.com">Ask a question</a>
</main>
```

style.css

CODE EXAMPLE Copy online ↗

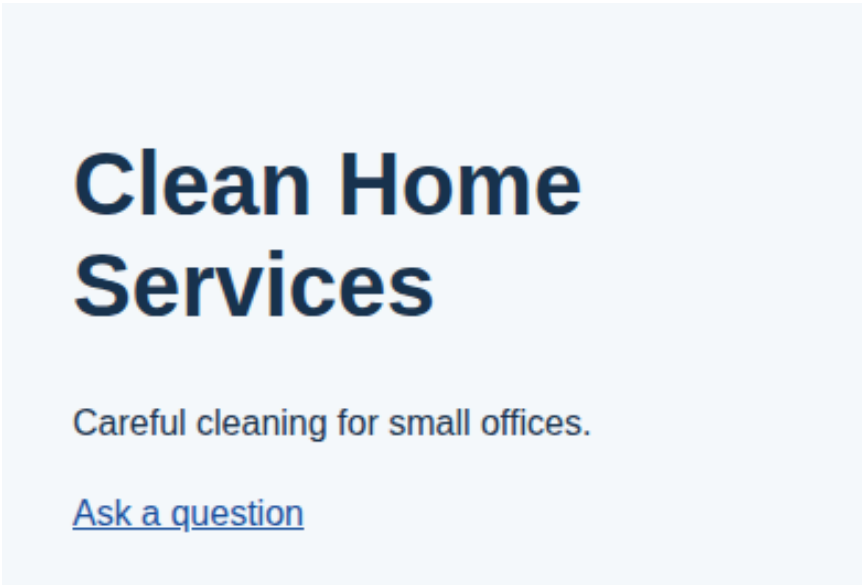
```
body {
  font-family: Arial, sans-serif;
  color: #17324d;
  background: #f4f8fb;
  line-height: 1.6;
  padding: 24px;
}
h1 { font-size: 2.5rem; line-height: 1.15; }
.intro { max-width: 700px; margin: auto; }
a { color: #164e9b; }
```

CSS 3/6

A clear business introduction

Starting screen

REAL WEBSITE OUTPUT



body, h1 and a selectors

Choose all matching elements.

CSS

A clear business introduction

.intro

Chooses the element with class="intro".

font-family and font-size

Set the font and text size. rem scales from the page’s base text size.

color and background

Set foreground and background colours.

line-height

Sets comfortable line spacing.

CSS

A clear business introduction

padding, max-width and margin

Add inner space, limit line length and centre the block.

HTML main, h1, p and a

Contain the introduction, heading, sentence and mail link; mailto opens a configured email app.

TRY

Change the brand colour in a and compare. Keep text contrast readable.

CSS 6/6

A clear business introduction

CHECK

The heading is larger than the paragraph. Content remains readable when the browser narrows.

FIX IT

Check the linked CSS filename first. Then check the selector spelling, braces and semicolons.

CSS 1/6

A product card with breathing room

YOUR GOAL

Use the box model to make a merchandise card.

Preview live ↗ Copy files ↗

Open css-2 in VS Code. Put the HTML below inside the body of index.html.

CSS 2/6

A product card with breathing room

In index.html: inside <body>

CODE EXAMPLE Copy online ↗

```
<article class="card">
  <h1>Canvas tote bag</h1>
  <p>Reusable bag for everyday essentials.</p>
  <p class="price">Sample price: NGN 6000</p>
</article>
```

style.css

CODE EXAMPLE Copy online ↗

```
* { box-sizing: border-box; }
body { font-family: Arial; padding: 24px; }
.card {
  max-width: 360px;
  margin: 24px auto;
  padding: 24px;
  border: 2px solid #164e9b;
  border-radius: 12px;
  background: #eef5fb;
}
.price { font-weight: bold; }
```

CSS 3/6

A product card with breathing room

Starting screen

REAL WEBSITE OUTPUT



CSS 4/6

A product card with breathing room

* and box-sizing

Apply border-box sizing to all elements so declared width includes padding and border.

body and .card

Choose the page body and the card class.

padding and margin

Space inside the card versus space outside it.

border and border-radius

Draw an edge and round its corners.

CSS 5/6

A product card with breathing room

max-width and background

Limit card width and fill its background.

.price and font-weight

Make the price paragraph bold.

HTML article, h1, p and class

Group a standalone product, name it, describe it and attach style hooks.

TRY

Change padding from 24px to 4px, observe, then restore it.

CSS 6/6

A product card with breathing room

CHECK

Text never touches the border.
The price is bold.

CSS 1/6

A flexible navigation bar

YOUR GOAL

Arrange a studio name and navigation without fixed screen widths.

Preview live ↗ Copy files ↗

Open css-3 in VS Code. Put the HTML below inside the body of index.html.

CSS 2/6

A flexible navigation bar

In index.html: inside <body>

CODE EXAMPLE Copy online ↗

```
<header class="bar">
  <strong>Thread Studio</strong>
  <nav aria-label="Main">
    <a href="#services">Services</a>
    <a href="#contact">Contact</a>
  </nav>
</header>
<main>
  <h1 id="services">Tailoring services</h1>
  <p id="contact">Practice contact: hello@example.com</p
>
</main>
```

CSS 3/6

A flexible navigation bar

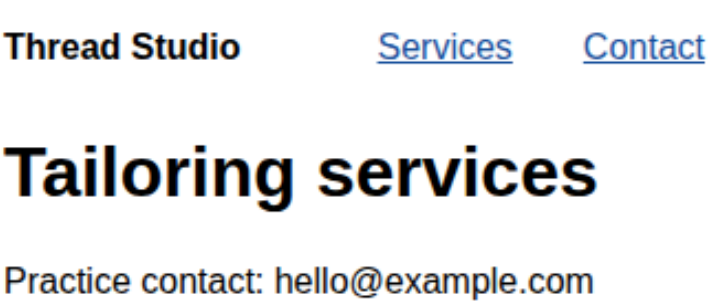
style.css

CODE EXAMPLE Copy online ↗

```
body { font-family: Arial; padding: 24px; }
.bar, nav { display: flex; gap: 16px; flex-wrap: wrap; }
.bar { justify-content: space-between; align-items: cent
er; }
a { color: #164e9b; padding: 8px; }
a:hover { background: #eef5fb; }
a:focus-visible { outline: 3px solid #164e9b; }
```

Starting screen

REAL WEBSITE OUTPUT



CSS

A flexible navigation bar

.bar, nav and comma

Apply the same rule to two selectors.

display: flex

Lay out direct children along an axis.

gap and flex-wrap

Add space and allow wrapping.

justify-content and align-items

Distribute items along the main axis and align them across it.

CSS

A flexible navigation bar

:hover and :focus-visible

Style pointer hover and visible keyboard focus.

outline, padding, color and background

Provide clear focus, space and colour cues.

HTML header, strong, nav, main, h1, p and a

Build a named studio header, navigation and two link targets; id matches each fragment.

CSS 6/6

A flexible navigation bar

TRY

Add a third navigation link and matching section id.

CHECK

Links wrap rather than overflow. Both links work using Tab then Enter.

CSS 1/6

A responsive course card grid

YOUR GOAL

Display training options in one column on a phone and three on a wide screen.

Preview live ↗ Copy files ↗

Open css-4 in VS Code. Put the HTML below inside the body of index.html.

CSS 2/6

A responsive course card grid

In index.html: inside <body>

CODE EXAMPLE Copy online ↗

```
<main>
  <h1>Weekend classes</h1>
  <div class="courses">
    <article><h2>Sewing</h2><p>Start with a simple bag.</p></article>
    <article><h2>Baking</h2><p>Make your first cupcakes.</p></article>
    <article><h2>Web design</h2><p>Build a first page.</p></article>
  </div>
</main>
```

style.css

CODE EXAMPLE Copy online ↗

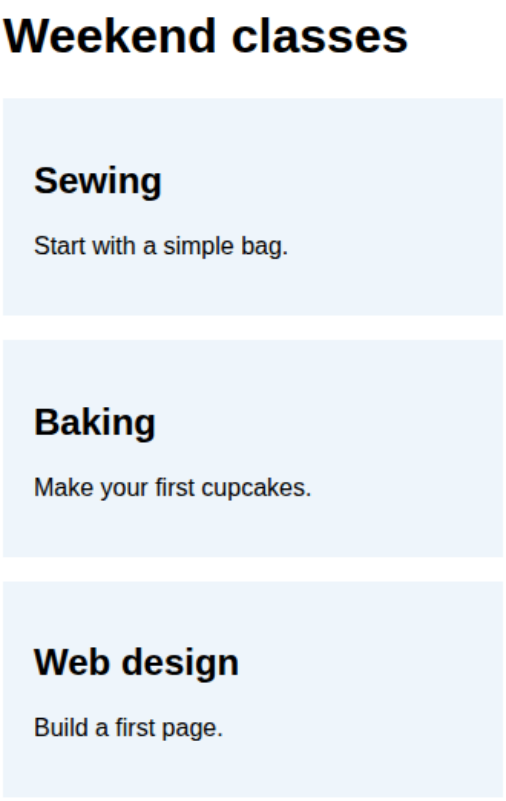
```
body { font-family: Arial; padding: 24px; }
.courses { display: grid; gap: 16px; }
article { padding: 20px; background: #eef5fb; }
@media (min-width: 700px) {
  .courses { grid-template-columns: repeat(3, 1fr); }
}
```

CSS 3/6

A responsive course card grid

Starting screen

REAL WEBSITE OUTPUT



CSS

A responsive course card grid

display: grid and gap

Create a grid with spacing between cells.

@media and min-width

Apply the enclosed rule when the viewport is at least 700 CSS pixels wide.

grid-template-columns

Define the columns.

repeat(3, 1fr)

Create three equal fractional columns.

CSS 5/6

A responsive course card grid

body and article rules

Set type and breathing room; give each card a fill.

HTML main, h1, div, article, h2 and p

Name the page, group the layout and describe each course; .courses selects the div.

TRY

Add a fourth course. Observe the second row at wide sizes.

CSS 6/6

A responsive course card grid

CHECK

Below 700px: one column. At 700px or wider: three columns.

CSS 1/7

A readable enquiry form

YOUR GOAL

Style labels and controls without changing their behaviour.

Preview live ↗ Copy files ↗

Open css-5 in VS Code. Put the HTML below inside the body of index.html.

CSS

A readable enquiry form

In index.html: inside <body>

CODE EXAMPLE

Copy online ↗

```
<main>
  <h1>Ask about a class</h1>
  <form action="index.html" method="get">
    <label for="name">First name</label>
    <input id="name" name="name" required>
    <label for="message">Your question</label>
    <textarea id="message" name="message" required></tex
tarea>
    <button>Try the form</button>
  </form>
</main>
```

CSS 3/7

A readable enquiry form

style.css

CODE EXAMPLE Copy online ↗

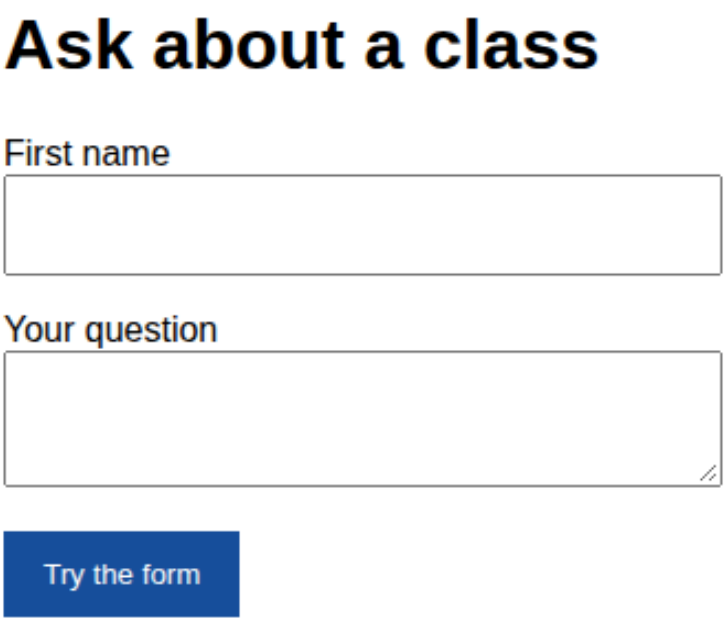
```
:root { --brand: #164e9b; }
* { box-sizing: border-box; }
body { font-family: Arial; padding: 24px; }
form { max-width: 480px; }
label { display: block; margin-top: 16px; }
input, textarea { width: 100%; padding: 12px; font: inhe
rit; }
button {
  margin-top: 16px; padding: 12px 18px;
  color: white; background: var(--brand); border: 0;
}
input:focus-visible, textarea:focus-visible,
button:focus-visible { outline: 3px solid #9c5c00; }
```

CSS 4/7

A readable enquiry form

Starting screen

REAL WEBSITE OUTPUT



CSS 5/7

A readable enquiry form

:root, --brand and var()

Define and reuse a custom colour property.

box-sizing, width and max-width

Keep controls within the form width including their padding.

display: block and margin-top

Put each label on its own line and separate fields.

font: inherit

Use the surrounding font in controls.

CSS

A readable enquiry form

padding, color, background and border

Give comfortable controls and a distinct button.

:focus-visible and outline

Show which control the keyboard is using.

HTML form, label, input, textarea and button

Use the same field-and-label rules from HTML 4; the default button type in a form is submit.

CSS 7/7

A readable enquiry form

TRY

Change --brand to #166649. Test keyboard focus and empty submission again.

CHECK

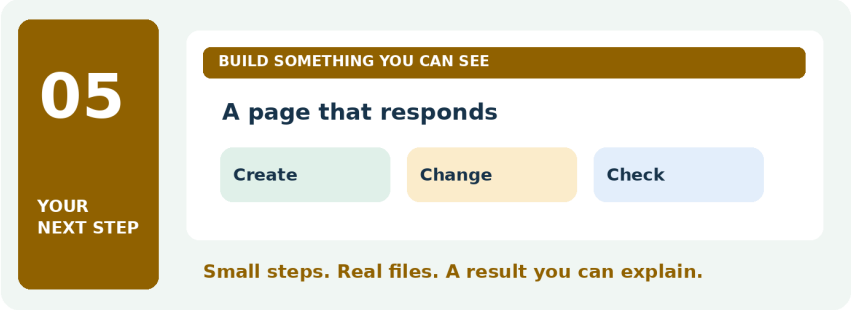
Labels are above their fields. Controls fit a narrow screen.

JavaScript 1/1

Part 05 JavaScript makes a page respond

Connect a visitor’s action to a visible result.

VISUAL EXPLANATION



Your finish line

Make a greeting, calculator, catalogue, answer toggle and search filter.

JavaScript 1/3

Find the first error

1. Save index.html and app.js. Confirm the HTML includes the script line shown in the practical.

2. Open your page in the browser. Press F12, then click Console. If F12 changes volume, try Fn + F12 or Ctrl + Shift + J.

3. Click the page’s button. If a red error appears, read the first one. Click its filename and line number to locate the problem.

JavaScript 2/3

Find the first error

4 Compare the spelling of the id in HTML with the #name used by querySelector. They must match.

Unexpected token

Missing quote, bracket or extra punctuation near the reported line.

Cannot read properties of null

The selector found no element. Check its spelling and script position.

JavaScript 3/3

Find the first error

Button does nothing

Confirm the script filename, save both files and refresh.

Code works only once

Repeat the action and check which value changes.

TRY

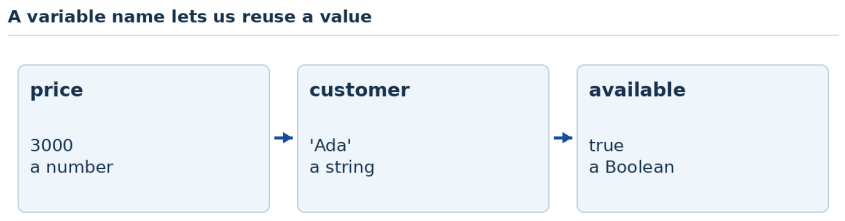
Temporarily misspell an id in a copy of your project. Observe the error, restore the spelling and confirm the button works. You have practised fixing a real bug.

JavaScript 1/4

Values decisions and actions

JavaScript can respond to a visitor inside the browser. Use app.js connected by a script element with defer. The script file itself does not contain <script> tags.

VISUAL EXPLANATION



JavaScript 2/4

Values decisions and actions

const holds a binding you do not reassign. let allows reassignment. Text uses quotes; numbers do not. true and false are Boolean values. = assigns; === compares strictly.

A function groups instructions. Calling it runs them. if chooses whether to run a block. Arrays hold lists; objects hold named details. Events such as click and input tell us when a visitor acts.

JavaScript 3/4

Values decisions and actions

Braces group blocks or define objects depending on context. Parentheses hold function arguments or conditions. Square brackets define arrays or access values. Semicolons end the statements in these examples.

Practise before the projects

Open a practice page, press F12 and choose Console. Type console.log(2 + 3), then Enter. Expect 5. Type console.log('2' + 3). Expect 23 because a string was joined. Close the Console when done.

JavaScript 4/4

Values decisions and actions

Our five levels are a greeting, a calculator, an array-driven list, state that toggles, and live filtering. In each case, inspect the real page result. Source: N7.

JavaScript 1/8

A visitor greeting on a page

YOUR GOAL

Read a name from a field and update the page safely.

Preview live ↗ Copy files ↗

Open javascript-1 in VS Code. Put the HTML below inside the body of index.html.

Just before </body>, add <script src="app.js" defer></script>. Create app.js beside index.html.

JavaScript 2/8

A visitor greeting on a page

In index.html: inside <body>

CODE EXAMPLE Copy online ↗

```
<main>
  <h1>Welcome to our workshop</h1>
  <label for="name">First name</label>
  <input id="name">
  <button id="greet" type="button">Greet me</button>
  <p id="result" aria-live="polite"></p>
</main>
```

JavaScript 3/8

A visitor greeting on a page

app.js

CODE EXAMPLE Copy online ↗

```
const nameField = document.querySelector('#name');
const button = document.querySelector('#greet');
const result = document.querySelector('#result');
button.addEventListener('click', function () {
  const name = nameField.value.trim();
  if (name === '') {
    result.textContent = 'Please enter your first name.'
;
    return;
  }
  result.textContent = 'Hello, ' + name + '!';
});
```

JavaScript 4/8

A visitor greeting on a page

Starting screen

REAL WEBSITE OUTPUT



After trying the example

REAL WEBSITE OUTPUT



JavaScript 5/8

A visitor greeting on a page

const and =

Give a value a name. This example does not assign a new value to that name.

document.querySelector and

Find part of the page using its id. The DOM is the browser's working model of the page.

addEventListener, click and function

Run the enclosed instructions when the button is clicked.

JavaScript 6/8

A visitor greeting on a page

value and trim()

Read the typed text. Remove spaces from its beginning and end.

if, === and return

Check whether the field is empty. If it is, stop this action.

textContent and +

Join words and show them as text. A visitor’s typed HTML stays text.

JavaScript 7/8

A visitor greeting on a page

HTML label, input, button, p and aria-live

Collect a name, trigger an action and announce changing text politely; main and h1 identify the page.

TRY

Change the greeting wording. Try a name containing <b> and observe literal text.

CHECK

Ada produces Hello, Ada!. Spaces only produce the missing-name message.

JavaScript 8/8

A visitor greeting on a page

FIX IT

Open Console and read its first error. Check the script filename and the matching HTML ids.

JavaScript 1/8

A printing cost calculator

YOUR GOAL

Multiply a valid quantity by a fixed classroom unit price.

Preview live ↗ Copy files ↗

Open javascript-2 in VS Code. Put the HTML below inside the body of index.html.

JavaScript 2/8

A printing cost calculator

In index.html: inside <body>

CODE EXAMPLE Copy online ↗

```
<main>
  <h1>Print estimate</h1>
  <label for="quantity">Number of flyers</label>
  <input id="quantity" type="number" min="1" step="1">
  <button id="calculate" type="button">Calculate</button
>
  <p id="result" aria-live="polite"></p>
</main>
```

JavaScript 3/8

A printing cost calculator

app.js

CODE EXAMPLE Copy online ↗

```
const field = document.querySelector('#quantity');
const result = document.querySelector('#result');
function totalFor(quantity, price) {
  return quantity * price;
}
document.querySelector('#calculate').addEventListener('c
lick', function () {
  const quantity = Number(field.value);
  if (!Number.isInteger(quantity) || quantity < 1 || qua
ntity > 1000) {
    result.textContent = 'Enter a whole number from 1 to
 1000.';
    return;
  }
  result.textContent = 'Sample estimate: NGN ' + totalFo
r(quantity, 100);
});
```

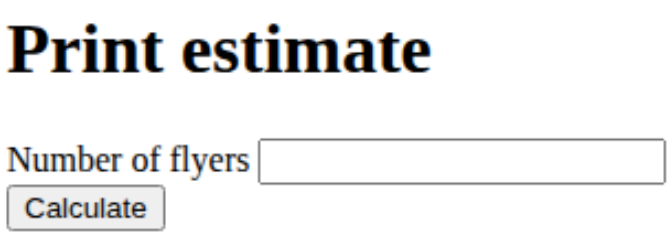
For the second screenshot, enter 10 and click Calculate. The result comes from the original JavaScript.

JavaScript 4/8

A printing cost calculator

Starting screen

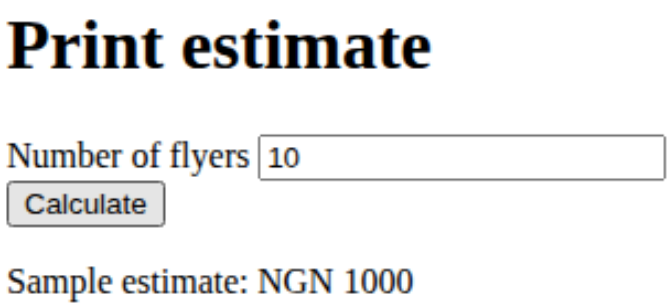
REAL WEBSITE OUTPUT



A printing cost calculator

After trying the example

REAL WEBSITE OUTPUT



querySelector, const and addEventListener

Find controls, retain references and respond to a click.

JavaScript 6/8

A printing cost calculator

function, parameters and return

Define a reusable calculation receiving quantity and price.

Number() and field.value

Convert a field string to a number; an empty string becomes 0 and is rejected by the range test.

Number.isInteger(), ! and / /

Reject a non-integer or any value outside the range.

JavaScript 7/8

A printing cost calculator

<, > and *

Compare lower/upper limits and multiply values.

textContent and +

Show an error or join the label to the calculated total.

HTML number input, min and step

Suggest whole positive values; JavaScript still checks them. Label, button and live paragraph provide accessible interaction.

JavaScript 8/8

A printing cost calculator

TRY

Change the sample price to 150 and calculate 20 flyers. Expected: 3000.

CHECK

50 gives 5000. Empty, 0, 1.5 and 1001 are rejected.

JavaScript 1/5

A small service catalogue

YOUR GOAL

Render an array of service objects as page content.

Preview live ↗ Copy files ↗

Open javascript-3 in VS Code. Put the HTML below inside the body of index.html.

In index.html: inside <body>

CODE EXAMPLE Copy online ↗

```
<main>
  <h1>Our services</h1>
  <ul id="services"></ul>
</main>
```

JavaScript 2/5

A small service catalogue

app.js

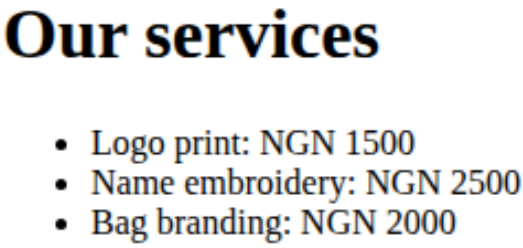
CODE EXAMPLE Copy online ↗

```
const services = [
  { name: 'Logo print', price: 1500 },
  { name: 'Name embroidery', price: 2500 },
  { name: 'Bag branding', price: 2000 }
];
const list = document.querySelector('#services');
for (const service of services) {
  const item = document.createElement('li');
  item.textContent = service.name + ': NGN ' + service.p
rice;
  list.append(item);
}
```

A small service catalogue

Starting screen

REAL WEBSITE OUTPUT



[ ] and commas

Define an array containing several values.

{ name, price } and colon

Define an object with named properties for one service.

JavaScript 4/5

A small service catalogue

const, querySelector and #services

Keep the catalogue and find the destination list.

for...of

Repeat the block once for each service object.

createElement, li and append

Create an HTML list item and attach it to the list.

service.name and service.price

Read object properties using dot notation.

JavaScript 5/5

A small service catalogue

textContent and +

Construct and insert plain text, not interpreted HTML.

HTML main, h1 and ul

Provide a page heading and an initially empty unordered list.

TRY

Add a fourth object. The loop should show it without another append statement.

CHECK

Three list items appear. Changing one price changes only that item.

JavaScript 1/6

A frequently asked questions control

YOUR GOAL

Toggle extra information and keep its button state understandable.

Preview live ↗ Copy files ↗

Open javascript-4 in VS Code. Put the HTML below inside the body of index.html.

JavaScript 2/6

A frequently asked questions control

In index.html: inside <body>

CODE EXAMPLE Copy online ↗

```
<main>
  <h1>Delivery information</h1>
  <button id="toggle" type="button"
          aria-expanded="false" aria-controls="answer">S
how answer</button>
  <p id="answer" hidden>Sample delivery time: 3 working
days.</p>
</main>
```

app.js

CODE EXAMPLE Copy online ↗

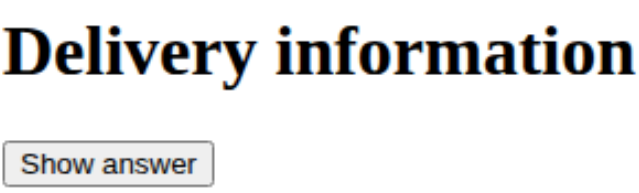
```
const button = document.querySelector('#toggle');
const answer = document.querySelector('#answer');
let open = false;
button.addEventListener('click', function () {
  open = !open;
  answer.hidden = !open;
  button.setAttribute('aria-expanded', String(open));
  button.textContent = open ? 'Hide answer' : 'Show answ
er';
});
```

JavaScript 3/6

A frequently asked questions control

Starting screen

REAL WEBSITE OUTPUT



After trying the example

REAL WEBSITE OUTPUT



JavaScript 4/6

A frequently asked questions control

const versus let

References stay bound; open is reassigned after every click.

false, true and !

Boolean state and logical negation switch between closed and open.

hidden

Hide or reveal the answer element.

setAttribute and String()

Update the accessibility state as the text true or false.

JavaScript 5/6

A frequently asked questions control

? : conditional operator

Choose one button label according to the current state.

querySelector, event listener and textContent

Connect the existing HTML to the click behaviour and label.

HTML button, aria-controls, aria-expanded and hidden

Identify the controlled answer and starting closed state. A real button supports keyboard activation.

JavaScript 6/6

A frequently asked questions control

TRY

Write a different question and answer. Test using Tab and Enter only.

CHECK

First click reveals; second click hides. aria-expanded matches the visible state.

JavaScript 1/8

A catalogue search page

YOUR GOAL

Filter visible services as the visitor types.

Preview live ↗ Copy files ↗

Open javascript-5 in VS Code. Put the HTML below inside the body of index.html.

JavaScript 2/8

A catalogue search page

In index.html: inside <body>

CODE EXAMPLE Copy online ↗

```
<main>
  <h1>Find a class</h1>
  <label for="search">Search classes</label>
  <input id="search" type="search">
  <ul id="results" aria-live="polite"></ul>
</main>
```

JavaScript 3/8

A catalogue search page

app.js

CODE EXAMPLE Copy online ↗

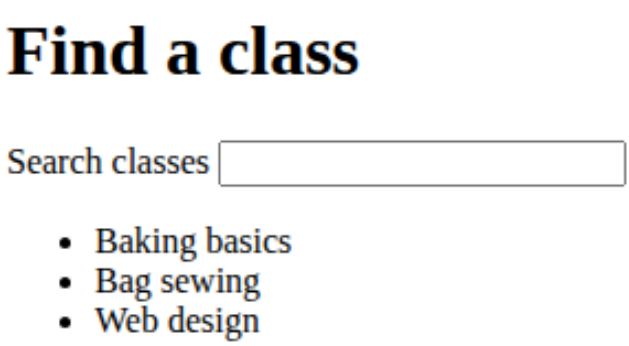
```
const classes = ['Baking basics', 'Bag sewing', 'Web des
ign'];
const field = document.querySelector('#search');
const list = document.querySelector('#results');
function showClasses() {
  const query = field.value.trim().toLowerCase();
  const matches = classes.filter(function (name) {
    return name.toLowerCase().includes(query);
  });
  list.textContent = '';
  for (const name of matches) {
    const item = document.createElement('li');
    item.textContent = name;
    list.append(item);
  }
  if (matches.length === 0) list.textContent = 'No match
ing class.';
}
field.addEventListener('input', showClasses);
showClasses();
```

JavaScript 4/8

A catalogue search page

Starting screen

REAL WEBSITE OUTPUT

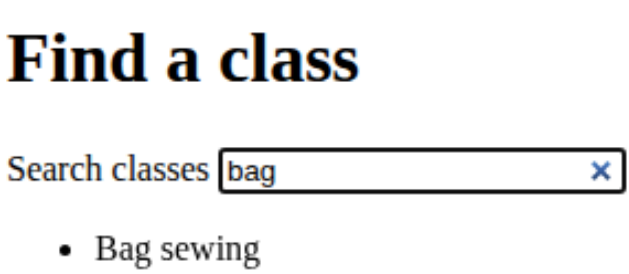


JavaScript 5/8

A catalogue search page

After trying the example

REAL WEBSITE OUTPUT



array, const and selectors

Store names and locate the search field and result list.

JavaScript 6/8

A catalogue search page

function showClasses and its final call

Define the update routine and run it once for the initial full list.

trim and toLowerCase

Normalise the query; normalise each name too.

filter, callback, return and includes

Keep names containing the query and return a new array.

JavaScript 7/8

A catalogue search page

textContent = empty string

Clear old results before creating the next set.

for...of, createElement and append

Render each match as a list item.

length, === and if

Show useful feedback when there are zero matches.

JavaScript 8/8

A catalogue search page

input event and HTML type="search"

Update while the visitor types; label and aria-live explain the field and changing results.

TRY

Add Photography to the array. Search with upper-case letters and verify it still works.

CHECK

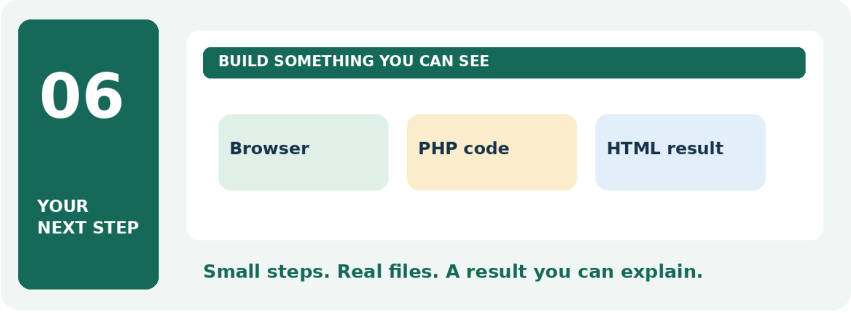
Empty search shows all three. web shows only Web design. zzz shows the empty-result message.

PHP 1/1

Part 06 PHP works on the server

Generate pages, validate forms and remember a session.

VISUAL EXPLANATION



Your finish line

Create five PHP pages and check their server-generated results.

PHP 1/5

Meet the terminal slowly

YOU NEED

VS Code. Read a command first, type it once, then compare the result.

The terminal is a place to give the computer written instructions. Each instruction is called a command.

1 Open VS Code. Choose File → Open Folder and open Website Class.

PHP 2/5

Meet the terminal slowly

2 Choose Terminal → New Terminal. A panel opens at the bottom.

3 Click beside the blinking cursor in that panel.

4 Type the command below and press Enter once.

CODE EXAMPLE Copy online ↗

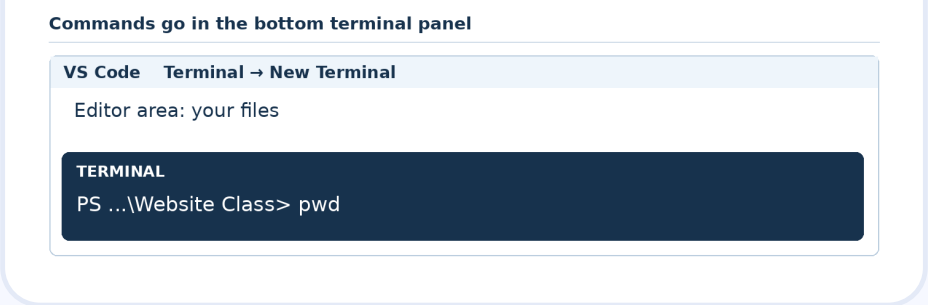
```
pwd
```

PHP 3/5

Meet the terminal slowly

In Windows PowerShell, this shows your current folder. If your terminal is Command Prompt instead, use cd with no extra words.

VISUAL EXPLANATION



PHP 4/5

Meet the terminal slowly

Three useful habits

Type one command at a time. Wait for it to finish before typing the next one. Commands go in the terminal, not inside index.html.

When a command starts a local website, it may keep running. Leave that terminal open. To stop the website, click inside that terminal and press Ctrl + C.

PHP 5/5

Meet the terminal slowly

An address such as http://127.0.0.1:8000 points to a service on your own computer. It is not a public link you can send to a customer.

TRY

close the terminal panel using its panel controls. Open a new terminal again. Find the folder name it shows.

PHP 1/5

Install and check PHP

YOU NEED

VS Code, the preceding terminal practice and Laravel Herd for Windows. No Laravel project is needed yet.

PHP needs a program that can run it. Live Server previews static files; it does not execute PHP. We will use PHP's local development server for these exercises.

PHP 2/5

Install and check PHP

1. Open https://herd.laravel.com/windows. Check the Windows requirements, download the installer and open it. Follow setup, then launch Herd and finish its welcome screens. Approve an administrator prompt only on a computer you are allowed to manage.

2. Close and reopen VS Code after installation. Choose Terminal → New Terminal.

PHP 3/5

Install and check PHP

3 Type the following, then press Enter:

CODE EXAMPLE Copy online ↗

```
php --version
```

4 You should see a PHP version. If PHP is not recognised, close VS Code, open Herd once, reopen VS Code and create a new terminal. Try php --version again. Use the PHP troubleshooting page if it still fails.

PHP 4/5

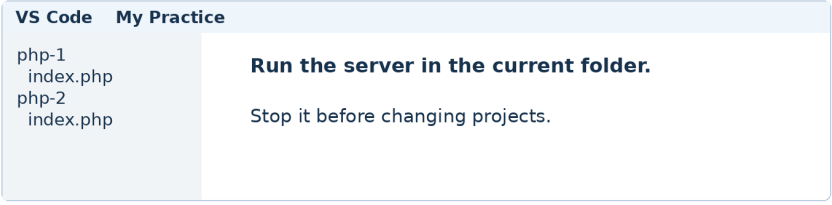
Install and check PHP

5 In File Explorer, create Website Class → php-bakery. This folder will be useful later. For the five separate PHP practicals, create php-1 through php-5 inside My Practice.

6 In VS Code, choose File → Open Folder → My Practice → php-1.

VISUAL EXPLANATION

Separate practice projects contain their own PHP page



PHP 5/5

Install and check PHP

Our separate PHP practicals use one public demonstration file each. The combined projects later separate public files from the database.

If you already have a working PHP installation, use it. Keep one active setup. Do not install XAMPP, WampServer and Herd all together just for this lesson.

Ready to move on: php --version succeeds and you can open php-1 as the project folder.

References: S28, N8, N14.

PHP 1/5

Start a server and fix setup errors

Check the tools before the code

In VS Code choose Terminal → New Terminal. Type each command on its own line and press Enter:

CODE EXAMPLE Copy online ↗

```
php --version
php -m
php --ini
```

PHP 2/5

Start a server and fix setup errors

The first prints a version. The second lists modules; the complete websites need PDO and pdo_sqlite. The third identifies the configuration file PHP uses.

php is not recognised

Close VS Code. Open Herd and finish setup. Reopen VS Code and a new terminal. Run php --version again.

PHP 3/5

Start a server and fix setup errors

Could not find driver

Run php -m and check pdo_sqlite. Use your PHP provider’s extension instructions to enable the matching SQLite driver; restart the terminal and recheck.

Address already in use

Stop your earlier server with Ctrl + C. Or choose port 8001 and use 8001 in the browser too.

Browser says connection refused

Keep the terminal server running. Match its address and port exactly.

PHP 4/5

Start a server and fix setup errors

PHP source appears in browser

You opened a file directly or used Live Server. Open the http://127.0.0.1 address printed by PHP instead.

Open the official Herd Windows guide ↗

Read the PHP SQLite driver guide ↗

PHP 5/5

Start a server and fix setup errors

CHECK

You are ready for PHP when its version command works. You are ready for the combined websites when PDO and pdo_sqlite also appear. Do not paste later commands over an unresolved setup error.

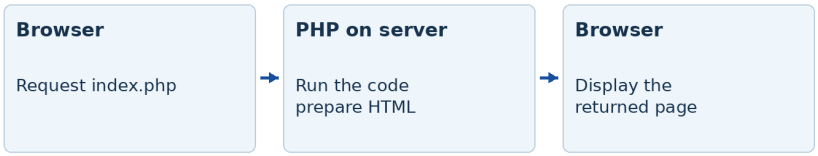
PHP

Where the code runs

The browser sends a request. PHP runs on the server. The server returns generated HTML. Your browser does not execute a PHP source file opened by double-clicking it.

VISUAL EXPLANATION

The browser receives PHP output instead of its source



PHP

Where the code runs

For each separate practical, open its folder in VS Code. Create index.php. Choose Terminal → New Terminal and run:

CODE EXAMPLE

Copy online ↗

```
php -S 127.0.0.1:8000
```

Open http://127.0.0.1:8000/index.php. Leave the terminal running. To change projects, click the terminal, press Ctrl + C, open the next folder and restart. If port 8000 is busy, use 8001 in both command and browser address.

PHP 3/4

Where the code runs

PHP variables start with $. Text uses quotes. A dot joins text. Most statements end with a semicolon. <?php begins code; ?> returns to surrounding HTML. <?= outputs a value.

Our five levels are a safe greeting, a calculation, arrays and loops, POST validation, and sessions. SQL is taught separately after these.

PHP 4/4

Where the code runs

The built-in server is for local development. Later website projects run it with -t public so private code and databases are outside the served folder. Sources: N8, N14.

PHP 1/6

A server generated welcome page

YOUR GOAL

Use variables and safe output inside an HTML document.

Open code online ↗

YOU NEED

VS Code, PHP and your php-1 folder. Create index.php with this complete code. In Terminal → New Terminal, run php -S 127.0.0.1:8000. Open http://127.0.0.1:8000/index.php. Stop any earlier server first.

PHP 2/6

A server generated welcome page

This single-file practice folder contains no database or secret files. Later complete projects use a separate public web root.

index php

CODE EXAMPLE Copy online ↗

```
<?php
$name = 'Ada';
function e(string $text): string {
    return htmlspecialchars($text, ENT_QUOTES, 'UTF-8');
}
?>
<!doctype html>
<html lang="en">
<head><meta charset="UTF-8"><title>Welcome</title></head
>
<body>
  <h1>Welcome, <?= e($name) ?></h1>
  <p>This greeting was prepared by PHP.</p>
</body>
</html>
```

PHP 3/6

A server generated welcome page

Starting screen

REAL WEBSITE OUTPUT



<?php, ?> and <?= ?>

Enter PHP, return to HTML and echo an expression in the short output form.

PHP 4/6

A server generated welcome page

$name, =, quotes and semicolon

Name a text value, assign it and finish the statement.

function e, string and return

Define a reusable text-escaping function with string input and output.

htmlspecialchars, ENT_QUOTES and UTF-8

Escape special characters and both quote types for HTML text with this encoding.

PHP 5/6

A server generated welcome page

HTML doctype, html, head, meta and title

Declare HTML, language, character encoding and browser-tab title.

body, h1 and p

Show the greeting and explanation in the page body.

TRY

Change $name to your first name. Then try Ada and check it appears as literal text.

PHP 6/6

A server generated welcome page

CHECK

The server URL shows the heading. View Source does not contain the PHP assignment.

FIX IT

Use the local server address, not a file address. Read errors in the terminal and check the named line.

PHP 1/5

A server calculated quotation page

YOUR GOAL

Calculate a fixed example and use a condition and reusable function.

Open code online ↗

YOU NEED

VS Code, PHP and your php-2 folder. Create index.php with this complete code. In Terminal → New Terminal, run php -S 127.0.0.1:8000. Open http://127.0.0.1:8000/index.php. Stop any earlier server first.

PHP 2/5

A server calculated quotation page

index php

CODE EXAMPLE Copy online ↗

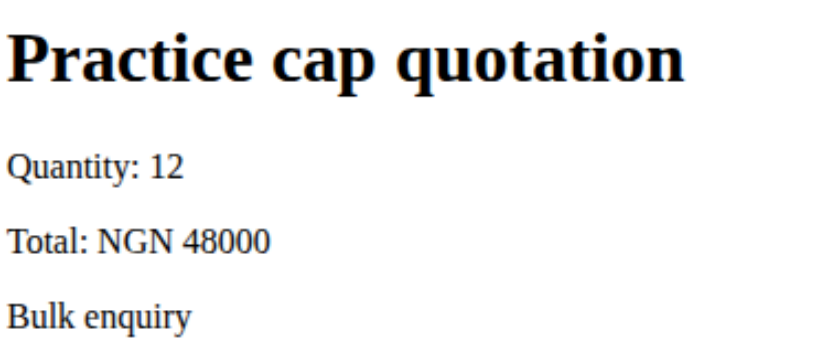
```
<?php
$quantity = 12;
$unitPrice = 4000;
function subtotal(int $quantity, int $price): int {
    return $quantity * $price;
}
$total = subtotal($quantity, $unitPrice);
if ($quantity >= 10) {
    $note = 'Bulk enquiry';
} else {
    $note = 'Small enquiry';
}
?>
<!doctype html>
<html lang="en">
<head><meta charset="UTF-8"><title>Cap quote</title></he
ad>
<body>
  <h1>Practice cap quotation</h1>
  <p>Quantity: <?= $quantity ?></p>
  <p>Total: NGN <?= $total ?></p>
  <p><?= $note ?></p>
</body>
</html>
```

PHP 3/5

A server calculated quotation page

Starting screen

REAL WEBSITE OUTPUT



$quantity, $unitPrice and $total

Store two whole-number inputs and the calculated result.

PHP 4/5

A server calculated quotation page

function, int parameters and return

Define a reusable whole-number multiplication and return its result.

*, = and semicolons

Multiply, assign and end statements.

if, >=, else and braces

Choose the note according to the quantity.

<?= ?>

Output controlled numeric values and fixed strings; user-provided strings would need escaping.

PHP 5/5

A server calculated quotation page

HTML structure, title, h1 and p

Create a valid page and display the three results.

TRY

Change quantity to 5. Predict 20000 and Small enquiry, then verify.

CHECK

12 produces 48000. 5 selects the small-enquiry branch.

PHP 1/5

A dynamic price list page

YOUR GOAL

Use an array and loop to generate repeated HTML.

Open code online ↗

YOU NEED

VS Code, PHP and your php-3 folder. Create index.php with this complete code. In Terminal → New Terminal, run php -S 127.0.0.1:8000. Open http://127.0.0.1:8000/index.php. Stop any earlier server first.

PHP 2/5

A dynamic price list page

index php

CODE EXAMPLE Copy online ↗

```
<?php
$services = [
    ['name' => 'Haircut', 'price' => 3000],
    ['name' => 'Hair wash', 'price' => 2000],
    ['name' => 'Beard trim', 'price' => 1500]
];
function e(string $text): string {
    return htmlspecialchars($text, ENT_QUOTES, 'UTF-8');
}
?>
<!doctype html>
<html lang="en">
<head><meta charset="UTF-8"><title>Service prices</title
></head>
<body>
<h1>Practice salon prices</h1>
<ul>
<?php foreach ($services as $service): ?>
  <li><?= e($service['name']) ?>: NGN <?= $service['pric
e'] ?></li>
<?php endforeach; ?>
</ul>
</body>
</html>
```

PHP 3/5

A dynamic price list page

Starting screen

REAL WEBSITE OUTPUT



[ ], commas and =>

Build an array of associative arrays; each key maps to a value.

PHP 4/5

A dynamic price list page

foreach...as, colon and endforeach

Repeat a template once per service; this is PHP alternative control syntax.

$service['name'] and ['price']

Read the current row values by key.

e(), htmlspecialchars and <?= ?>

Escape name text and output the template values; prices are controlled integers here.

PHP 5/5

A dynamic price list page

HTML doctype, head, meta, title and body

Provide the basic page document.

h1, ul and li

Name the page and display each service in the generated list.

TRY

Add a fourth service. Do not copy another li; let the loop do the work.

CHECK

Three list items render. The added service appears after refreshing.

PHP 1/8

A validated estimate form

YOUR GOAL

Process a real form on the server without storing customer data.

Open code online ↗

YOU NEED

VS Code, PHP and your php-4 folder. Create index.php with this complete code. In Terminal → New Terminal, run php -S 127.0.0.1:8000. Open http://127.0.0.1:8000/index.php. Stop any earlier server first.

PHP 2/8

A validated estimate form

index php

CODE EXAMPLE Copy online ↗

```
<?php
$message = 'Enter a quantity from 1 to 100.';
if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $raw = $_POST['quantity'] ?? '';
    $quantity = is_string($raw) ? filter_var($raw,
        FILTER_VALIDATE_INT, ['options' => [
            'min_range' => 1, 'max_range' => 100]]) : fa
lse;
    if ($quantity === false) {
        $message = 'Please enter a valid whole quantity.
';
    } else {
        $message = 'Server estimate: NGN ' . $quantity *
 3000;
    }
}
?>
<!doctype html>
<html lang="en">
<head><meta charset="UTF-8"><title>Bakery estimate</titl
e></head>
<body>
<h1>Cupcake box estimate</h1>
<form method="post" action="index.php">
  <label for="qty">Boxes</label>
  <input id="qty" name="quantity" type="number" min="1"
max="100" required>
```

PHP 3/8

A validated estimate form

CODE CONTINUED Copy online ↗

```
  <button>Calculate on server</button>
</form>
<p><?= htmlspecialchars($message, ENT_QUOTES, 'UTF-8') ?
></p>
</body>
</html>
```

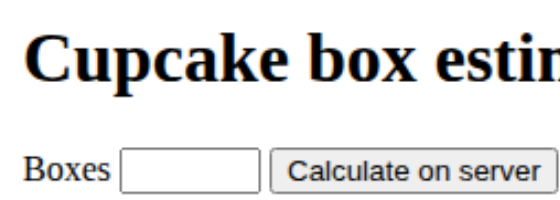
For the second screenshot, enter 2 and click Calculate on server. PHP returns an estimate of NGN 6000.

PHP 4/8

A validated estimate form

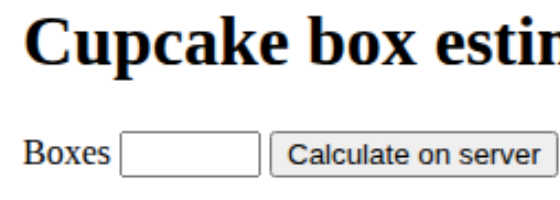
Starting screen

REAL WEBSITE OUTPUT



After trying the example

REAL WEBSITE OUTPUT



PHP 5/8

A validated estimate form

$_SERVER and REQUEST_METHOD

Detect whether the request is a form POST.

$_POST and ??

Read the submitted quantity or an empty default.

is_string and ? :

Reject unexpected array-shaped input before validating the scalar value.

PHP 6/8

A validated estimate form

filter_var and FILTER_VALIDATE_INT

Validate a whole number within min_range and max_range.

if, === false and else

Distinguish validation failure from valid numeric values.

dot, * and htmlspecialchars

Join the fixed message to a server calculation and escape output.

form method, action and name

POST to this PHP file and identify the submitted field.

PHP 7/8

A validated estimate form

label, id, number type, min, max and required

Provide associated labels and browser checks; PHP checks again.

button, h1, p and document shell

Submit, name the page and show the resulting message.

TRY

Change the PHP unit price to 3500 and test two boxes. Do not add a client-controlled price.

PHP 8/8

A validated estimate form

CHECK

2 produces 6000. A direct invalid POST is rejected by PHP.

PHP 1/5

A session based visit counter page

YOUR GOAL

Learn how a server remembers state across requests.

Open code online ↗

YOU NEED

VS Code, PHP and your php-5 folder. Create index.php with this complete code. In Terminal → New Terminal, run php -S 127.0.0.1:8000. Open http://127.0.0.1:8000/index.php. Stop any earlier server first.

PHP 2/5

A session based visit counter page

index php

CODE EXAMPLE Copy online ↗

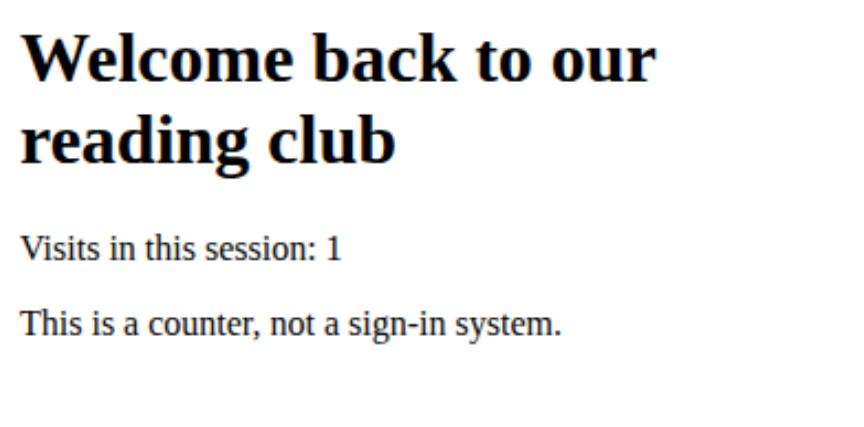
```
<?php
session_start();
$_SESSION['visits'] = ($_SESSION['visits'] ?? 0) + 1;
$visits = $_SESSION['visits'];
?>
<!doctype html>
<html lang="en">
<head><meta charset="UTF-8"><title>Session practice</tit
le></head>
<body>
<h1>Welcome back to our reading club</h1>
<p>Visits in this session: <?= $visits ?></p>
<p>This is a counter, not a sign-in system.</p>
</body>
</html>
```

PHP 3/5

A session based visit counter page

Starting screen

REAL WEBSITE OUTPUT



session_start()

Start or resume a session before any HTML output.

PHP 4/5

A session based visit counter page

$_SESSION and visits key

Keep a server-side value associated with this session.

?? 0, + 1 and assignment

Use zero on the first visit, increment and save.

$visits and <?= ?>

Read and display the controlled integer counter.

Cookie relationship

A browser cookie usually carries an identifier, while this value is stored server-side.

PHP 5/5

A session based visit counter page

HTML document shell, h1 and p

Provide a valid page with a heading and two paragraphs.

TRY

Refresh three times. Compare a normal window and a private window, then explain why this is not authentication.

CHECK

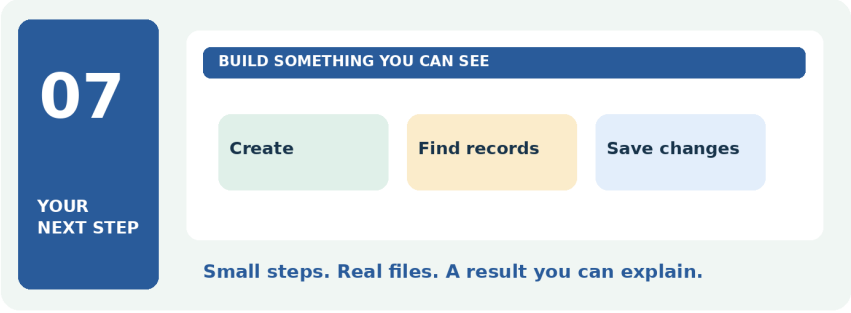
Counter increases in one session. PHP starts the session before HTML output.

SQL 1/1

Part 07 SQL works with stored data

Create tables, select records and understand related data.

VISUAL EXPLANATION



Your finish line

Complete five database practicals before connecting SQL to a website.

SQL 1/4

Understand records and relationships

SQL is the language used to define, read and change data in a relational database. It does not colour headings or draw buttons.

VISUAL EXPLANATION

A database table stores one record per row

Database products

id	name	price
1	Polo shirt	7500
2	Canvas bag	6000
3	Cap	4000

SQL 2/4

Understand records and relationships

A table holds rows and columns. A row is one record, such as a product. A column is one kind of detail, such as its price. A primary key identifies a row. A foreign key links a row to a record in another table.

We use SQLite for these beginner exercises because the database is a local file. MySQL and PostgreSQL are other database systems that use SQL. Their setup, types and some syntax differ, so do not assume every SQLite command works unchanged in them.

SQL 3/4

Understand records and relationships

Our five levels are creating and inserting, selecting and filtering, updating and deleting, joining related records, and reporting with transactions.

Use invented products and customer names. Prices are whole-number NGN examples. For money requiring fractional units, decide on a smallest-unit integer representation such as kobo.

SQL 4/4

Understand records and relationships

When SQL is connected to PHP later, the server will use prepared statements to bind values. A browser does not get direct database access.

Sources: N10, N12.

SQL 1/5

Open DB Browser and run a query

1. Visit https://sqlitebrowser.org/dl/. In Windows, open Settings → System → About and read System type. Choose the matching Windows installer, download it, open it and follow setup. Launch DB Browser for SQLite from Start.

SQL 2/5

Open DB Browser and run a query

2 Open DB Browser for SQLite. Click New Database. Save class-practice.sqlite inside My Practice → sql. If a Create Table dialog appears, cancel it; we will create the table using SQL.

3 Click the Execute SQL tab. This is where the commands go. Do not put SQL into index.html.

SQL 3/5

Open DB Browser and run a query

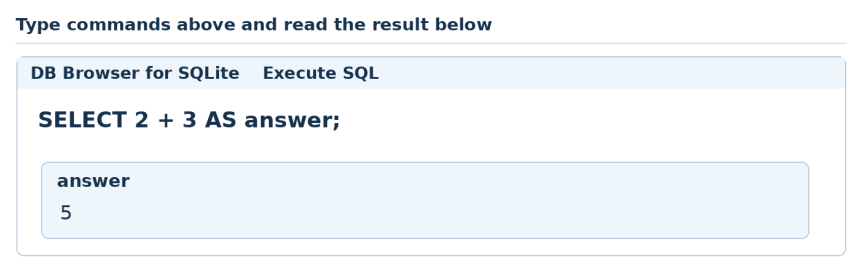
4 Type SELECT 2 + 3 AS answer; and click the execute control, usually a play triangle. You should see one result cell: 5.

5 Replace that query with the SQL 1 code. Run it once. Use Browse Data to inspect products and Write Changes to save pending changes.

SQL 4/5

Open DB Browser and run a query

VISUAL EXPLANATION



Before the next practical, clear only the query editor text. Keep the database tables. Run SQL 1 to SQL 5 in order. Do not run the table-creation scripts repeatedly on the same database.

SQL 5/5

Open DB Browser and run a query

For a fresh start, create a new empty database and run the scripts again in order. This is safer than deleting tables you do not recognise. Click Write Changes before beginning the transaction exercise in SQL 5.

The query result grid is the output for this separate SQL section. The combined websites later turn database results into HTML pages.

Sources: N9, N10.

SQL 1/4

Run one statement and read the result

Open DB Browser → Open Database and choose class-practice.sqlite. Choose Execute SQL. This is the box where you type SQL; the result table appears below it.

1. Type SELECT 2 + 3 AS answer; and run it. Expect one column named answer with the value 5.

SQL 2/4

Run one statement and read the result

2 If an exercise has several SELECT statements, highlight one complete statement, including its semicolon, then use Execute selected SQL. Read that result before selecting the next statement.

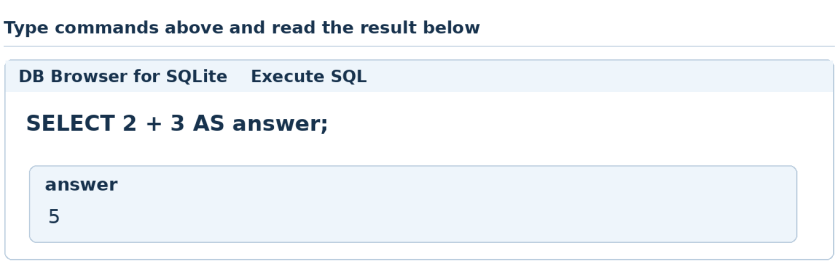
3 Use Write Changes after completing changes. Before SQL practical 5, use Write Changes so no earlier transaction remains open.

SQL 3/4

Run one statement and read the result

4 Run practicals 1–5 in order, once each, in the same fresh database. If you want to restart, create a new database with a new filename.

VISUAL EXPLANATION



SQL 4/4

Run one statement and read the result

FIX IT

“Table already exists” means the table was created earlier. Do not keep rerunning the setup. Continue with the next statement, or start a new practice database. “No such table” usually means a different database is open or an earlier practical was skipped.

TRY

Change 2 + 3 to 10 + 5. Predict 15, run the query and compare. A query is simply a request you write to the database.

SQL 1/6

Create a product table

YOUR GOAL

Build the data behind a small merchandise catalogue.

Open code online ↗

YOU NEED

DB Browser for SQLite and the same classroom database from SQL 1. Run these practicals in order. Open Execute SQL, clear the previous query text and type this exercise. Run selected statements when comparing more than one SELECT. Write Changes after each completed practical.

SQL 2/6

Create a product table

Start with a new empty database named class-practice.sqlite. Do not rerun CREATE TABLE on the already-created tables.

practical 1 sql

CODE EXAMPLE Copy online ↗

```
PRAGMA foreign_keys = ON;
CREATE TABLE products (
  id INTEGER PRIMARY KEY,
  name TEXT NOT NULL,
  category TEXT NOT NULL,
  price INTEGER NOT NULL CHECK (price >= 0)
);
INSERT INTO products (id, name, category, price) VALUES
  (1, 'Polo shirt', 'Clothing', 7500),
  (2, 'Canvas bag', 'Bags', 6000),
  (3, 'Cap', 'Clothing', 4000);
SELECT id, name, category, price FROM products;
```

Create a product table

Actual SQLite results

These values were returned by running the supplied SQL files in order. SQL returns records; HTML and CSS decide how a website displays them.

id	name	category	price
1	Polo shirt	Clothing	7500
2	Canvas bag	Bags	6000
3	Cap	Clothing	4000

SQL 4/6

Create a product table

PRAGMA foreign_keys

Enables SQLite foreign-key enforcement for this connection.

CREATE TABLE, parentheses and commas

Define a table and separate its columns.

INTEGER, TEXT and PRIMARY KEY

Choose whole numbers or text and identify each row uniquely.

SQL 5/6

Create a product table

NOT NULL and CHECK

Require a value and reject negative prices.

INSERT INTO, column list and VALUES

Add three records in the specified column order.

SELECT and FROM

Choose columns and the table to read.

Quotes and semicolons

Single quotes enclose SQL text; semicolons end statements.

SQL 6/6

Create a product table

TRY

Add a fictional fourth product with a new id. Then remove it or reset from the supplied starting file before SQL 2.

CHECK

Compare every row with the expected output. Write Changes after completing the practical.

FIX IT

Check the open database and exercise order. Run one selected statement when comparing results.

SQL 1/4

Find the products a visitor needs

YOUR GOAL

Select only products that meet a condition and order the result.

Open code online ↗

practical 2 sql

CODE EXAMPLE Copy online ↗

```
SELECT name, price
FROM products
WHERE price >= 6000
ORDER BY price DESC;
SELECT name
FROM products
WHERE name LIKE '%bag%';
```

Find the products a visitor needs

Actual SQLite results

name	price
Polo shirt	7500
Canvas bag	6000

name
Canvas bag

The second SELECT returns Canvas bag. Run it separately to see that result.

SQL 3/4

Find the products a visitor needs

SELECT and FROM

Return only the named columns from products.

WHERE and >=

Include only rows whose price is at least 6000.

ORDER BY and DESC

Sort from the highest price to the lowest.

LIKE and %

Match a text pattern; % stands for any sequence of characters.

SQL 4/4

Find the products a visitor needs

Two statements

Each SELECT has its own result grid; run one at a time to inspect both.

TRY

Change the minimum price to 7000. Expect only Polo shirt. Restore the original query.

SQL 1/4

Correct and remove specific records

YOUR GOAL

Practise changes with a WHERE clause on a local classroom database.

Open code online ↗

practical 3 sql

CODE EXAMPLE Copy online ↗

```
UPDATE products
SET price = 4500
WHERE id = 3;
INSERT INTO products (id, name, category, price)
VALUES (4, 'Practice only', 'Test', 0);
DELETE FROM products WHERE id = 4;
SELECT id, name, price FROM products ORDER BY id;
```

Correct and remove specific records

Actual SQLite results

id	name	price
1	Polo shirt	7500
2	Canvas bag	6000
3	Cap	4500

UPDATE and SET

Change the selected record value.

WHERE id = 3

Limit the update to the cap row; without WHERE, all product prices would change.

SQL 3/4

Correct and remove specific records

INSERT INTO and VALUES

Create a disposable classroom record.

DELETE FROM and WHERE

Remove only that disposable row. DELETE removes rows, not the table definition.

SELECT and ORDER BY

Inspect the remaining rows in a predictable order.

SQL 4/4

Correct and remove specific records

TRY

Before running a new UPDATE, use SELECT with the same WHERE condition to inspect the target.

SQL 1/5

Join customer and order information

YOUR GOAL

Connect separate records instead of repeating a customer name in every order.

Open code online ↗

SQL 2/5

Join customer and order information

practical 4 sql

CODE EXAMPLE Copy online ↗

```
CREATE TABLE customers (
  id INTEGER PRIMARY KEY,
  name TEXT NOT NULL
);
CREATE TABLE orders (
  id INTEGER PRIMARY KEY,
  customer_id INTEGER NOT NULL REFERENCES customers(id),
  product_id INTEGER NOT NULL REFERENCES products(id),
  quantity INTEGER NOT NULL CHECK (quantity > 0)
);
INSERT INTO customers VALUES (1, 'Ada'), (2, 'Tunde');
INSERT INTO orders VALUES (1, 1, 2, 5), (2, 2, 1, 2);
SELECT c.name, p.name AS product, o.quantity
FROM orders AS o
JOIN customers AS c ON c.id = o.customer_id
JOIN products AS p ON p.id = o.product_id
ORDER BY o.id;
```

Join customer and order information

Actual SQLite results

name	product	quantity
Ada	Canvas bag	5
Tunde	Polo shirt	2

REFERENCES

Declare links from each order to an existing customer and product.

CHECK quantity > 0

Require positive quantities.

Join customer and order information

INSERT without a column list

Uses the table column order; explicit column lists are clearer in longer applications.

AS and aliases o, c and p

Give tables short names and rename the product result column.

JOIN and ON

Combine matching rows using their related keys.

c.name, p.name and o.quantity

Qualify which table each column comes from.

SQL 5/5

Join customer and order information

ORDER BY o.id

Present orders in their id order.

TRY

Try adding an order using a nonexistent customer id with foreign keys enabled. Expect a constraint error, not a saved order.

SQL 1/4

Build a report and undo a trial change

YOUR GOAL

Summarise orders and see how a transaction protects an experiment.

Open code online ↗

practical 5 sql

CODE EXAMPLE Copy online ↗

```
SELECT c.name, SUM(o.quantity * p.price) AS total
FROM orders AS o
JOIN customers AS c ON c.id = o.customer_id
JOIN products AS p ON p.id = o.product_id
GROUP BY c.id, c.name
ORDER BY total DESC;
BEGIN TRANSACTION;
UPDATE products SET price = 1 WHERE id = 2;
ROLLBACK;
SELECT name, price FROM products WHERE id = 2;
```

SQL 2/4

Build a report and undo a trial change

Actual SQLite results

name	total
Ada	30000
Tunde	15000

name	price
Canvas bag	6000

After ROLLBACK, the final SELECT returns Canvas bag with price 6000. Write Changes before BEGIN if DB Browser has pending edits.

SQL 3/4

Build a report and undo a trial change

SUM, multiplication and AS total

Calculate each line value and add values into a named result.

GROUP BY

Produce one summary per customer rather than one row per order.

ORDER BY total DESC

Show the largest summary first.

BEGIN TRANSACTION and ROLLBACK

Start a group of changes and cancel them; COMMIT would keep them.

SQL 4/4

Build a report and undo a trial change

Final SELECT

Proves the bag price returned to 6000.

Reporting limitation

This teaching report uses current catalogue prices. The combined websites save quoted unit prices so later price changes do not rewrite old totals.

TRY

Explain when to choose COMMIT and when to choose ROLLBACK. Do not experiment on a real business database.

SQL 1/4

Foundation checkpoint Before combining the languages

Demonstrate these tasks without asking AI to replace the whole project.

Internet

Explain a browser request, DNS and a server response

HTML

Create a labelled field and a working link

CSS

Change spacing and make a layout adapt to a narrow screen

SQL 2/4

Foundation checkpoint Before combining the languages

JavaScript

Read input, validate it and update visible text

PHP

Process POST and reject an invalid value on the server

SQL

Insert a record and retrieve it with a condition

SQL 3/4

Foundation checkpoint Before combining the languages

A useful way to explain your work

“This tag creates the input. This rule styles it. This event reacts to typing. This PHP code checks the submitted value. This SQL statement saves the record.”

VISUAL EXPLANATION



SQL 4/4

Foundation checkpoint Before combining the languages

A page working in the browser is a good start. A saved record that you can read back is stronger evidence that the whole flow works.

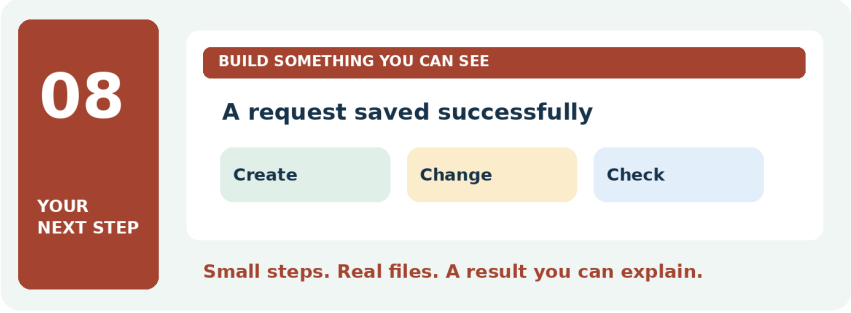
Keep a backup of your successful exercises. The five websites next reuse the same small application structure so you learn how to adapt it, not simply copy five unrelated piles of code.

Complete websites 1/1

Part 08 Put the five subjects together

Follow one enquiry from a page to a saved database record.

VISUAL EXPLANATION



Your finish line

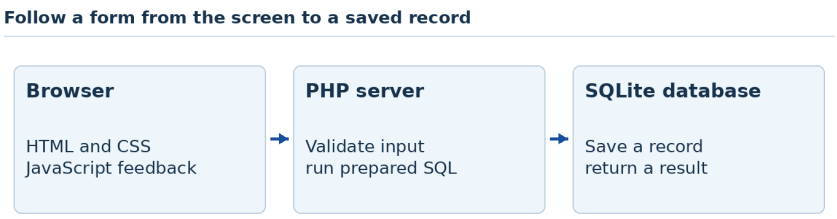
Build the bakery site, then adapt it into four more complete enquiry websites.

Complete websites 1/5

Five complete websites How the parts work together

Each website has Home, Catalogue, Request and Result pages. Its catalogue comes from SQLite. JavaScript filters the cards. PHP validates submissions and saves records using SQL. The result page confirms the saved request.

VISUAL EXPLANATION



Complete websites 2/5

Five complete websites How the parts work together

These are five complete small enquiry websites, with different subjects and rules. They are not full e-commerce, school management or hotel reservation systems. Their forms save real local records; they do not send email, take payment or confirm availability.

Five complete websites How the parts work together

Build one then adapt it carefully

Website 1 teaches the complete shared source code. Websites 2–5 reuse that code and change a configuration file, content and business rules. Each has its own standalone folder, database, test cases and student task. Reuse is deliberate: it is a practical professional skill.

Complete websites 4/5

Five complete websites How the parts work together

Type Website 1's files in the shown folders, or open the supplied website-1 folder to compare. Never serve the whole project folder. The command must include -t public because private PHP files and the database live above that folder.

Complete websites 5/5

Five complete websites How the parts work together

The code uses a CSRF token, server validation, escaped HTML, prepared SQL and a redirect after POST. These are useful foundations. Real public deployment still requires a production environment, secure sessions, operational checks and experienced review.

Complete websites 1/4

Website 1 Build a bakery website

YOUR GOAL

create a bakery catalogue with a searchable product list and a saved enquiry.

1 Make a folder named website-1. Inside it, create public and data. Open website-1 in VS Code.

2 Create the named files on the following pages. Root-level files sit beside public, not inside it. Only names beginning public/ go inside public.

Complete websites 2/4

Website 1 Build a bakery website

3 Check php --version and php -m in Terminal. The modules must include PDO and pdo_sqlite. If a driver is missing, follow the PHP troubleshooting page before continuing.

4 After saving all files, run php setup.php from website-1. Expect “Setup complete. Start the local server.”

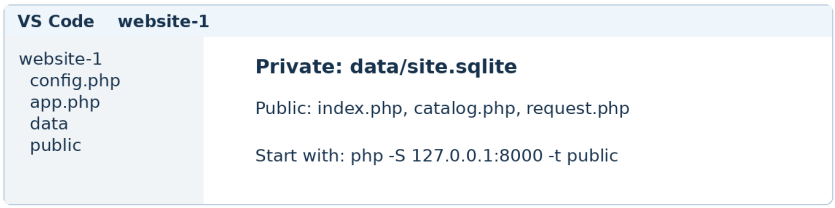
Complete websites 3/4

Website 1 Build a bakery website

5 Run php -S 127.0.0.1:8000 -t public. Visit http://127.0.0.1:8000/index.php. Keep the terminal running.

VISUAL EXPLANATION

Only the public folder is exposed by the local server



Use the supplied files as the answer copy if you lose a bracket. Read the explanation after the code before moving to Website 2.

Complete websites 4/4

Website 1 Build a bakery website

Sample acceptance test: choose Vanilla box, enter Ada and quantity 2. The result should say NGN 6000. A row should appear in the database. This will be a request, not a real order.

Complete websites 1/1

Website 1 · config php

File: config.php. Create this file at the exact path shown.

CODE EXAMPLE Copy online ↗

```
<?php
return [
    'name' => 'Ada Bakes',
    'headline' => 'Cakes for your next celebration',
    'intro' => 'Explore our fictional bakery catalogue a
nd practise making an enquiry.',
    'quantityLabel' => 'Number of boxes',
    'min' => 1,
    'max' => 100,
    'notice' => 'Sample bakery enquiry. No payment or de
livery is arranged.',
    'items' => [
        [1, 'Vanilla box', 3000],
        [2, 'Chocolate box', 3500],
        [3, 'Fruit box', 4000],
    ]
];
```

Complete websites 1/1

Website 1 · schema sql

File: schema.sql. Create this file at the exact path shown.

CODE EXAMPLE Copy online ↗

```
CREATE TABLE IF NOT EXISTS items (
  id INTEGER PRIMARY KEY,
  name TEXT NOT NULL,
  price INTEGER NOT NULL CHECK (price >= 0)
);
CREATE TABLE IF NOT EXISTS requests (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  customer TEXT NOT NULL,
  item_id INTEGER NOT NULL REFERENCES items(id),
  quantity INTEGER NOT NULL CHECK (quantity > 0),
  unit_price INTEGER NOT NULL,
  total INTEGER NOT NULL
);
```

Complete websites 1/1

Website 1 · setup php

File: setup.php. Create this file at the exact path shown.

CODE EXAMPLE Copy online ↗

```
<?php
$config = require __DIR__ . '/config.php';
if (!is_dir(__DIR__ . '/data')) mkdir(__DIR__ . '/data')
;
$db = new PDO('sqlite:' . __DIR__ . '/data/site.sqlite')
;
$db->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPT
ION);
$db->exec('PRAGMA foreign_keys = ON');
$db->exec(file_get_contents(__DIR__ . '/schema.sql'));
$insert = $db->prepare(
    'INSERT OR IGNORE INTO items (id, name, price) VALUE
S (?, ?, ?)'
);
foreach ($config['items'] as $item) $insert->execute($it
em);
echo "Setup complete. Start the local server.\n";
```

Complete websites 1/1

Website 1 · app php

File: app.php. Create this file at the exact path shown.

CODE EXAMPLE Copy online ↗

```
<?php
session_start();
$config = require __DIR__ . '/config.php';
function e(string $text): string {
    return htmlspecialchars($text, ENT_QUOTES, 'UTF-8');
}
if (!is_file(__DIR__ . '/data/site.sqlite')) {
    exit('Run php setup.php from the project folder firs
t.');
}
$db = new PDO('sqlite:' . __DIR__ . '/data/site.sqlite')
;
$db->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPT
ION);
$db->exec('PRAGMA foreign_keys = ON');
if (!isset($_SESSION['token'])) {
    $_SESSION['token'] = bin2hex(random_bytes(32));
}
```

Complete websites 1/1

Website 1 · header php

File: header.php. Create this file at the exact path shown.

CODE EXAMPLE Copy online ↗

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport"
        content="width=device-width, initial-scale=1">
  <title><?= e($config['name']) ?></title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
<header>
  <p class="brand"><?= e($config['name']) ?></p>
  <nav aria-label="Main navigation">
    <a href="index.php">Home</a>
    <a href="catalog.php">Catalogue</a>
    <a href="request.php">Make a request</a>
  </nav>
</header>
<main>
```

Complete websites 1/1

Website 1 · footer php

File: footer.php. Create this file at the exact path shown.

CODE EXAMPLE Copy online ↗

```
</main>
<footer><p>Fictional classroom website. Requests are not
 confirmed orders.</p></footer>
<script src="app.js" defer></script>
</body>
</html>
```

Complete websites 1/1

Website 1 · public index php

File: public/index.php. Create this file at the exact path shown.

CODE EXAMPLE Copy online ↗

```
<?php
require __DIR__ . '/../app.php';
require __DIR__ . '/../header.php';
?>
<h1><?= e($config['headline']) ?></h1>
<p><?= e($config['intro']) ?></p>
<p><a class="button" href="catalog.php">Explore our opti
ons</a></p>
<section>
  <h2>How it works</h2>
  <ol>
    <li>Choose an option from our catalogue.</li>
    <li>Send a fictional practice request.</li>
    <li>Review the saved request confirmation.</li>
  </ol>
</section>
<?php require __DIR__ . '/../footer.php'; ?>
```

Complete websites 1/1

Website 1 · public catalog php

File: public/catalog.php. Create this file at the exact path shown.

CODE EXAMPLE Copy online ↗

```
<?php
require __DIR__ . '/../app.php';
$items = $db->query('SELECT * FROM items ORDER BY id')
    ->fetchAll(PDO::FETCH_ASSOC);
require __DIR__ . '/../header.php';
?>
<h1>Choose an option</h1>
<label for="search">Search the catalogue</label>
<input id="search" type="search">
<p id="empty" hidden>No matching option.</p>
<div class="cards">
<?php foreach ($items as $item): ?>
  <article data-name="<?= e(strtolower($item['name'])) ?
>">
    <h2><?= e($item['name']) ?></h2>
    <p>Sample unit price: NGN <?= (int) $item['price'] ?
></p>
    <a href="request.php?item=<?= (int) $item['id'] ?>">
Request this</a>
  </article>
<?php endforeach; ?>
</div>
<?php require __DIR__ . '/../footer.php'; ?>
```

Complete websites 1/5

Website 1 · public request php

File: public/request.php. Create this file at the exact path shown.

Complete websites 2/5

Website 1 · public request php

CODE EXAMPLE Copy online ↗

```
<?php
require __DIR__ . '/../app.php';
$items = $db->query('SELECT * FROM items ORDER BY id')
    ->fetchAll(PDO::FETCH_ASSOC);
$error = '';
if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $token = $_POST['token'] ?? '';
    if (!is_string($token) || !hash_equals($_SESSION['to
ken'], $token)) {
        http_response_code(403);
        exit('Form expired. Return to the request page.'
);
    }
    $name = $_POST['customer'] ?? '';
    $itemId = $_POST['item'] ?? '';
    $raw = $_POST['quantity'] ?? '';
    if (!is_string($name) || !is_string($itemId) || !is_
string($raw)) {
        http_response_code(400);
        exit('Invalid input.');
    }
    $name = trim($name);
    $qty = filter_var($raw, FILTER_VALIDATE_INT, ['optio
ns' => [
        'min_range' => $config['min'], 'max_range' => $c
onfig['max']]]);
    $find = $db->prepare('SELECT * FROM items WHERE id =
```

Complete websites 3/5

Website 1 · public request php

CODE CONTINUED Copy online ↗

```
?');
    $find->execute([$itemId]);
    $item = $find->fetch(PDO::FETCH_ASSOC);
    if ($name === '' || strlen($name) > 60
        || $qty === false || !$item) {
        $error = 'Check your name, option and quantity r
ange.';
```

File: public/request.php. Append these lines below the previous part in the same file.

Complete websites 4/5

Website 1 · public request php

CODE EXAMPLE Copy online ↗

```
} else {
        $total = $qty * (int) $item['price'];
        $save = $db->prepare('INSERT INTO requests
          (customer, item_id, quantity, unit_price, tota
l)
          VALUES (?, ?, ?, ?, ?)');
        $save->execute([$name, $item['id'], $qty, $item[
'price'], $total]);
        $_SESSION['receipt'] = ['name' => $name, 'total'
 => $total,
            'id' => $db->lastInsertId()];
        $_SESSION['token'] = bin2hex(random_bytes(32));
        header('Location: thanks.php', true, 303);
        exit;
    }
}
$selected = filter_input(INPUT_GET, 'item', FILTER_VALID
ATE_INT);
require __DIR__ . '/../header.php';
?>
<h1>Make a practice request</h1>
<p><?= e($config['notice']) ?></p>
<p role="alert"><?= e($error) ?></p>
<form method="post" action="request.php">
  <input type="hidden" name="token" value="<?= e($_SESSI
ON['token']) ?>">
  <label for="customer">First name</label>
```

Complete websites 5/5

Website 1 · public request php

CODE CONTINUED Copy online ↗

```
  <input id="customer" name="customer" maxlength="60" re
quired>
  <label for="item">Choose an option</label>
  <select id="item" name="item">
  <?php foreach ($items as $row): ?>
    <option value="<?= (int) $row['id'] ?>"
```

CODE EXAMPLE Copy online ↗

```
<?= $selected === (int) $row['id'] ? 'selected' : '' ?>>
      <?= e($row['name']) ?>
    </option>
  <?php endforeach; ?>
  </select>
  <label for="quantity"><?= e($config['quantityLabel'])
?></label>
  <input id="quantity" name="quantity" type="number" ste
p="1"
    min="<?= $config['min'] ?>" max="<?= $config['max']
?>" required>
  <button type="submit">Save practice request</button>
</form>
<?php require __DIR__ . '/../footer.php'; ?>
```

Complete websites 1/1

Website 1 · public thanks php

File: public/thanks.php. Create this file at the exact path shown.

CODE EXAMPLE Copy online ↗

```
<?php
require __DIR__ . '/../app.php';
$receipt = $_SESSION['receipt'] ?? null;
require __DIR__ . '/../header.php';
?>
<h1>Request result</h1>
<?php if ($receipt): ?>
  <p>Saved request <?= (int) $receipt['id'] ?></p>
  <p>Thank you, <?= e($receipt['name']) ?>.</p>
  <p>Recorded estimate: NGN <?= (int) $receipt['total']
?></p>
  <p>This is a request, not a confirmed booking or payme
nt.</p>
<?php else: ?>
  <p>No request has been submitted in this session.</p>
<?php endif; ?>
<?php require __DIR__ . '/../footer.php'; ?>
```

Complete websites 1/3

Website 1 · public style css

File: public/style.css. Create this file at the exact path shown.

Complete websites 2/3

Website 1 · public style css

CODE EXAMPLE Copy online ↗

```
* { box-sizing: border-box; }
body {
  margin: 0; color: #17324d; background: #f4f8fb;
  font-family: Arial, sans-serif; line-height: 1.6;
}
header, main, footer { max-width: 1050px; margin: auto;
padding: 24px; }
.brand { font-weight: bold; font-size: 1.4rem; }
nav { display: flex; gap: 20px; flex-wrap: wrap; }
a { color: #164e9b; }
h1 { font-size: 2.3rem; line-height: 1.2; }
.cards { display: grid; gap: 20px; }
article { padding: 24px; background: white; border: 1px
solid #bdcddd; }
form { max-width: 500px; }
label { display: block; margin-top: 16px; }
input, select { width: 100%; padding: 12px; font: inheri
t; }
button, .button {
  display: inline-block; padding: 12px 18px; margin-top:
 16px;
  background: #164e9b; color: white; border: 0; font: in
herit;
}
a:focus-visible, input:focus-visible,
select:focus-visible, button:focus-visible {
  outline: 3px solid #9c5c00; outline-offset: 3px;
```

Complete websites 3/3

Website 1 · public style css

CODE CONTINUED Copy online ↗

```
}
@media (min-width: 720px) {
  .cards { grid-template-columns: repeat(3, 1fr); }
}
```

Complete websites 1/1

Website 1 · public app js

File: public/app.js. Create this file at the exact path shown.

CODE EXAMPLE Copy online ↗

```
const search = document.querySelector('#search');
if (search) {
  const cards = document.querySelectorAll('[data-name]')
;
  const empty = document.querySelector('#empty');
  search.addEventListener('input', function () {
    const query = search.value.trim().toLowerCase();
    let visible = 0;
    for (const card of cards) {
      const match = card.dataset.name.includes(query);
      card.hidden = !match;
      if (match) visible += 1;
    }
    empty.hidden = visible !== 0;
  });
}
```

Complete websites 1/3

Website 1 Inspect the saved record

Create inspect.php in website-1, outside public:

CODE EXAMPLE Copy online ↗

```
<?php
require __DIR__ . '/app.php';
$rows = $db->query('SELECT id, customer, quantity, unit_
price, total
                    FROM requests ORDER BY id')->fetchAll
(PDO::FETCH_ASSOC);
foreach ($rows as $row) echo implode(' | ', $row) . PHP_
EOL;
```

Open a second terminal in website-1. Run php inspect.php. A clean database after the Ada test should print:

Complete websites 2/3

Website 1 Inspect the saved record

CODE EXAMPLE Copy online ↗

```
1 | Ada | 2 | 3000 | 6000
```

The columns are request id, customer, quantity, saved unit price and total. Each successful submission adds another row. Refreshing thanks.php should not add a row because the handler redirected after the POST.

You can also open data/site.sqlite in DB Browser and inspect the requests table. Do not edit it while PHP is writing. Close DB Browser before changes if it holds a lock.

Complete websites 3/3

Website 1 Inspect the saved record

Stop and restart the local server. Run php inspect.php again. The saved row should remain. Session receipt data and persistent database records are different: the database is the lasting evidence.

TRY

deliberately submit an invalid quantity with browser limits removed using Inspect. Confirm the server rejects it and the database row count stays unchanged.

Complete websites 1/6

Website 1 Explain the HTML CSS and JavaScript

Follow the visible side

header.php and footer.php

Share the document shell, navigation and closing content on every page. require loads them.

lang, charset, viewport and title

Declare document language, encoding, device viewport and browser title.

Complete websites 2/6

Website 1 Explain the HTML CSS and JavaScript

header, nav, main, section, article and footer

Describe the roles of the content regions. h1 and h2 create a heading hierarchy.

a and href

Move between pages; ?item= passes an item id to preselect the request option.

label, for, id and name

Give controls readable labels and submitted field names. select and option list database products.

Complete websites 3/6

Website 1 Explain the HTML CSS and JavaScript

input type hidden and token value

Send the session form token back. Hidden fields are not secret or trusted by themselves.

min, max, step and required

Offer quick browser checks; the PHP handler also checks.

role alert and e($error)

Announce validation feedback and render it as escaped text.

Complete websites 4/6

Website 1 Explain the HTML CSS and JavaScript

style.css selectors and properties

Set readable typography, page width, spacing, form controls and focus outlines. The media rule changes the grid to three columns.

Complete websites 5/6

Website 1 Explain the HTML CSS and JavaScript

Read the catalogue filter

app.js first finds #search. if (search) prevents errors on pages without that field. querySelectorAll collects cards with data-name. An input listener normalises the search text, tests includes(), changes each hidden property and counts visible cards. The final comparison controls the empty-result message.

Complete websites 6/6

Website 1 Explain the HTML CSS and JavaScript

textContent is used for text in the separate language lessons; this catalogue already has escaped server-rendered text. dataset.name reads the data-name attribute. No database query is needed for this small browser filter.

Complete websites 1/5

Website 1 Explain PHP SQL and the request flow

Follow one submission

config.php return array

Holds trusted site content, quantity limits and seed items. Each later website changes this file.

setup.php and schema.sql

Create a private database and tables. INSERT OR IGNORE seeds each id once; it does not update later config price changes.

Complete websites 2/5

Website 1 Explain PHP SQL and the request flow

PDO, attributes and PRAGMA

Connect to SQLite, turn SQL errors into exceptions and enable foreign keys.

app.php session_start and e

Start server-side session state and escape text for HTML using htmlspecialchars.

random_bytes, bin2hex and token

Generate an unpredictable form token. hash_equals compares the submitted token with the session token.

Complete websites 3/5

Website 1 Explain PHP SQL and the request flow

POST checks, is_string and trim

Require scalar text inputs and clean outer name whitespace. strlen limits bytes to 60 in this demo.

filter_var and config min/max

Validate a whole quantity on the server, including each website business limit.

prepare, ? and execute

Keep SQL separate from values. Fetch the selected item and its trusted database price.

Complete websites 4/5

Website 1 Explain PHP SQL and the request flow

INSERT and saved unit_price

Store an enquiry with a price snapshot; later price changes do not rewrite that quoted amount.

lastInsertId and receipt

Keep a short receipt for this session; no public customer list is created.

header with 303 and exit

Redirect after saving, then stop. Refreshing the result performs GET, not another insertion.

Complete websites 5/5

Website 1 Explain PHP SQL and the request flow

SELECT reads items. ORDER BY gives a predictable sequence. fetchAll produces arrays for foreach to render. PRIMARY KEY identifies a row; REFERENCES relates a request to its product. CHECK and NOT NULL add database constraints.

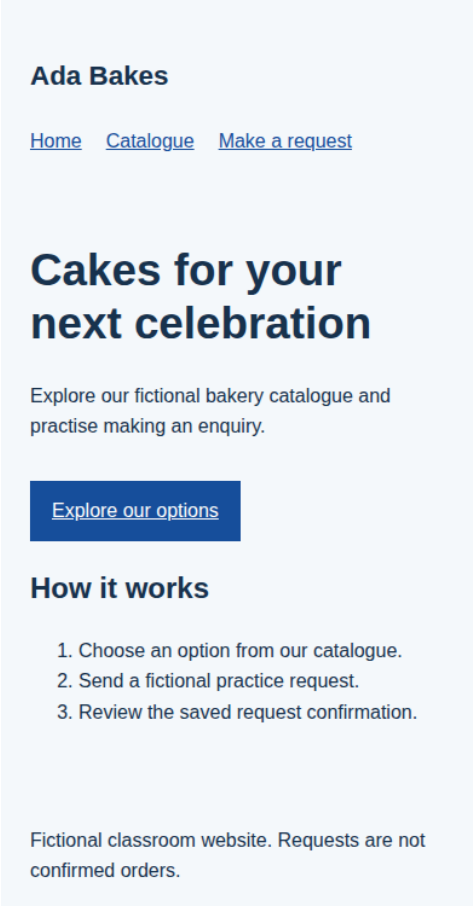
The database is outside public. The HTML escaping helper is for text/attribute contexts here; SQL safety comes from bound parameters, not HTML escaping.

Complete websites 1/8

Website 1 See the result & test it

Home page

REAL WEBSITE OUTPUT

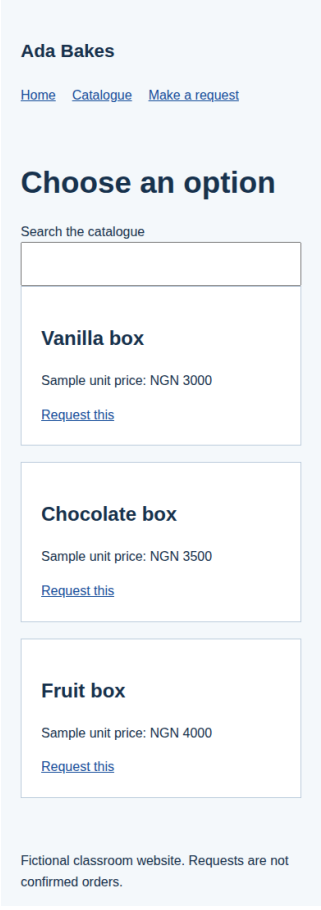


Complete websites 2/8

Website 1 See the result & test it

Catalogue page

REAL WEBSITE OUTPUT

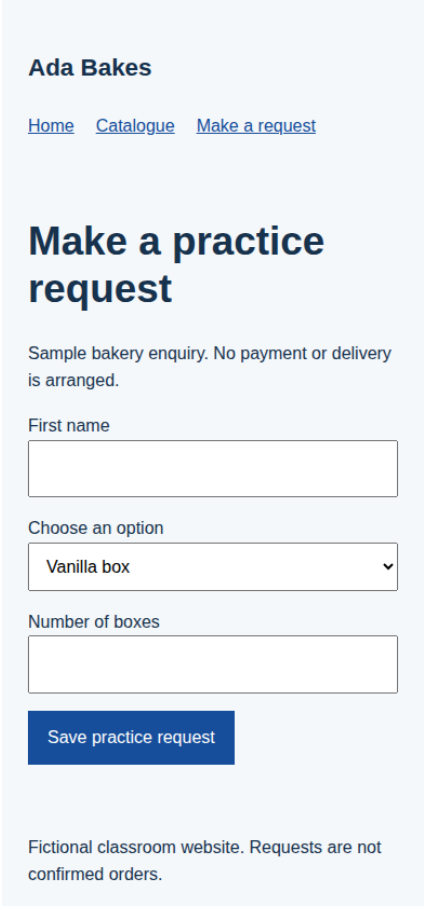


Complete websites 3/8

Website 1 See the result & test it

Request page

REAL WEBSITE OUTPUT

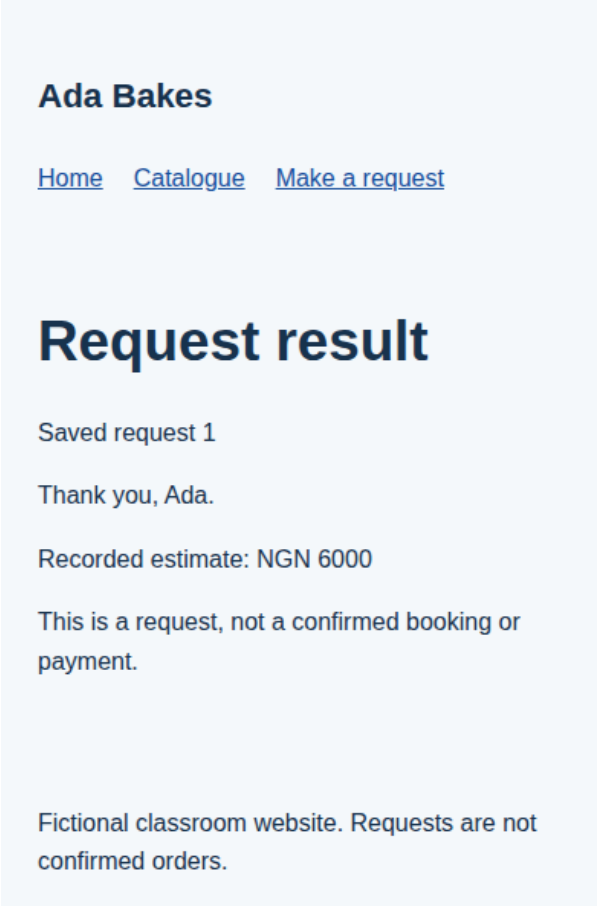


Complete websites 4/8

Website 1 See the result & test it

Saved practice request

REAL WEBSITE OUTPUT



Complete websites 5/8

Website 1 See the result & test it

Use the site's Home, Catalogue, Request and Result pages in that order.

Search chocolate

Only the matching catalogue option remains.

Search zzz

No matching option is shown.

First item with Ada and quantity 2

Recorded estimate: NGN 6000.

Website 1 See the result & test it

Quantity 0 or 101

Browser blocks it; server also rejects a bypassed invalid submission.

Run php inspect.php

A saved row shows the same quantity and total.

Refresh the Result page

No duplicate request is inserted.

Narrow the window and use Tab

Cards stack; links and form controls remain usable.

Complete websites 7/8

Website 1 See the result & test it

Explain the combined result

HTML provides the pages and labelled form. CSS lays them out. JavaScript filters cards. PHP validates the name, item and the 1–100 quantity range. SQL selects catalogue records and saves the checked request. The shared handler always reads the price from the database.

Complete websites 8/8

Website 1 See the result & test it

The code walkthrough after Website 1 explains every shared file. Use it while tracing this site; changing a config value does not replace the validation or database code.

Your demonstration: open one file from each language and explain its role. Then show the saved record. Sample bakery enquiry. No payment or delivery is arranged.

Complete websites 1/4

Website 2 A tailoring and branding enquiry website

YOUR GOAL

adapt the complete application to Thread Studio.

1. Stop the running PHP server with Ctrl + C.

2. Copy the source files from Website 1 into a new website-2 folder, including public. Start with an EMPTY data folder. Do not copy the earlier site.sqlite database.

Complete websites 2/4

Website 2 A tailoring and branding enquiry website

3 Replace config.php with the complete code on the next page. Alternatively, open the supplied website-2 answer folder.

4 Open website-2 in VS Code. Run php setup.php, then php -S 127.0.0.1:8000 -t public.

5 Open http://127.0.0.1:8000/index.php. Check the name on all pages and click every navigation link.

Complete websites 3/4

Website 2 A tailoring and branding enquiry website

This is a separate working website with its own database. Its shared source files are exactly those explained in Website 1. The change is its content and enforced business rules, not just a colour swap.

VISUAL EXPLANATION



Complete websites 4/4

Website 2 A tailoring and branding enquiry website

Student task: Set a new minimum of 20 in config.php. Confirm both the HTML field and server reject 10. Do not change only the browser field.

Keep all data fictional. Minimum 10 garments. This is an enquiry, not an accepted order.

Complete websites 1/3

Website 2 Configuration code

CODE EXAMPLE Copy online ↗

```
<?php
return [
    'name' => 'Thread Studio',
    'headline' => 'Branded clothing for your team',
    'intro' => 'Compare sample garment services and subm
it a bulk enquiry.',
    'quantityLabel' => 'Number of garments',
    'min' => 10,
    'max' => 500,
    'notice' => 'Minimum 10 garments. This is an enquiry
, not an accepted order.',
    'items' => [
        [1, 'Polo branding', 2500],
        [2, 'Apron branding', 1800],
        [3, 'Cap embroidery', 1500],
    ]
];
```

name, headline and intro

Set site identity and Home content.

Complete websites 2/3

Website 2 Configuration code

quantityLabel, min and max

Name the quantity and set browser plus server limits.

notice

Explain what a request does and does not confirm.

items

Seed rows as id, name and whole-number price. The shared SQL uses these values.

return and PHP array syntax

Return the settings to require; => maps each key to its value.

Complete websites 3/3

Website 2 Configuration code

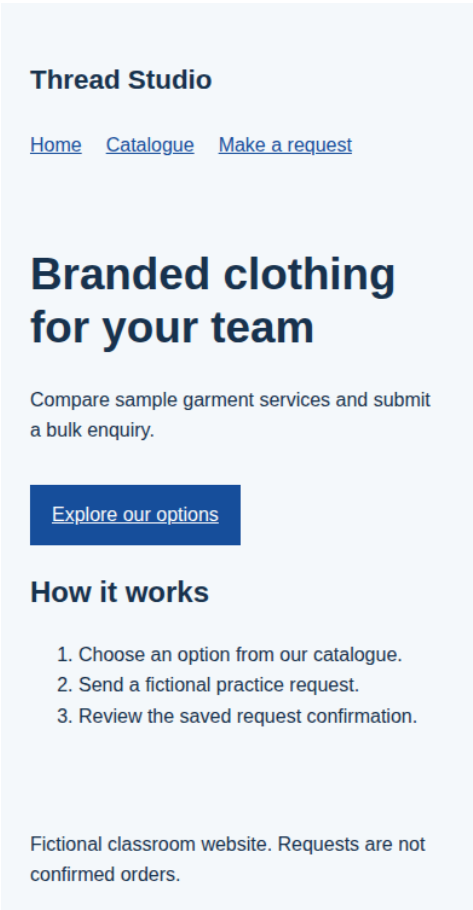
Use a fresh database for this exercise. setup.php deliberately does not overwrite existing rows or erase earlier requests.

Complete websites 1/8

Website 2 See the result & test it

Home page

REAL WEBSITE OUTPUT

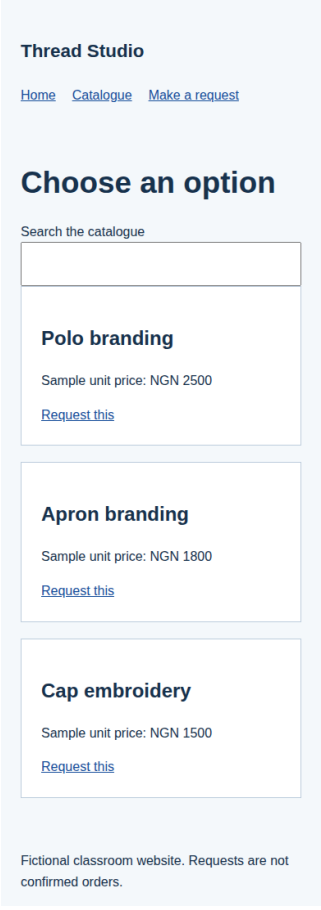


Complete websites 2/8

Website 2 See the result & test it

Catalogue page

REAL WEBSITE OUTPUT

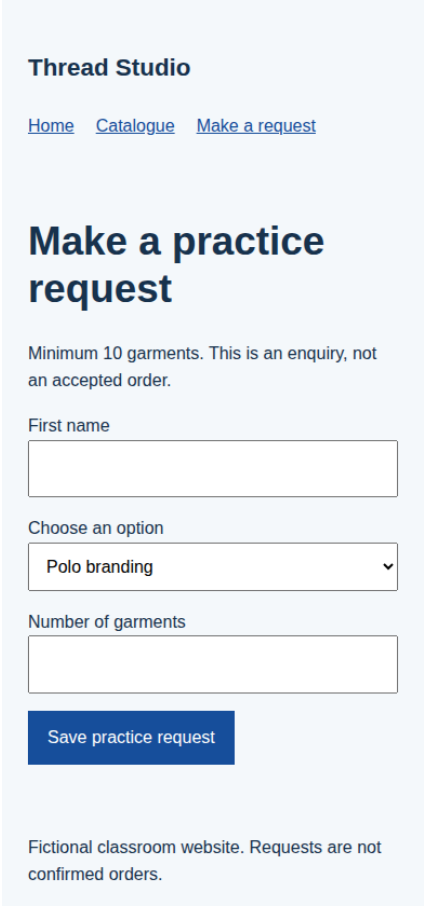


Complete websites 3/8

Website 2 See the result & test it

Request page

REAL WEBSITE OUTPUT

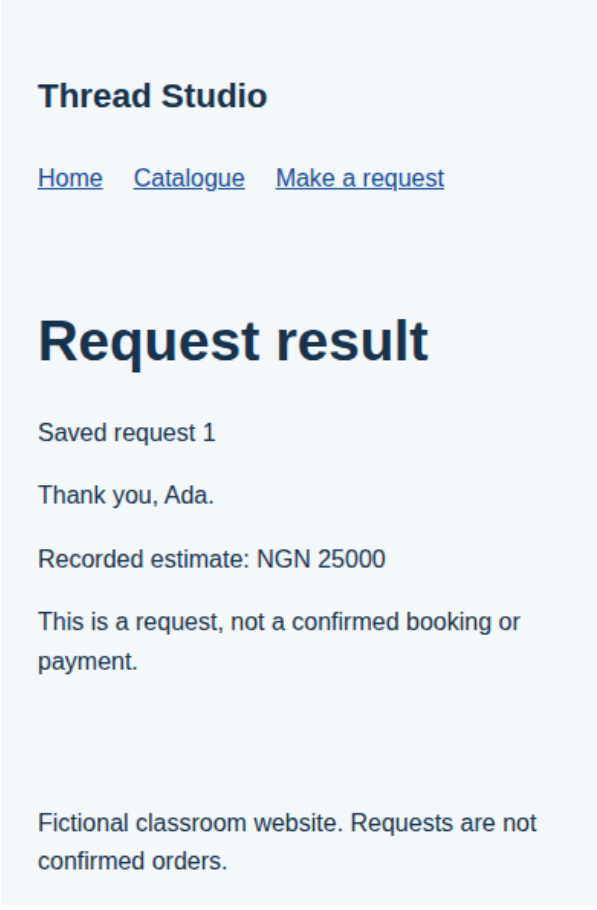


Complete websites 4/8

Website 2 See the result & test it

Saved practice request

REAL WEBSITE OUTPUT



Complete websites 5/8

Website 2 See the result & test it

Search cap

Only the matching catalogue option remains.

Search zzz

No matching option is shown.

First item with Bola and quantity 10

Recorded estimate: NGN 25000.

Quantity 9 or 501

Browser blocks it; server also rejects a bypassed invalid submission.

Complete websites 6/8

Website 2 See the result & test it

Run php inspect.php

A saved row shows the same quantity and total.

Refresh the Result page

No duplicate request is inserted.

Narrow the window and use Tab

Cards stack; links and form controls remain usable.

Website 2 See the result & test it

Explain the combined result

HTML provides the pages and labelled form. CSS lays them out. JavaScript filters cards. PHP validates the name, item and the 10–500 quantity range. SQL selects catalogue records and saves the checked request. The shared handler always reads the price from the database.

Complete websites 8/8

Website 2 See the result & test it

Your demonstration: open one file from each language and explain its role. Then show the saved record. Minimum 10 garments. This is an enquiry, not an accepted order.

Complete websites 1/3

Website 3 A training centre course website

YOUR GOAL

adapt the complete application to SkillStart Academy.

1 Stop the running PHP server with Ctrl + C.

2 Copy the source files from Website 1 into a new website-3 folder, including public. Start with an EMPTY data folder. Do not copy the earlier site.sqlite database.

Complete websites 2/3

Website 3 A training centre course website

3 Replace config.php with the complete code on the next page. Alternatively, open the supplied website-3 answer folder.

4 Open website-3 in VS Code. Run php setup.php, then php -S 127.0.0.1:8000 -t public.

5 Open http://127.0.0.1:8000/index.php. Check the name on all pages and click every navigation link.

Complete websites 3/3

Website 3 A training centre course website

VISUAL EXPLANATION



Student task: Add a page describing course prerequisites and link it from the shared header. State what is required without claiming a place is reserved.

Keep all data fictional. One place per request. The request does not confirm admission.

Complete websites 1/2

Website 3 Configuration code

CODE EXAMPLE Copy online ↗

```
<?php
return [
    'name' => 'SkillStart Academy',
    'headline' => 'Practical weekend learning',
    'intro' => 'Browse sample workshops and request one
learner place.',
    'quantityLabel' => 'Learner places',
    'min' => 1,
    'max' => 1,
    'notice' => 'One place per request. The request does
 not confirm admission.',
    'items' => [
        [1, 'Baking basics', 15000],
        [2, 'Bag sewing', 12000],
        [3, 'Web design basics', 18000],
    ]
];
```

name, headline and intro

Set site identity and Home content.

Complete websites 2/2

Website 3 Configuration code

quantityLabel, min and max

Name the quantity and set browser plus server limits.

notice

Explain what a request does and does not confirm.

items

Seed rows as id, name and whole-number price. The shared SQL uses these values.

return and PHP array syntax

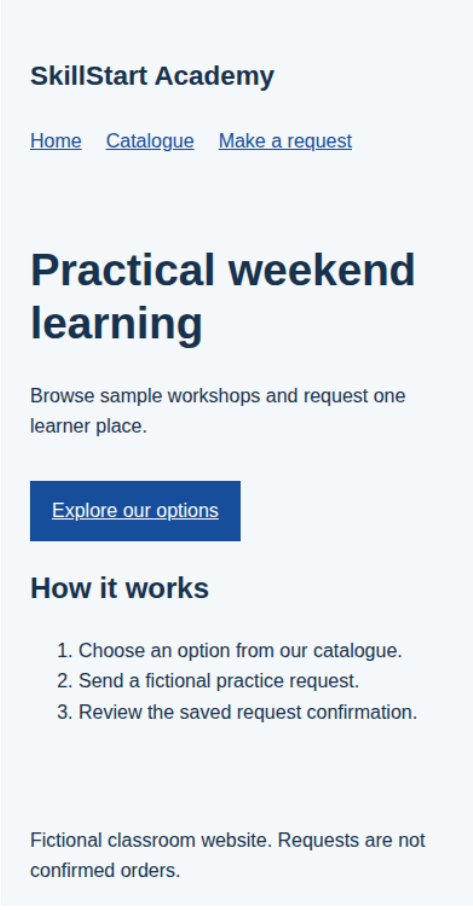
Return the settings to require; => maps each key to its value.

Complete websites 1/8

Website 3 See the result & test it

Home page

REAL WEBSITE OUTPUT

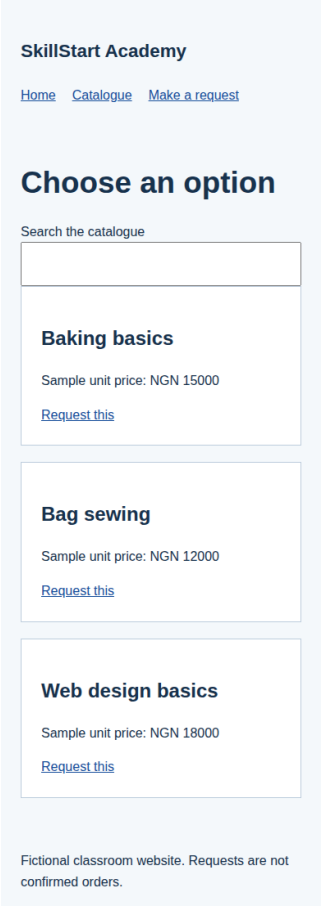


Complete websites 2/8

Website 3 See the result & test it

Catalogue page

REAL WEBSITE OUTPUT

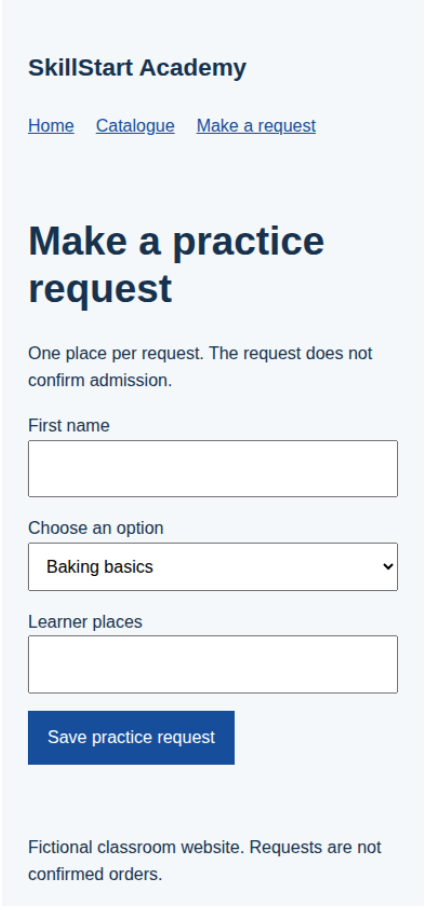


Complete websites 3/8

Website 3 See the result & test it

Request page

REAL WEBSITE OUTPUT

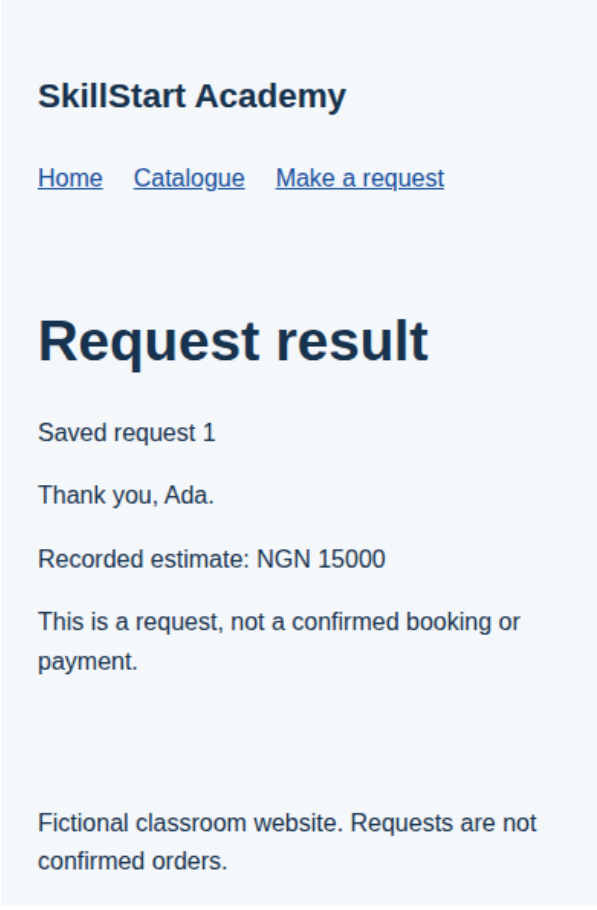


Complete websites 4/8

Website 3 See the result & test it

Saved practice request

REAL WEBSITE OUTPUT



Complete websites 5/8

Website 3 See the result & test it

Search web

Only the matching catalogue option remains.

Search zzz

No matching option is shown.

First item with Chika and quantity 1

Recorded estimate: NGN 15000.

Quantity 0 or 2

Browser blocks it; server also rejects a bypassed invalid submission.

Complete websites 6/8

Website 3 See the result & test it

Run php inspect.php

A saved row shows the same quantity and total.

Refresh the Result page

No duplicate request is inserted.

Narrow the window and use Tab

Cards stack; links and form controls remain usable.

Complete websites 7/8

Website 3 See the result & test it

Explain the combined result

HTML provides the pages and labelled form. CSS lays them out. JavaScript filters cards. PHP validates the name, item and the 1–1 quantity range. SQL selects catalogue records and saves the checked request. The shared handler always reads the price from the database.

Complete websites 8/8

Website 3 See the result & test it

Your demonstration: open one file from each language and explain its role. Then show the saved record. One place per request. The request does not confirm admission.

Complete websites 1/3

Website 4 A guesthouse room enquiry website

YOUR GOAL

adapt the complete application to Palm Rest Guesthouse.

1 Stop the running PHP server with Ctrl + C.

2 Copy the source files from Website 1 into a new website-4 folder, including public. Start with an EMPTY data folder. Do not copy the earlier site.sqlite database.

Complete websites 2/3

Website 4 A guesthouse room enquiry website

3 Replace config.php with the complete code on the next page. Alternatively, open the supplied website-4 answer folder.

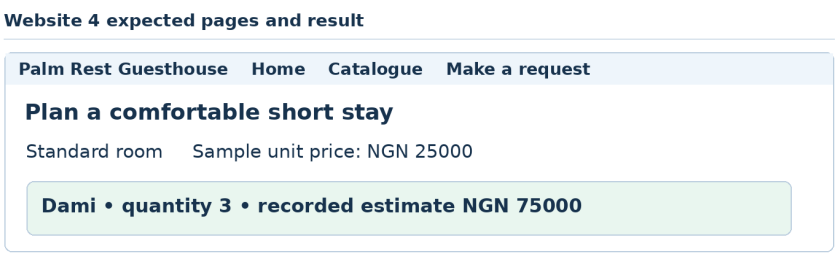
4 Open website-4 in VS Code. Run php setup.php, then php -S 127.0.0.1:8000 -t public.

5 Open http://127.0.0.1:8000/index.php. Check the name on all pages and click every navigation link.

Complete websites 3/3

Website 4 A guesthouse room enquiry website

VISUAL EXPLANATION



Student task: Add a facilities section to Home. Do not label the result as a confirmed reservation: no date or availability system has been built.

Keep all data fictional. Dates and availability are not checked. This is a quote request only.

Complete websites 1/2

Website 4 Configuration code

CODE EXAMPLE Copy online ↗

```
<?php
return [
    'name' => 'Palm Rest Guesthouse',
    'headline' => 'Plan a comfortable short stay',
    'intro' => 'Compare sample nightly room rates and re
quest a stay estimate.',
    'quantityLabel' => 'Number of nights',
    'min' => 1,
    'max' => 14,
    'notice' => 'Dates and availability are not checked.
 This is a quote request only.',
    'items' => [
        [1, 'Standard room', 25000],
        [2, 'Family room', 40000],
        [3, 'Garden room', 30000],
    ]
];
```

name, headline and intro

Set site identity and Home content.

Complete websites 2/2

Website 4 Configuration code

quantityLabel, min and max

Name the quantity and set browser plus server limits.

notice

Explain what a request does and does not confirm.

items

Seed rows as id, name and whole-number price. The shared SQL uses these values.

return and PHP array syntax

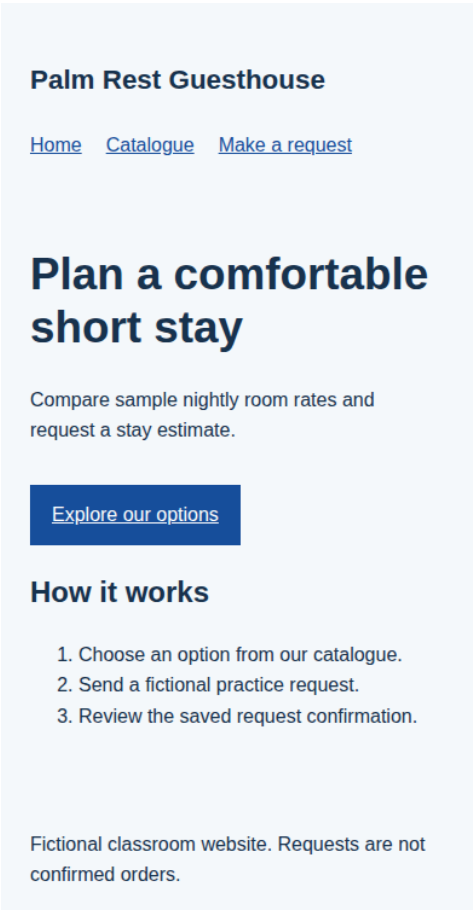
Return the settings to require; => maps each key to its value.

Complete websites 1/8

Website 4 See the result & test it

Home page

REAL WEBSITE OUTPUT

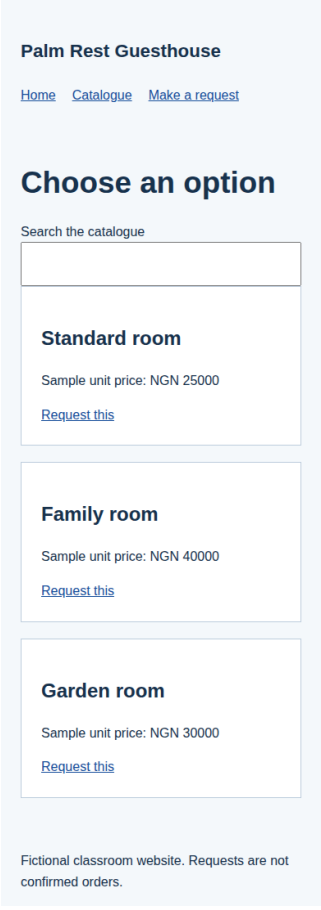


Complete websites 2/8

Website 4 See the result & test it

Catalogue page

REAL WEBSITE OUTPUT

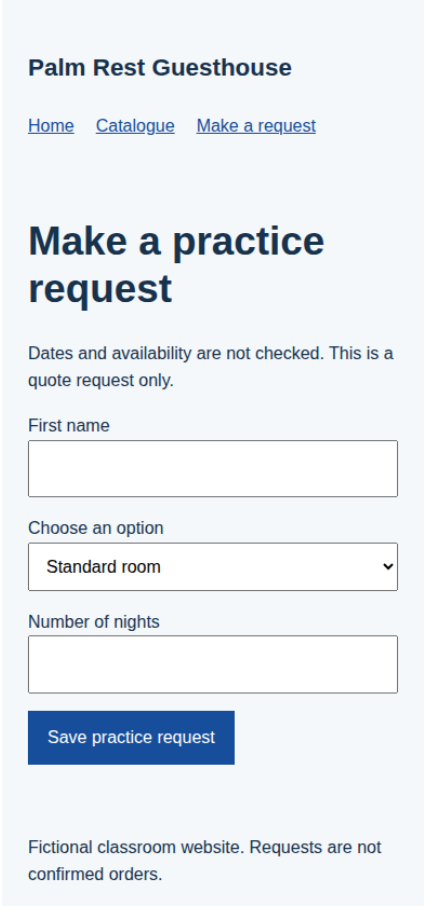


Complete websites 3/8

Website 4 See the result & test it

Request page

REAL WEBSITE OUTPUT

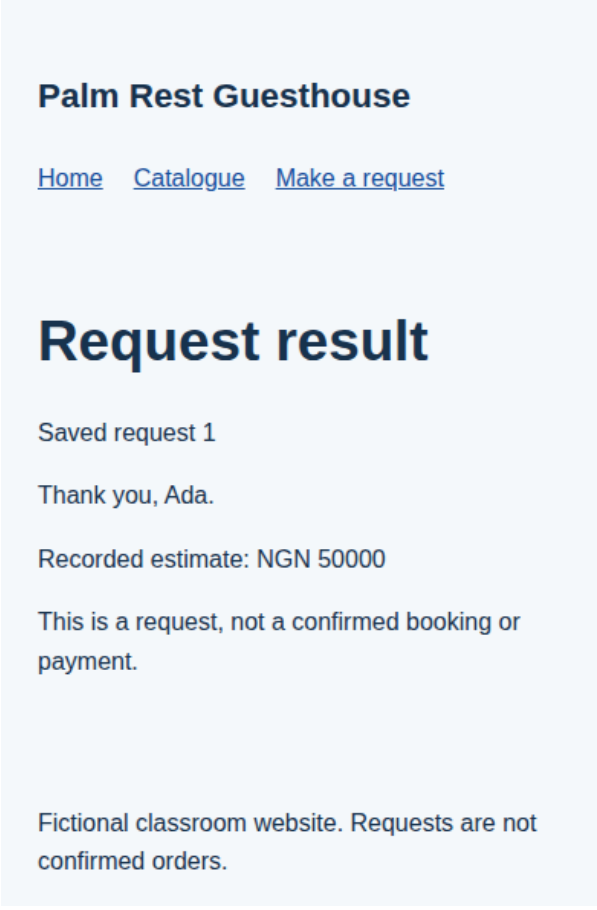


Complete websites 4/8

Website 4 See the result & test it

Saved practice request

REAL WEBSITE OUTPUT



Complete websites 5/8

Website 4 See the result & test it

Search family

Only the matching catalogue option remains.

Search zzz

No matching option is shown.

First item with Dami and quantity 3

Recorded estimate: NGN 75000.

Quantity 0 or 15

Browser blocks it; server also rejects a bypassed invalid submission.

Complete websites 6/8

Website 4 See the result & test it

Run php inspect.php

A saved row shows the same quantity and total.

Refresh the Result page

No duplicate request is inserted.

Narrow the window and use Tab

Cards stack; links and form controls remain usable.

Complete websites 7/8

Website 4 See the result & test it

Explain the combined result

HTML provides the pages and labelled form. CSS lays them out. JavaScript filters cards. PHP validates the name, item and the 1–14 quantity range. SQL selects catalogue records and saves the checked request. The shared handler always reads the price from the database.

Complete websites 8/8

Website 4 See the result & test it

Your demonstration: open one file from each language and explain its role. Then show the saved record. Dates and availability are not checked. This is a quote request only.

Complete websites 1/3

Website 5 A community event registration website

YOUR GOAL

adapt the complete application to Community Learning Day.

1. Stop the running PHP server with Ctrl + C.

2. Copy the source files from Website 1 into a new website-5 folder, including public. Start with an EMPTY data folder. Do not copy the earlier site.sqlite database.

Complete websites 2/3

Website 5 A community event registration website

3 Replace config.php with the complete code on the next page. Alternatively, open the supplied website-5 answer folder.

4 Open website-5 in VS Code. Run php setup.php, then php -S 127.0.0.1:8000 -t public.

5 Open http://127.0.0.1:8000/index.php. Check the name on all pages and click every navigation link.

Complete websites 3/3

Website 5 A community event registration website

VISUAL EXPLANATION



Student task: Add an event schedule page using an HTML table. Link it from every page through header.php and test the links.

Keep all data fictional. No payment, seat allocation or email ticket is issued.

Complete websites 1/2

Website 5 Configuration code

CODE EXAMPLE Copy online ↗

```
<?php
return [
    'name' => 'Community Learning Day',
    'headline' => 'Choose a workshop for the event',
    'intro' => 'Explore event sessions and submit a samp
le ticket request.',
    'quantityLabel' => 'Tickets requested',
    'min' => 1,
    'max' => 6,
    'notice' => 'No payment, seat allocation or email ti
cket is issued.',
    'items' => [
        [1, 'Creative workshop', 2000],
        [2, 'Business talk', 1500],
        [3, 'Digital skills session', 2500],
    ]
];
```

name, headline and intro

Set site identity and Home content.

Complete websites 2/2

Website 5 Configuration code

quantityLabel, min and max

Name the quantity and set browser plus server limits.

notice

Explain what a request does and does not confirm.

items

Seed rows as id, name and whole-number price. The shared SQL uses these values.

return and PHP array syntax

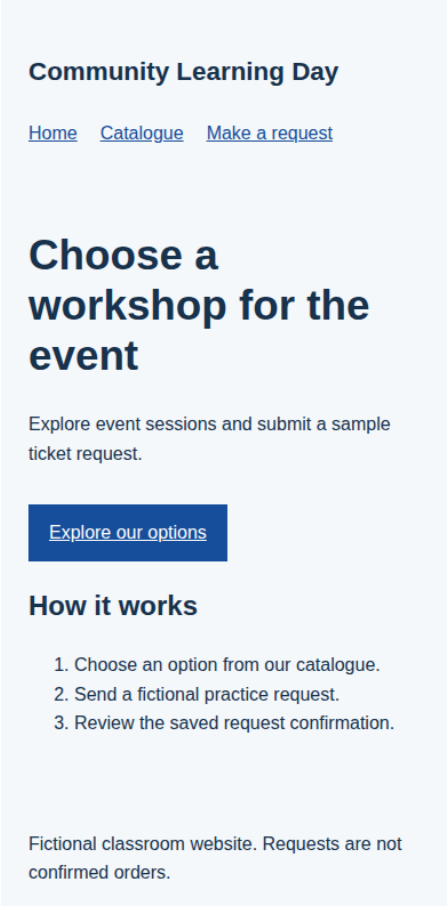
Return the settings to require; => maps each key to its value.

Complete websites 1/7

Website 5 See the result & test it

Home page

REAL WEBSITE OUTPUT

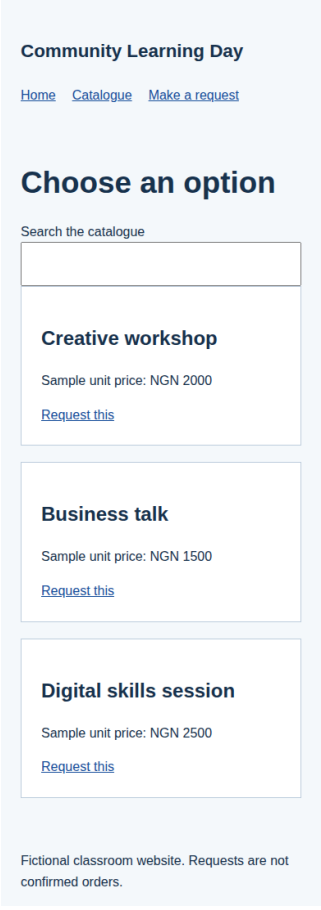


Complete websites 2/7

Website 5 See the result & test it

Catalogue page

REAL WEBSITE OUTPUT

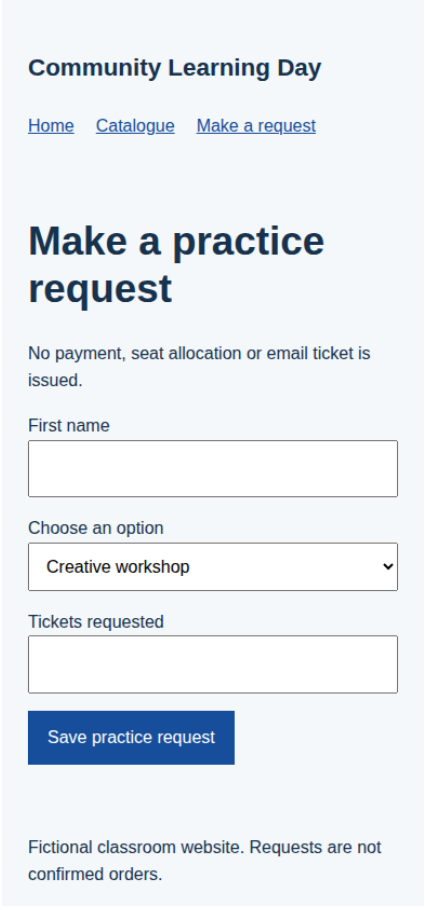


Complete websites 3/7

Website 5 See the result & test it

Request page

REAL WEBSITE OUTPUT

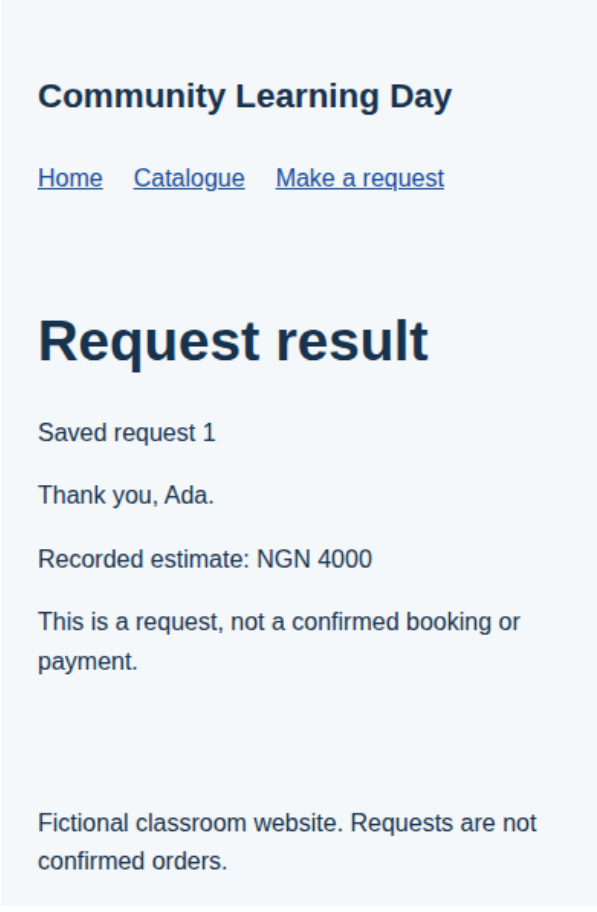


Complete websites 4/7

Website 5 See the result & test it

Saved practice request

REAL WEBSITE OUTPUT



Complete websites 5/7

Website 5 See the result & test it

Search digital

Only the matching catalogue option remains.

Search zzz

No matching option is shown.

First item with Efe and quantity 3

Recorded estimate: NGN 6000.

Quantity 0 or 7

Browser blocks it; server also rejects a bypassed invalid submission.

Complete websites 6/7

Website 5 See the result & test it

Run php inspect.php

A saved row shows the same quantity and total.

Refresh the Result page

No duplicate request is inserted.

Narrow the window and use Tab

Cards stack; links and form controls remain usable.

Complete websites 7/7

Website 5 See the result & test it

Explain the combined result

HTML provides the pages and labelled form. CSS lays them out. JavaScript filters cards. PHP validates the name, item and the 1–6 quantity range. SQL selects catalogue records and saves the checked request. The shared handler always reads the price from the database.

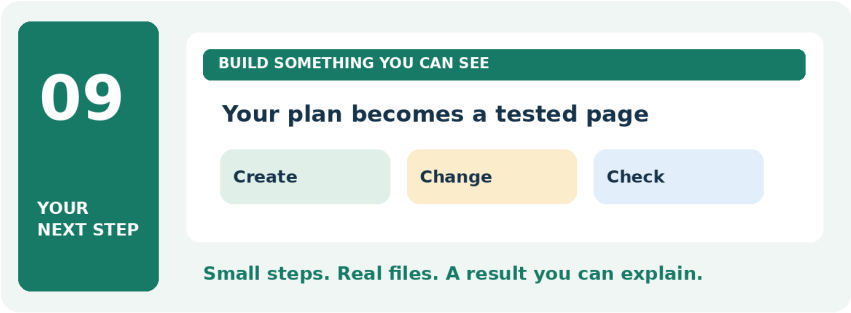
Your demonstration: open one file from each language and explain its role. Then show the saved record. No payment, seat allocation or email ticket is issued.

AI & publishing 1/1

Part 09 Build with AI and explore more tools

Use your foundation skills to guide, inspect and test AI-assisted work.

VISUAL EXPLANATION



Your finish line

Try Codex, publishing platforms and frameworks; finish with a Qoder project.

AI & publishing 1/4

Prepare the next AI practice project

Starting bakery project

REAL WEBSITE OUTPUT

Ada Bakes

Fresh cakes for your celebration.

Our cakes Contact

Our cakes



- Vanilla box: NGN 3000
- Chocolate box: NGN 3500
- Fruit box: NGN 4000

Show a message

Contact

Classroom practice: no real orders.

Ada Bakes learning project

AI & publishing 2/4

Prepare the next AI practice project

You have now built with the languages yourself. We will use AI to help improve a small familiar project.

1. Copy the ai-bakery folder from the practice files into Website Class.

2. In VS Code, choose File → Open Folder → ai-bakery.

AI & publishing 3/4

Prepare the next AI practice project

3 Open index.html with Live Server. Read the page and click its message button.

4 Open styles.css and app.js. Identify the styles and click handler before asking an AI to change them.

5 Make a backup of ai-bakery. In the next lessons, “your bakery project” means this folder.

AI & publishing 4/4

Prepare the next AI practice project

The five complete PHP websites remain separate. Do not upload their private database files to a static host.

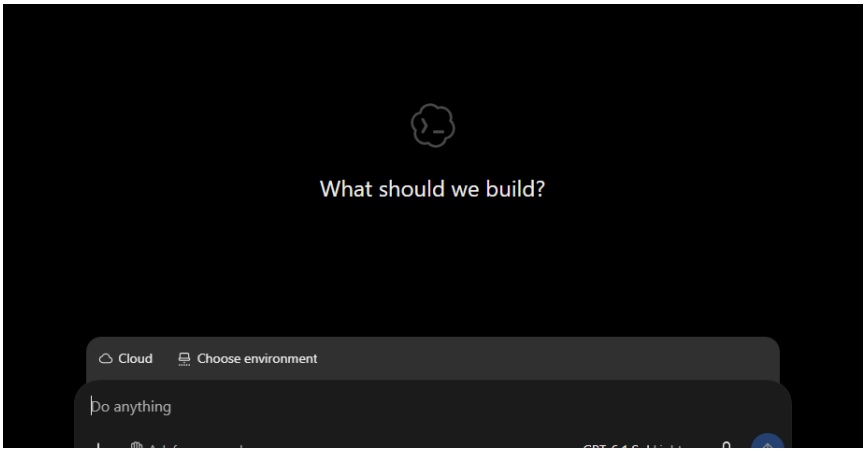
The following lessons retain the earlier practical visits to Codex, WordPress, Blogger, Google Sites, Dreamweaver, ChatGPT Sites, Laravel, Django, Flutter, React and Qoder. They come after the foundation work, not before it.

AI & publishing 1/5

Ask ChatGPT for help in ordinary English

Codex web workspace

REAL TOOL INTERFACE



The web workspace is separate from VS Code. Choose the working environment, type one clear task, and review what the assistant changes.

AI & publishing 2/5

Ask ChatGPT for help in ordinary English

YOU NEED

your browser and ChatGPT. Internet and an account may be required.

A prompt is simply the message you give an AI tool. You do not need special words. Describe the small thing you want help with.

Try your first request

1. Open a new browser tab with Ctrl + T.

AI & publishing 3/5

Ask ChatGPT for help in ordinary English

2 In the address bar, enter https://chatgpt.com. Sign in with your own approved account.

3 Start a new chat. Click the message box near the bottom.

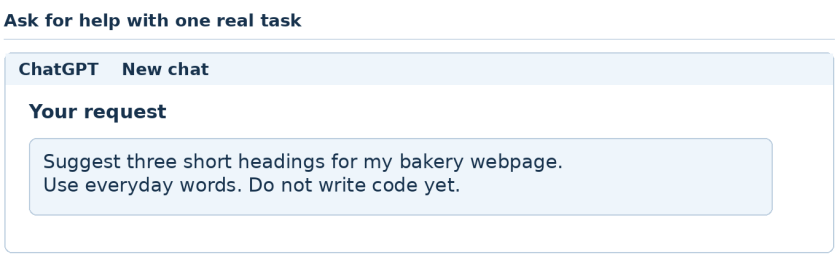
4 Type the message below and send it using the arrow button.

AI & publishing 4/5

Ask ChatGPT for help in ordinary English

“Help me improve my bakery website. Give me three short headings for a small bakery. Use simple English. Do not invent reviews or awards.”

VISUAL EXPLANATION



Read the suggestions. Choose one you like. Return to VS Code and change the words inside your h1. Save and view the page.

AI & publishing 5/5

Ask ChatGPT for help in ordinary English

You remain the designer: you choose what fits the business. The AI may suggest words that sound good but are not true.

If you copy code later, ask: “Which file does this go in? Am I adding it or replacing something?” Do not paste code until that is clear.

TRY

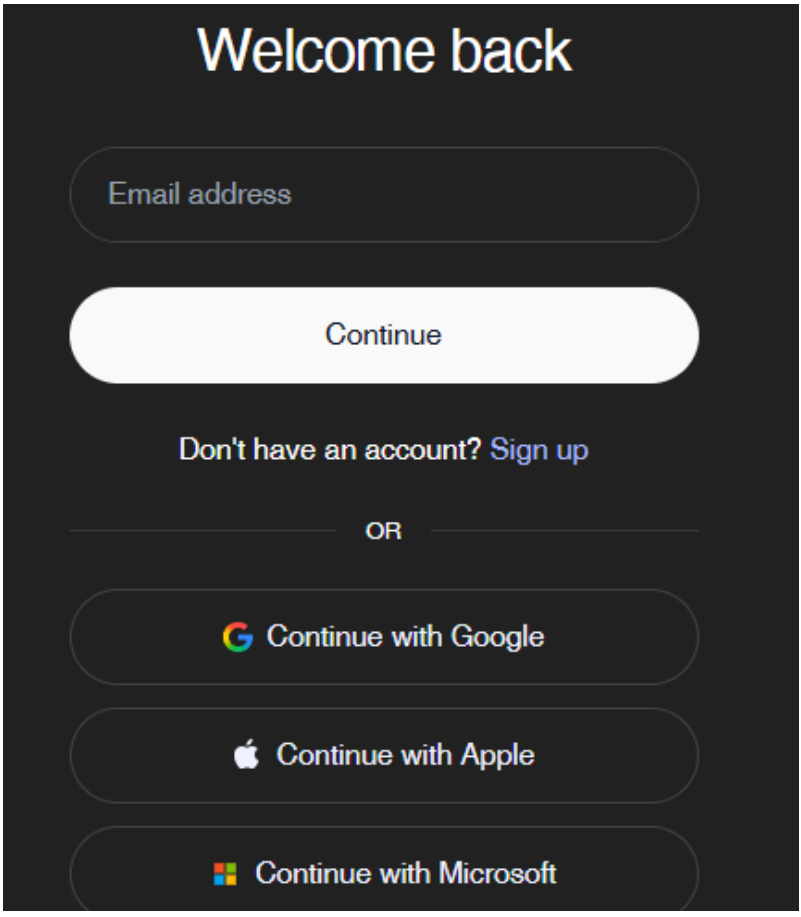
ask for a one-sentence description of a product. Edit the answer so it sounds natural to you.

AI & publishing 1/7

Install Codex in VS Code

Account sign-in screen

REAL TOOL INTERFACE



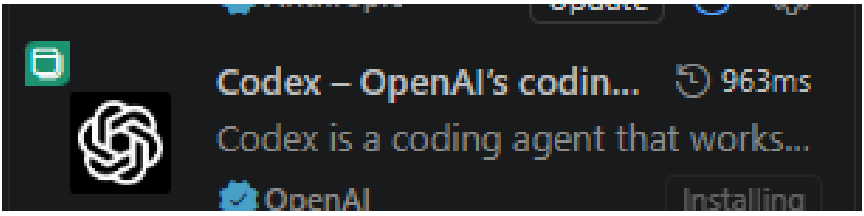
AI & publishing 2/7

Install Codex in VS Code

Use your own account. Follow the sign-in page, then return to VS Code. Never paste your password into a chat prompt.

The Codex extension in VS Code

REAL TOOL INTERFACE



AI & publishing 3/7

Install Codex in VS Code

Find Codex from OpenAI in Extensions. Install it, open its panel, and follow the sign-in instructions.

YOU NEED

VS Code, internet and an account with access to Codex.

Codex can work beside your files. It can explain code and help change it. Account access and usage limits vary; check your account before starting.

AI & publishing 4/7

Install Codex in VS Code

Set it up

1. Open ai-bakery in VS Code.

2. Press Ctrl + Shift + X to open Extensions.

3. Search for Codex. Choose the official extension published by OpenAI. Use the official installation link in S6 if you are unsure which result is correct.

AI & publishing 5/7

Install Codex in VS Code

4 Click Install. Click the Codex icon when it appears.

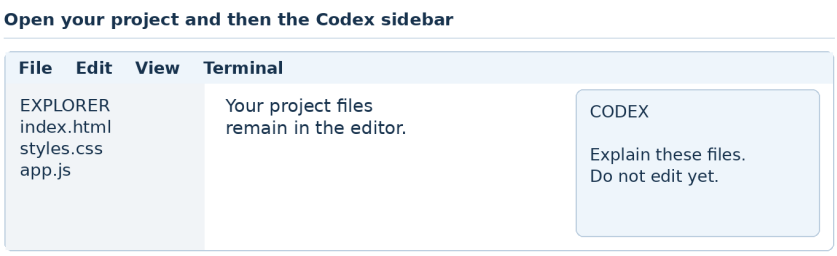
5 If you cannot see it, press Ctrl + Shift + P. Type Codex: Open Codex Sidebar and select that command.

6 Follow the sign-in instructions. Choose Sign in with ChatGPT when offered and return to VS Code after completing the browser sign-in.

AI & publishing

Install Codex in VS Code

VISUAL EXPLANATION



Begin with a question

Type in the Codex panel:

“Look at index.html, styles.css and app.js. Explain what each file does in three short sentences. Do not change any files.”

AI & publishing 7/7

Install Codex in VS Code

You should receive an explanation before making an edit. If the tool cannot see the files, check that the correct folder is open.

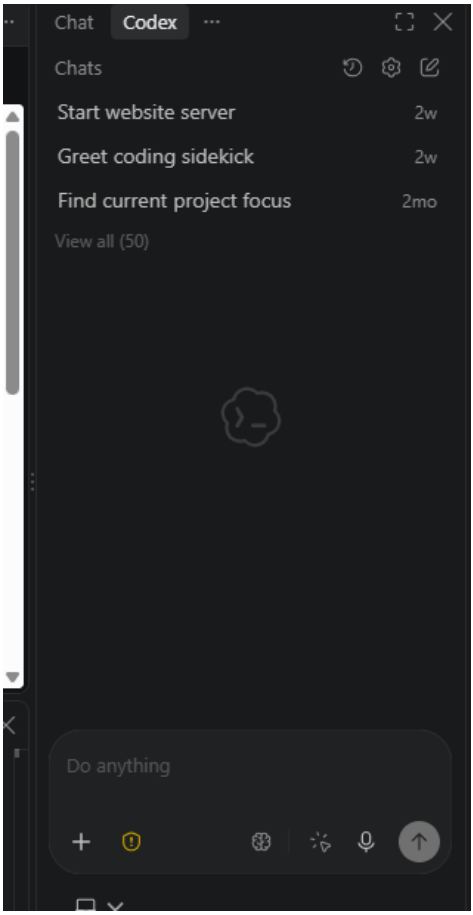
Do not buy API credits for this lesson by accident. If a screen requires unexpected payment, cancel it and check your account plan. You can continue by asking ChatGPT to explain code and making edits yourself.

AI & publishing 1/6

Let Codex make one small improvement

Codex chat panel

REAL TOOL INTERFACE



AI & publishing 2/6

Let Codex make one small improvement

Click the message box near the bottom. Ask for one small change. Read the proposed changes and test the page before keeping them.

YOU NEED

VS Code, Codex and the working bakery project.

First make another backup using the folder-copy steps in Setup lesson 4. A small change is easier to judge than an entire new website.

AI & publishing 3/6

Let Codex make one small improvement

Ask for this change

In the Codex panel, type:

“Make the existing message button blue with white text and comfortable padding. Keep the words and button behaviour. Change only styles.css. Explain the change briefly.”

1. Read requests to edit files or run commands. If a command is unfamiliar, ask the tool to explain what it changes before you approve it.

AI & publishing 4/6

Let Codex make one small improvement

2 When Codex finishes, open its changed-file view or compare styles.css with your backup.

3 Save if the editor shows unsaved changes.

4 Look at the browser. The button should look different.

5 Click it. The original thank-you message should still appear.

AI & publishing 5/6

Let Codex make one small improvement

VISUAL EXPLANATION

Read the change before accepting the result

Button style before

background: white;

Button style after

background: #164e9b;

If the result is wrong

Tell Codex exactly what you observed: “The colour changed, but clicking no longer shows the message. Restore the original click behaviour. Do not redesign the page.”

AI & publishing 6/6

Let Codex make one small improvement

You should keep a change only after trying it yourself. “Done” in a chat does not prove that a button works.

TRY

ask for one spacing improvement. Compare the changed file with the visible result.

AI & publishing 1/5

Save versions with GitHub Desktop

YOU NEED

GitHub Desktop, VS Code and your finished static files. Work in your own practice repository.

A saved version is a point you can return to. Git calls it a commit. GitHub Desktop lets you create these versions using buttons.

AI & publishing 2/5

Save versions with GitHub Desktop

Make a practice repository

1. Download GitHub Desktop from https://desktop.github.com. Install it, open it and sign in to your GitHub account when prompted.

2. Choose File → New repository. Name it bakery-practice. Pick Documents as its location. Select the option to create a README, then create the repository.

AI & publishing 3/5

Save versions with GitHub Desktop

3 In File Explorer, copy index.html, styles.css, app.js and images from ai-bakery into the new bakery-practice folder. Do not copy your backup folder.

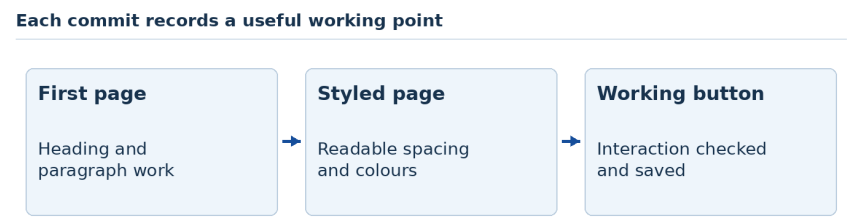
4 In GitHub Desktop, look at Changes. Click a filename to see its changes.

5 In Summary, type Add my bakery website. Click Commit to the displayed branch name.

AI & publishing

Save versions with GitHub Desktop

VISUAL EXPLANATION



You have saved a local version. It is not a public website. Publishing a repository online is a separate action.

Open bakery-practice in VS Code. Change a heading and save. GitHub Desktop should show that change. Commit it with a clear summary.

AI & publishing 5/5

Save versions with GitHub Desktop

In your own practice repository, choose History → right-click a practice commit → Revert Changes in Commit where available. This creates a new commit that reverses that change. Do not experiment with undoing shared work until you understand the effect.

Source: S7.

AI & publishing 1/5

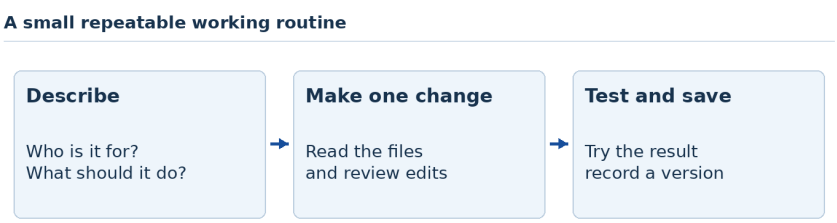
Work like a careful website designer

YOU NEED

VS Code, your browser and either Codex or Qoder later.

A useful workflow makes the work clearer. You do not need ten AI tools open at once.

VISUAL EXPLANATION



AI & publishing 2/5

Work like a careful website designer

Make a brief before asking AI to build

Create a file called brief.txt in your project. Type:

CODE EXAMPLE Copy online ↗

```
Business: Ada Bakes, a fictional bakery.
Visitor: someone choosing a celebration cake.
Pages: Home, Cakes and Contact.
Style: white background, blue details, easy reading.
Main action: read the contact information.
No payments, accounts or real order submission yet.
```

Build in small rounds

1 Ask AI for a short plan based on brief.txt. Read it.

AI & publishing 3/5

Work like a careful website designer

2 Build only the Home page. Open it and check it.

3 Save a version. Add Cakes next. Test its link.

4 Add Contact. Test all three pages.

AI & publishing 4/5

Work like a careful website designer

5 Open the page in a fresh browser tab. Can you find a cake and the contact information without reading the code? If possible, ask someone else to try too.

A professional habit: write down what should happen before testing. For example, “Clicking Cakes opens the Cakes page.”

AI & publishing 5/5

Work like a careful website designer

If two AI tools work on the same folder, use one at a time. Finish, review and save before switching. Otherwise, one tool may overwrite the other's work.

TRY

change the brief for a fictional barber, school club or clothing shop.

AI & publishing 1/5

Choose the right kind of website tool

YOU NEED

this comparison to choose your next exercise.

You have built with files. Some tools let you arrange content with buttons and blocks. They are useful too.

Notepad and VS Code

Understand and edit our website files

WordPress.com

Make a hosted website without arranging a server ourselves

AI & publishing 2/5

Choose the right kind of website tool

WordPress.org software

Run WordPress with our own local setup or chosen hosting

Blogger

Publish short articles and updates

Google Sites

Make a simple information site using visual controls

Dreamweaver

Briefly compare code and visual editing

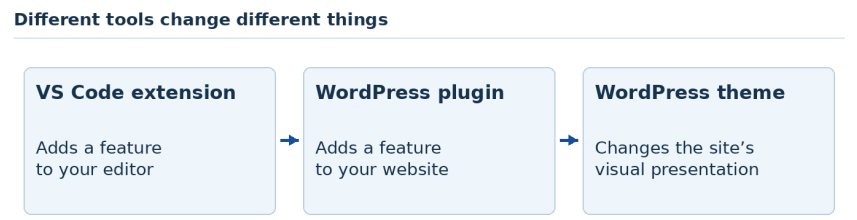
AI & publishing 3/5

Choose the right kind of website tool

Codex and Qoder

Help plan, edit and check our code

VISUAL EXPLANATION



WordPress.com and WordPress.org are not two spellings of the same service. WordPress.org provides the software. WordPress.com provides a hosted service built around WordPress.

AI & publishing 4/5

Choose the right kind of website tool

A theme changes a WordPress site's design. A plugin adds a feature. A VS Code extension adds a feature to the editor. These are different kinds of add-on.

TRY

choose a tool for a class news blog. Then choose a tool for a custom student-results application. Explain why they may need different tools.

AI & publishing 5/5

Choose the right kind of website tool

Suggested answers: Blogger or WordPress can suit a news blog. A custom application may suit Laravel or Django, with proper access controls.

Source: S8.

AI & publishing 1/5

Make a page on WordPress com

YOU NEED

your browser, internet and an approved WordPress.com account.

Create a classroom site

1. Go to https://wordpress.com. Choose the option to start a site and follow account setup.

AI & publishing 2/5

Make a page on WordPress com

2 Use a fictional site name such as Ada Bakes Practice. Choose a free site address and free option where offered. If the flow requires unexpected payment, go back and check the free option. Skip paid features for this exercise.

3 Open the site's dashboard. Find Pages → Add new page.

AI & publishing 3/5

Make a page on WordPress com

4 Enter About Ada Bakes as the title. In the writing area, type two short sentences.

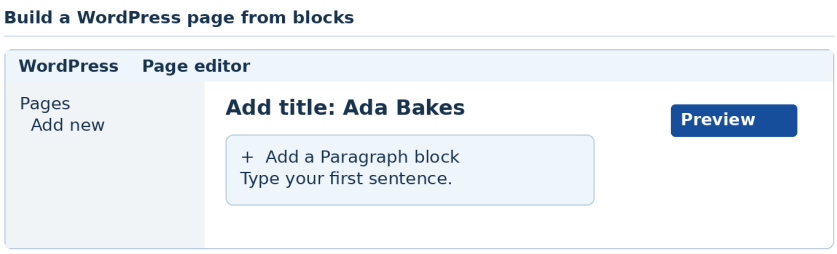
5 Click + to add a block. Choose Heading and type Our cakes. Add a Paragraph block below it.

6 Use Preview to inspect the page. Save as a draft. Before publishing, check the words, images and any personal details.

AI & publishing 4/5

Make a page on WordPress com

VISUAL EXPLANATION



A block is one piece of content, such as a heading, paragraph or image. Click a block to change it.

To explore design: dashboard → Appearance → Themes. Preview a theme before activating it. With a compatible block theme, Appearance → Editor opens the site-wide design editor.

AI & publishing 5/5

Make a page on WordPress com

TRY

add a Contact page containing fictional details. Check it on a narrow preview.

Publishing a page and launching the whole site can be separate steps. Confirm the site's visibility before sharing a link. Installing plugins on WordPress.com requires an eligible paid plan; this brief lab does not need them.

Sources: S9–S11.

AI & publishing 1/5

Try WordPress software on your computer

YOU NEED

WordPress Studio for a local WordPress site.

This is our practical introduction to the WordPress.org route. WordPress Studio runs the WordPress software locally. You do not have to buy hosting to do this lesson.

Create a local site

1. Visit https://wordpress.com/studio or follow the download link in S12.

AI & publishing 2/5

Try WordPress software on your computer

2 Download the Windows installer, open it and complete setup. Compare the stated requirements with Settings → System → About first.

3 Open Studio. Name the first site Ada Bakes Local and click Continue. If Studio already has sites, use Add site.

4 Select the site. Start it if it is stopped.

AI & publishing 3/5

Try WordPress software on your computer

5 Choose WP Admin to open its WordPress dashboard.

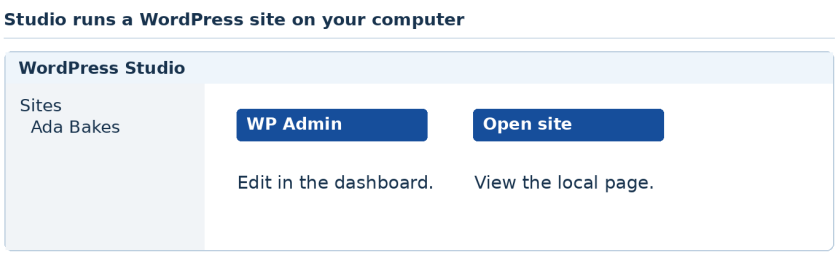
6 Open Pages → Add New. Create About Us, add two sentences, then publish it locally.

7 Use Open site from Studio to see the result. A new page may need to be added to navigation separately.

AI & publishing 4/5

Try WordPress software on your computer

VISUAL EXPLANATION



Local means running on this computer. Publishing inside this local WordPress installation does not automatically publish a public internet site.

AI & publishing 5/5

Try WordPress software on your computer

If installation is unavailable, open https://playground.wordpress.net in your browser. It is a useful WordPress sandbox. Use its save or export options before closing; do not assume a temporary session keeps your work forever.

TRY

change the page heading and view the updated local site.

Sources: S12–S14.

AI & publishing 1/6

Change a WordPress theme and try a plugin

YOU NEED

the local WordPress site from Further lesson 8. Do this locally, not on a client's live site.

Change the design

1. Open WP Admin from Studio.

2. Click Appearance → Themes. Look at the installed themes.

AI & publishing 2/6

Change a WordPress theme and try a plugin

3 Preview a theme, then Activate it. Open the site and compare the result.

4 If you need another theme, use Add New Theme and the official directory. Choose a simple block theme from the official directory and preview it before activating it.

AI & publishing 3/6

Change a WordPress theme and try a plugin

Try one small plugin

1. Open Plugins → Add New Plugin.

2. Search for Hello Dolly, the demonstration plugin by Matt Mullenweg.

3. If available, Install Now, then Activate. It displays a short lyric in the administration area; it does not create a website page.

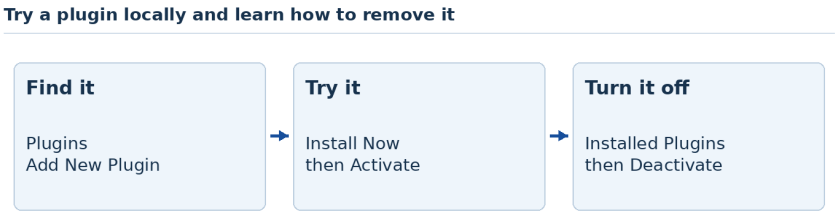
AI & publishing

Change a WordPress theme and try a plugin

4 Look for that change. Then Deactivate the plugin and compare.

5 Leave it deactivated or delete it after the demonstration.

VISUAL EXPLANATION



AI & publishing 5/6

Change a WordPress theme and try a plugin

The lesson is how installation, activation and deactivation work. A real site's plugins should solve a real need. Check maintenance, compatibility and the author before adding them.

Before changing more, open Studio's Import / Export area and export your local site using the offered export option. Save the archive in Documents → Website Class → Backups. Create Backups if it does not exist. A complete site backup differs from WordPress Tools → Export, which mainly exports content.

AI & publishing 6/6

Change a WordPress theme and try a plugin

TRY

explain the difference between a theme and a plugin.

Sources: S13, S15, S16.

AI & publishing 1/6

Publish a short article with Blogger

YOU NEED

your browser and an approved Google account.

A blog is useful for articles, news and regular updates.

Create a practice blog

1. Visit https://www.blogger.com and sign in.

AI & publishing 2/6

Publish a short article with Blogger

2 For an existing dashboard, open the blog selector at the top left and choose New blog. First-time users may see a direct Create your blog option.

3 Enter a title such as My Website Class. Continue and choose an available blog address.

4 In the dashboard, click New Post.

AI & publishing 3/6

Publish a short article with Blogger

5 Enter My first website lesson as the post title. Write three sentences about what you made.

6 Click Preview. Correct any mistakes. Save as a draft. Publish only after previewing and checking that the content is yours to share.

AI & publishing 4/6

Publish a short article with Blogger

VISUAL EXPLANATION



To change the appearance, open Theme in the left menu. Preview a theme before applying it. Layout controls where supported blog items appear.

AI & publishing 5/6

Publish a short article with Blogger

A post is an article. A page is better for information that stays useful, such as About Me. Use Pages → New Page for that kind of content.

TRY

make an About page with only information you are comfortable sharing. Do not publish a child's private details or class contact list.

CHECK

open the published link in a signed-out browser if public viewing is intended.

AI & publishing 6/6

Publish a short article with Blogger

Source: S17.

AI & publishing 1/5

Build a simple site with Google Sites

Make a class information site

1. Go to https://sites.google.com. Choose Blank under Start a new site.

2. Enter Website Class as the site document name. Click the page title and type Welcome to our class.

3. On the right, choose Insert → Text box. Type a welcome sentence.

AI & publishing 2/5

Build a simple site with Google Sites

4 Choose Insert → Images and add an image you created or have permission to use.

5 Open Pages on the right. Click + and create an About page.

6 Use the Preview button near the top. Try the phone and computer preview sizes.

AI & publishing 3/5

Build a simple site with Google Sites

7 When ready, click Publish. Choose an available address and check who can view the published site before confirming.

VISUAL EXPLANATION



AI & publishing 4/5

Build a simple site with Google Sites

Google Sites saves editing work as you go, but later changes still need publishing before visitors see the updated public version.

If an account cannot create or publicly share a site, it may be restricted by a school, organisation or account type. Read the restriction message. Use an account you own only if the service permits it; otherwise continue with a local project.

AI & publishing 5/5

Build a simple site with Google Sites

TRY

add a Learning Goals page with three goals. Open the published site and test its navigation.

This is Google Sites. The ChatGPT Sites demonstration later is a different product.

Source: S18.

AI & publishing 1/5

Briefly try Dreamweaver and a design tool

YOU NEED

Dreamweaver only if you already have licensed access. This optional comparison is not required to finish the course.

Compare another editor

1. In Dreamweaver, choose File → New.

2. Select an HTML document and click Create.

AI & publishing 2/5

Briefly try Dreamweaver and a design tool

3 In Code view, place <h1>Hello from our class</h1> between the body tags.

4 Choose File → Save as. Save the document as dreamweaver-practice.html in a separate practice folder.

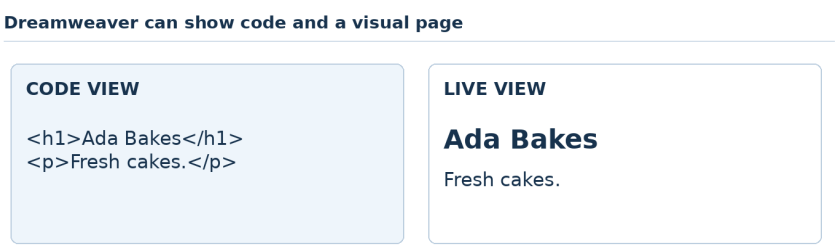
5 Switch to Split or Live view using the document view controls. Compare the code with its visible result.

AI & publishing 3/5

Briefly try Dreamweaver and a design tool

You should recognise the HTML even though the editor looks different. An editor helps you work; the HTML file still matters.

VISUAL EXPLANATION



AI & publishing 4/5

Briefly try Dreamweaver and a design tool

Sketch before building

Paper is enough. Draw a rectangle for a phone screen. Add a business name, a heading, three products and a contact button.

For a digital alternative, you can open Figma or Penpot, create a design file and draw the same arrangement with rectangles and text. These tools help plan a design; a drawing is not automatically a working website.

AI & publishing 5/5

Briefly try Dreamweaver and a design tool

TRY

point to the first thing you want a visitor to notice. Make it clearer without making every item large.

Sources: S19, S20.

AI & publishing 1/4

Try building with ChatGPT Sites

YOU NEED

your account if ChatGPT Sites is available. Skip this optional exercise if you cannot see it.

This is separate from Google Sites. It allows an AI-assisted site-building workflow inside ChatGPT.

Watch one small build

1. Open ChatGPT and start a new conversation with Sites available in that workspace.

AI & publishing 2/4

Try building with ChatGPT Sites

2 Send this request:

“Use Sites to build a simple fictional bakery website called Ada Bakes. Use a white background, blue buttons and three cake cards. Add Home, Cakes and Contact sections. Keep it private for review. Do not add payments or collect customer information.”

3 Open the resulting site preview or link. Read the actual text and try every link.

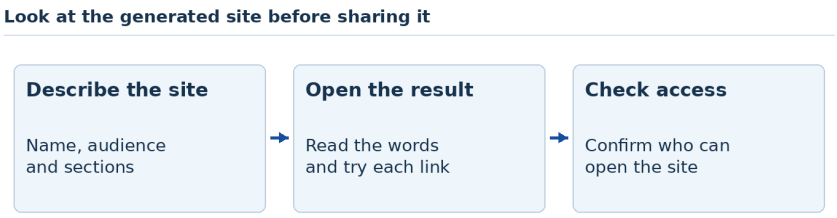
AI & publishing 3/4

Try building with ChatGPT Sites

4 Ask for one change: “Make the cake descriptions easier to read on a phone. Keep everything else.”

5 View the revised result. Check the site's actual sharing setting before sending its link to anyone.

VISUAL EXPLANATION



AI & publishing 4/4

Try building with ChatGPT Sites

Some generated sites may already have a hosted link while remaining private. A preview, a hosted site and a publicly accessible site are not automatically the same thing.

Discuss: what was quicker than hand-coding? What still needed a human decision? Look for wrong content, broken actions and small-screen problems.

Source: S21. Availability depends on the account and workspace.

AI & publishing

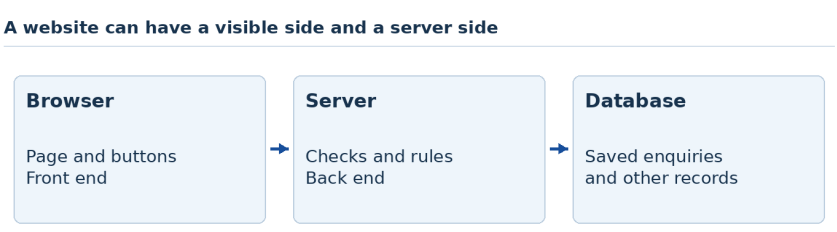
What is a stack

YOU NEED

paper and a pencil. No new installation yet.

A stack is a group of tools used together to make a working website. You choose it for the job you need to do.

VISUAL EXPLANATION



AI & publishing 2/5

What is a stack

The front end is what a visitor sees and uses. The back end handles work on a server, such as checking a sign-in. A database stores organised information, such as orders.

HTML, CSS and JavaScript

A small business information website

Laravel with PHP

A website that saves enquiries or bookings

AI & publishing 3/5

What is a stack

Django with Python

A data-driven website with an administration area

React with JavaScript

An interactive interface with changing screens

Flutter with Dart

An app-style interface for web and other platforms

AI & publishing 4/5

What is a stack

VS Code and Qoder are editors. They can help you work on different stacks. They are not replacements for PHP, Python or Dart.

Choose the job first

Circle one: a bakery brochure, a booking system or an interactive learning app. Write which tools you would start with and one reason for your choice.

AI & publishing 5/5

What is a stack

For a simple brochure, our existing HTML website is enough. For ordinary text-heavy websites, HTML or a web framework is usually a better first choice than Flutter. Flutter is useful when you want an app experience.

Sources: S22–S27.

AI & publishing 1/2

Meet the terminal slowly

1. Open VS Code. Choose File → Open Folder and open Website Class.
2. Choose Terminal → New Terminal. A panel opens at the bottom.
3. Click beside the blinking cursor in that panel.
4. Type the command below and press Enter once.

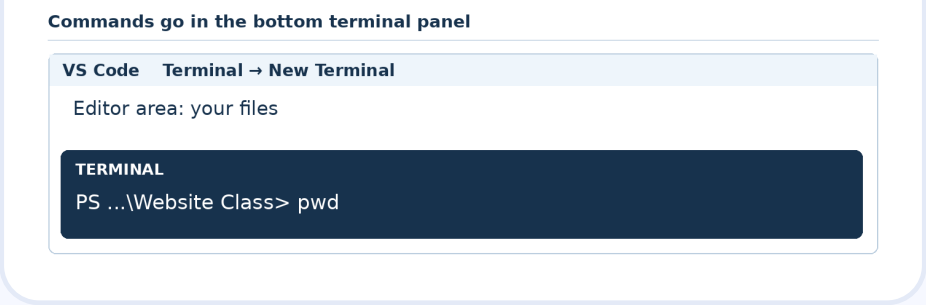
AI & publishing 2/2

Meet the terminal slowly

CODE EXAMPLE Copy online ↗

```
pwd
```

VISUAL EXPLANATION



Three useful habits

An address such as http://127.0.0.1:8000 points to a service on your own computer. It is not a public link you can send to a customer.

AI & publishing 1/6

Start a Laravel project

YOU NEED

VS Code and the working Laravel Herd installation from the PHP section.

PHP runs Laravel. Composer downloads the PHP packages it needs. Herd supplies these tools. Use supported versions of both tools; this exercise targets Laravel 13.

AI & publishing 2/6

Start a Laravel project

1. Visit herd.laravel.com. Choose the Windows download. Follow the same installation and verification steps used in the PHP section.

2. Restart VS Code after installation. Open Website Class, then Terminal → New Terminal.

AI & publishing

Start a Laravel project

3 Run each command separately. The first two should show version numbers. If either fails, close and reopen VS Code, check that Herd opens, then repeat only the failed command.

CODE EXAMPLE

Copy online ↗

```
php --version
composer --version
composer create-project laravel/laravel bakery-laravel "
^13.0"
cd bakery-laravel
php artisan serve
```

AI & publishing 4/6

Start a Laravel project

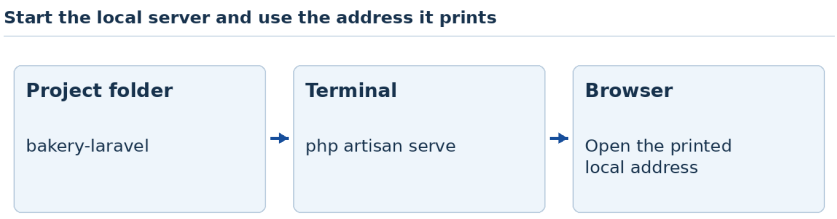
4 Wait while packages download. Open the local address printed after the final command. You should see Laravel's starting page.

5 In VS Code, choose File → Open Folder and open the new bakery-laravel folder. If the running terminal closes, open a terminal here and run php artisan serve again.

AI & publishing

Start a Laravel project

VISUAL EXPLANATION



This first page does not need a paid Herd service. We will use plain HTML in a Laravel template, so we do not need a separate front-end build process for this exercise.

AI & publishing 6/6

Start a Laravel project

Success: the page opens while the local server is running. If the address fails, check the terminal first.

Sources: S22, S28. Check the current PHP requirement in Laravel's installation guide before creating the project.

AI & publishing 1/4

Give Laravel your own page

YOU NEED
the bakery-laravel folder in VS Code.

A route tells Laravel what to show for an address. A view is the file containing the page you want to show.

1 In the Explorer, open resources, then views. Create a file named home.blade.php inside views.

2 Put this small page in the file. Save it.

AI & publishing 2/4

Give Laravel your own page

CODE EXAMPLE Copy online ↗

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport"
        content="width=device-width, initial-scale=1">
  <title>Ada Bakes</title>
</head>
<body>
  <h1>Ada Bakes</h1>
  <p>Our Laravel website is working.</p>
</body>
</html>
```

3 Open routes/web.php. Replace its contents with this and save:

AI & publishing 3/4

Give Laravel your own page

CODE EXAMPLE Copy online ↗

```
<?php
use Illuminate\Support\Facades\Route;
Route::get('/', function () {
    return view('home');
});
```

4 Refresh the local Laravel page. You should see Ada Bakes.

AI & publishing 4/4

Give Laravel your own page

Now ask Codex: “Style resources/views/home.blade.php with simple CSS inside a style tag. Use a blue heading and a readable centred layout. Keep the route and packages unchanged.” Review the change, refresh and narrow the browser.

Source: S22. A blank page or error means stop and read the terminal before adding more features.

AI & publishing 1/5

Let Laravel save an enquiry

YOU NEED

your working Laravel project and Codex. This is a supervised extension.

Until now, our pages have mainly displayed information. This task adds storage. Use invented details only.

Build one small feature

1. Save a Git checkpoint or copy the project folder before editing.

AI & publishing 2/5

Let Laravel save an enquiry

2 Open the Codex panel. Ask it to inspect the project and propose a short plan using this request:

“Add an enquiry form to this Laravel 13 project. Use SQLite. Fields: name, email and message. Validate them on the server. Use Laravel CSRF protection. Save valid enquiries in the database and show a clear success message. Do not add payments, email sending or a public list of enquiries. Explain the files and commands before changing them.”

AI & publishing 3/5

Let Laravel save an enquiry

3 Read the plan. Compare the proposed files with your requirements before allowing changes. Ask Codex to explain any database migration command before running it. A migration sets up or changes database tables.

4 Run the suggested tests. Then test the form yourself using the table below.

Submit an empty form

Helpful messages; nothing saved

AI & publishing 4/5

Let Laravel save an enquiry

Enter an invalid email

A useful email error

Submit fictional valid details

A success message and one saved record

Stop and restart the server

The record still exists in SQLite

AI & publishing 5/5

Let Laravel save an enquiry

Open the project database/database.sqlite in DB Browser, choose Browse Data, then choose the table named in your migration. Submit fictional data and confirm that its row exists. A success message alone does not prove that anything was saved.

Keep this exercise local. A real public enquiry service also needs spam controls, secure hosting, access controls, backups and a privacy review.

Sources: S22, S29.

AI & publishing 1/5

Start a Django project

YOU NEED

VS Code and Python 3.12 in a separate project folder.

Django is a web framework written in Python. We will use the supported Django 5.2 series for this classroom example.

AI & publishing 2/5

Start a Django project

1. Open https://www.python.org/downloads/windows/ and locate Python 3.12. Download the Windows installer matching your system. Run it and include the Python launcher if offered. Reopen VS Code and run py -3.12 --version. Expect Python 3.12.x.

2. Inside Website Class, make a new empty folder named bakery-django. Open that folder in VS Code.

AI & publishing 3/5

Start a Django project

3 Choose Terminal → New Terminal. Run these commands one at a time:

CODE EXAMPLE Copy online ↗

```
py -3.12 -m venv .venv
.venv\Scripts\python -m pip install "Django>=5.2,<5.3"
.venv\Scripts\python -m django startproject bakery .
.venv\Scripts\python manage.py migrate
.venv\Scripts\python manage.py runserver
```

4 Open the local address printed in the terminal. You should see Django's welcome page.

AI & publishing 4/5

Start a Django project

VISUAL EXPLANATION

Keep this Django project together in one folder

VS Code bakery-django	
bakery-django .venv bakery manage.py	.venv holds Python tools. bakery holds project settings. manage.py runs project tasks.

The .venv folder keeps this project's Python tools together. Using its Python path means we do not need a separate activation step here. The dot at the end of startproject bakery . means “put the project in this folder.” Do not leave it out.

AI & publishing 5/5

Start a Django project

If py is not recognised, reopen the official installer, choose Modify and check that the launcher is included, then reopen VS Code. Do not keep pasting the remaining commands after an error.

Sources: S23, S30.

AI & publishing 1/5

Give Django your own page

YOU NEED

bakery-django in VS Code. Keep the server running.

1 Inside the inner bakery folder, create views.py. This is beside settings.py and urls.py. Add:

CODE EXAMPLE Copy online ↗

```
from django.http import HttpResponse

def home(request):
    return HttpResponse(
        "<h1>Ada Bakes</h1>"
        "<p>Our Django page is working.</p>"
    )
```

AI & publishing 2/5

Give Django your own page

2 Open bakery/urls.py. Replace its contents with:

CODE EXAMPLE Copy online ↗

```
from django.contrib import admin
from django.urls import path
from .views import home

urlpatterns = [
    path("admin/", admin.site.urls),
    path("", home),
]
```

3 Save both files and refresh the local page. You should see your heading.

AI & publishing 3/5

Give Django your own page

Ask AI for the next small step

“Move my Django home page into a proper HTML template. Add a page title, viewport setting and simple readable CSS. Keep the home address working. Explain where each new file goes. Do not add packages.”

AI & publishing 4/5

Give Django your own page

Then request a Cake model with name and price, registered in Django admin. Read the generated migration, then run .venv\Scripts\python manage.py makemigrations followed by .venv\Scripts\python manage.py migrate. Run .venv\Scripts\python manage.py createsuperuser and answer its prompts. Open /admin/ on the local server, sign in, add a cake and confirm it appears.

AI & publishing 5/5

Give Django your own page

CHECK

sign in at /admin/, add one fictional cake and confirm it remains after restarting. Never paste the administrator password into AI chat.

Sources: S23 and the Django admin reference on the last source page. The template and admin extensions are guided AI exercises, not required before your first Django success.

AI & publishing 1/4

Start a Flutter web project

YOU NEED

VS Code, Git, Chrome and the Flutter extension. Allow setup time before class.

Flutter uses the Dart language. Today we are making a small web app, not a phone installation.

1. Install Git from https://git-scm.com/downloads/win and Chrome from https://www.google.com/chrome/. Use the VS Code installation you already prepared.

AI & publishing 2/4

Start a Flutter web project

2 In VS Code, open Extensions. Search Flutter. Install the official extension published by Dart Code.

3 Press Ctrl + Shift + P. Type Flutter: New Project and select it. If asked for the Flutter SDK, choose Download SDK. Choose a simple folder, such as C:\dev, and follow Clone Flutter and Add SDK to PATH when offered.

AI & publishing

Start a Flutter web project

4 Restart VS Code. Run flutter doctor -v. Read each message and follow the linked instructions for Flutter and Chrome. Missing Android tools can be ignored for this web-only exercise.

5 Open Website Class in VS Code. Run these commands separately:

CODE EXAMPLE

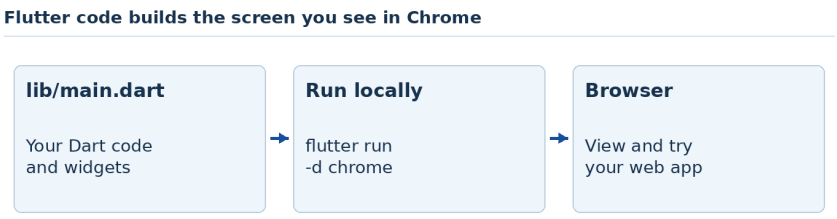
Copy online ↗

```
flutter create bakery_flutter
cd bakery_flutter
flutter run -d chrome
```

AI & publishing

Start a Flutter web project

VISUAL EXPLANATION



You should see Flutter's sample app in Chrome. Keep the terminal open. If Chrome is unavailable on Windows, check Edge support and use flutter run -d edge.

Source: S24. If the SDK download is interrupted, resume setup when the connection is stable. Keep practising HTML and CSS while it downloads.

AI & publishing 1/4

Change the Flutter screen

YOU NEED

the bakery_flutter folder in VS Code.

A widget is a piece of a Flutter screen. Text is a widget. A button is a widget. We put widgets together to make a page.

1 Open lib/main.dart. Replace its contents with the code below.

AI & publishing 2/4

Change the Flutter screen

2 Save. In the running Flutter terminal, press r to reload. If no server is running, use flutter run -d chrome again.

CODE EXAMPLE Copy online ↗

```
import 'package:flutter/material.dart';

void main() => runApp(const BakeryApp());

class BakeryApp extends StatelessWidget {
  const BakeryApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: const Text('Ada Bakes')),
        body: const Center(
          child: Text('Our Flutter web app works!'),
        ),
      ),
    );
  }
}
```

AI & publishing 3/4

Change the Flutter screen

3 You should see Ada Bakes and your message. Change the message and reload once more.

Ask Codex: “Add a simple cake quantity selector with plus and minus buttons. Do not allow a negative quantity. Keep all changes in lib/main.dart and add no packages. Explain the state that remembers the number.”

AI & publishing 4/4

Change the Flutter screen

CHECK

start at zero, press plus three times, then press minus four times. The number should finish at zero. This is local screen state; it is not a saved customer order.

Sources: S24, S25.

AI & publishing 1/5

A short visit to React

YOU NEED

VS Code and a current Node.js LTS release supported by Vite.

React helps build interactive interfaces from reusable pieces called components. Vite provides a development setup. Neither one automatically gives us a database or payment system.

AI & publishing 2/5

A short visit to React

1. Open https://nodejs.org, download the Windows LTS installer and follow setup. Restart VS Code. Run node --version and npm --version; both should print a number. Compare the Node version with https://vite.dev/guide/.

2. Open Website Class. In a new terminal, run each command:

AI & publishing 3/5

A short visit to React

CODE EXAMPLE Copy online ↗

```
npm create vite@latest bakery-react -- --template react
cd bakery-react
npm install
npm run dev
```

3 Open the address printed by Vite. Open bakery-react as the VS Code folder.

4 Replace src/App.jsx with this and save:

AI & publishing

A short visit to React

CODE EXAMPLE

Copy online ↗

```
import { useState } from 'react';

export default function App() {
  const [count, setCount] = useState(0);
  return <main>
    <h1>Ada Bakes</h1>
    <button onClick={() => setCount(count + 1)}>
      Cakes selected: {count}
    </button>
  </main>;
}
```

Click the button three times. You should see 3. Refresh: it returns to 0 because we have not added storage.

AI & publishing 5/5

A short visit to React

Ask AI to add a Reset button. Review its changes and test again. After these demonstrations, choose one stack for your next project rather than trying to master all four at once.

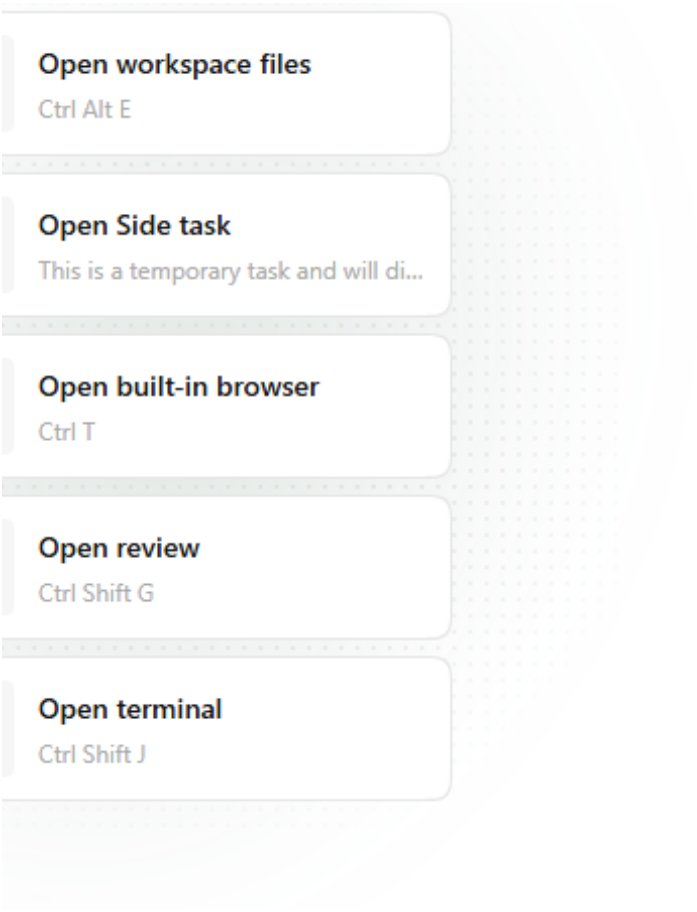
Sources: S26, S27.

AI & publishing 1/6

Meet Qoder IDE

Qoder workspace controls

REAL TOOL INTERFACE



AI & publishing 2/6

Meet Qoder IDE

Choose Open workspace files to inspect your project. Open terminal runs commands. Open review lets you check changes before keeping them.

YOU NEED

Qoder IDE in Editor mode. Sign in and check the plan shown in your account.

You already know the important habit: ask for a small change, inspect it and test it. We will use that same habit in Qoder.

AI & publishing 3/6

Meet Qoder IDE

1. Visit qoder.com. Download Qoder IDE for Windows from its official download page. Run the installer and sign in with an account you are allowed to use.

2. Open your existing bakery project folder using the welcome screen's open-project control or File → Open Folder. Confirm that index.html appears in the file list.

AI & publishing 4/6

Meet Qoder IDE

3 Stay in the Editor. Press Ctrl + L to open Chat. Select Ask mode in the chat controls.

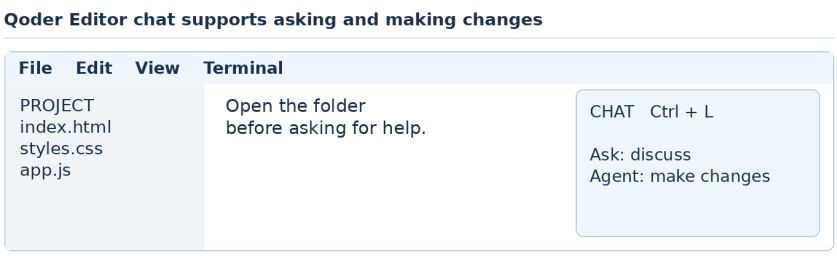
4 Ask: “Explain the files in this small website in everyday English. Do not change anything.”

5 Read the answer and point to the actual files it describes.

AI & publishing

Meet Qoder IDE

VISUAL EXPLANATION



Ask mode is useful for discussion. Agent mode can make changes across files. Before allowing changes, make a backup or Git checkpoint. Review requested commands and the changed files.

AI & publishing 6/6

Meet Qoder IDE

Qoder has a free plan, but it is not an unlimited promise. Agent access, credits and trial offers depend on the current plan. Check the account's plan and remaining allowance in your account settings. If access is limited, continue the same exercise with your available Codex access or manual edits.

Sources: S31, S32. Teach Editor rather than the retiring Quest window; see the tool notes.

AI & publishing 1/5

Your final Qoder website

YOU NEED

Qoder Editor and a fresh folder named ada-final inside Website Class.

We will make a complete small website in three rounds. Do not ask AI to build every possible feature in one message.

AI & publishing 2/5

Your final Qoder website

Round one Plan

Create brief.txt in the folder. Write the business name, visitors, three sections and one main action. Example: “Ada Bakes. Local customers. Home, Cakes, Contact. Main action: read how to enquire. This is a fictional classroom demo.”

In Ask mode, say: “Read brief.txt. Suggest a simple page layout using HTML, CSS and JavaScript. Use index.html, styles.css and app.js. Explain your plan before editing.”

AI & publishing 3/5

Your final Qoder website

Round two Build

Switch to Agent mode after reviewing the plan. Say: “Build the agreed page. Use three fictional cakes, readable text and a blue-and-white design. Make it work on a narrow screen. Use no external packages. Do not add a working order form or payments. Explain how to open it.”

Open index.html in your browser. If Qoder provides a local preview, use its printed address. You can also open this folder in VS Code and use Live Server.

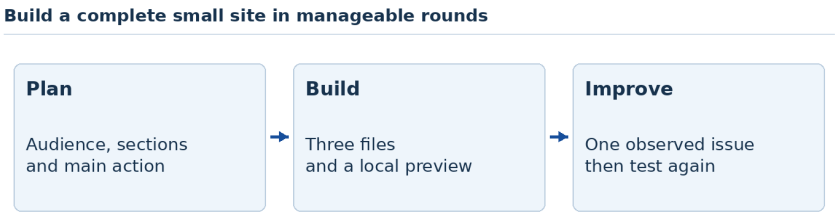
AI & publishing 4/5

Your final Qoder website

Round three Improve

Take a screenshot using Windows + Shift + S. Select the page area. If image attachments are available in your Qoder chat, attach it and ask for one specific improvement. Otherwise describe the problem and the screen width.

VISUAL EXPLANATION



AI & publishing 5/5

Your final Qoder website

Success: all three files are present, the contact section is reachable and the page reads clearly on a narrow screen. Save a checkpoint before the next lesson.

AI & publishing 1/6

Give AI a clear finished test

YOU NEED

the same final project. Choose the task that matches your progress.

Everyone Add a useful interaction

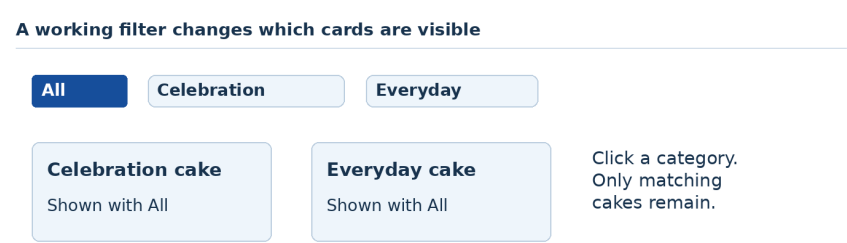
Ask Qoder Agent:

AI & publishing

Give AI a clear finished test

“Add buttons labelled All, Celebration and Everyday above the cake cards. Clicking a button should show only the matching cakes. Keep All selected on first load. Use real buttons, show which filter is selected, and keep keyboard use working. Use only our three existing files.”

VISUAL EXPLANATION



AI & publishing 3/6

Give AI a clear finished test

Test before accepting the work:

1. Refresh. All cakes should be visible.
2. Click Celebration. Only cakes in that category should remain.
3. Click Everyday, then All. Check both results.

AI & publishing

Give AI a clear finished test

4 Press Tab to reach a filter button. Press Enter. It should work without the mouse.

5 Narrow the browser and repeat the test.

If it fails, tell Qoder: “I clicked Celebration. Everyday cakes still showed. I expected only Celebration cakes. Find the cause and fix only this behaviour.”

AI & publishing 5/6

Give AI a clear finished test

Advanced option Use your Laravel project

Open bakery-laravel in Qoder. Ask it to inspect the enquiry feature from Further lesson 18 and propose tests for valid, empty and invalid submissions. Read each test, run it and check the database yourself. Keep the project local until you have completed the publishing checklist and obtained experienced review for real customer data.

AI & publishing 6/6

Give AI a clear finished test

Your achievement is not the size of the AI reply. It is a feature that you can explain, use and test.

AI & publishing 1/5

Put the simple website online

YOU NEED

a GitHub account and the browser. This route is for the HTML project only.

GitHub Pages can host our static files. It does not run Laravel, Django or WordPress's PHP server. Use fictional public content for this exercise.

AI & publishing 2/5

Put the simple website online

1. Sign in at github.com. Click + near the top, then New repository. Name it ada-bakes-demo. Choose Public and create the repository. Public means other people can see these files.

2. On the empty repository page, choose uploading an existing file. If files already exist, choose Add file → Upload files.

AI & publishing 3/5

Put the simple website online

3 Drag in index.html, styles.css, app.js and your images folder. Make sure index.html is at the top level, not inside another folder. Write “First website” as the commit message and commit the files.

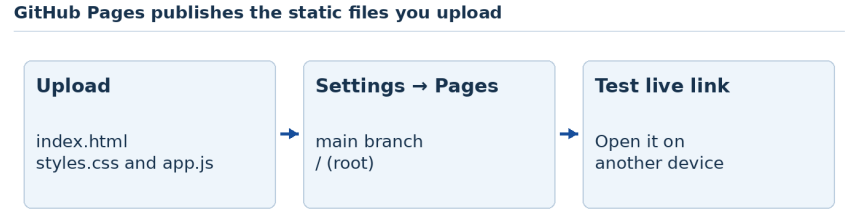
4 Open Settings → Pages. Under Build and deployment, choose Deploy from a branch. Select main and / (root), then Save.

AI & publishing

Put the simple website online

5 Wait for GitHub's deployment to finish. Return to Pages and open the site address shown there.

VISUAL EXPLANATION



Use paths such as images/cake.jpg, not C:\Users\... paths. Filenames and capital letters must match exactly.

AI & publishing 5/5

Put the simple website online

Open the public link on another device or in a private browser window. Check pictures, filters and contact links. If an image is missing online but works locally, compare its exact spelling and folder path.

Do not upload passwords, private files or .env files. For changes, upload the corrected files and wait for the new deployment. Keep a local copy too.

Source: S33. A GitHub account's eligible plan and repository visibility affect Pages availability.

AI & publishing 1/5

Show what you can do

YOU NEED

your finished website, the browser and your project folder.

Open your website and say: “This is who it is for. This is what a visitor should do first.” Then demonstrate the work instead of reading a long report.

Your final check

I can find the project folder and open the right file.

AI & publishing 2/5

Show what you can do

I can change a heading, save it and see the change.

I can explain what HTML, CSS and JavaScript do here.

The page is readable on a narrow screen and at 200% zoom.

I can reach buttons and links with Tab and activate them.

Pictures have useful alternative text, or empty alt text if decorative.

AI & publishing 3/5

Show what you can do

My links and filters do what their labels promise.

I can show a backup or saved Git version.

I can explain one AI mistake that I found and corrected.

The public website uses appropriate, accurate content.

AI & publishing 4/5

Show what you can do

Hand over the project

Create README.txt in your project folder. Write its name, how to open it, what works, what is only a demonstration, and how to change the main text. Include the live address if you published it.

If possible, invite someone to try the website without your help. When working alone, close your notes and repeat the main task as a visitor. Write down one confusing moment and improve it.

AI & publishing 5/5

Show what you can do

You began by learning where to click. You can now create files, build a page, improve its design and use AI with judgment. Your next goal is one small real project, completed carefully.

Your completion check: do each practical task without copying the answer. Revisit any task you can describe but cannot yet perform.

AI & publishing 1/3

Paul’s workflow: from an idea to a live website

1 **IDEA**
Explain the job clearly

↓

2 **AI + PROJECT FOLDER**
Build, review and correct

↓

3 **LOCAL BROWSER**
Try the working result

↓

4 **GITHUB**
Save a tested checkpoint

↓

5 **SERVER**
Deploy and check the live site

AI & publishing 2/3

Paul’s workflow: from an idea to a live website

You do not have to type every line yourself. Your job is to explain the idea, inspect what AI makes and try the website yourself.

Paul uses Codex and Kilo inside VS Code, and also works directly in Qoder. Sometimes he asks ChatGPT to turn an idea into clear instructions first. Either route is useful.

AI & publishing 3/3

Paul’s workflow: from an idea to a live website

Paul describes using GitHub to store the code for BelovedSubP.com, then pulling updates from his server terminal. We will practise that pattern on a small training project. We have not inspected that business website’s private code or server.

Start with the local project. Move to the server lessons after you can explain and test it.

AI & publishing 1/6

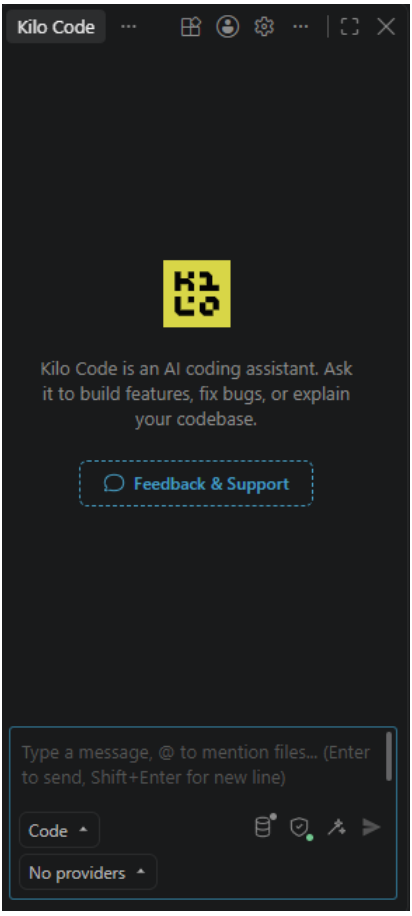
AI setup 1: open a folder and choose a helper

A project folder is the home for one website. Your AI helper needs to work inside that folder.

AI & publishing 2/6

AI setup 1: open a folder and choose a helper

REAL TOOL INTERFACE



AI & publishing 3/6

AI setup 1: open a folder and choose a helper

In this screenshot, No providers means a model connection still needs to be chosen. Open that control and follow Kilo’s current sign-in/provider instructions before sending your task.

1. In Windows File Explorer, open Documents. Right-click an empty space → New → Folder. Name it beloved-ai-practice.

AI & publishing 4/6

AI setup 1: open a folder and choose a helper

2 In VS Code choose File → Open Folder. Select beloved-ai-practice and click Select Folder. Trust it only because you created it.

3 Press Ctrl + Shift + X. This opens Extensions. Search for Codex, Kilo Code or Cline. Use the official installation links below to check the publisher before clicking Install.

AI & publishing 5/6

AI setup 1: open a folder and choose a helper

4 Click the helper’s icon in the side bar. Follow its sign-in or model-provider setup. Return to VS Code when sign-in finishes.

5 Choose one helper for this task. Start a new conversation and ask: “Which folder can you see? Do not change files yet.”

AI & publishing 6/6

AI setup 1: open a folder and choose a helper

CHECK

The answer should name beloved-ai-practice. If it names another project, stop and open the correct folder.

Codex: official extension setup

Kilo: official setup and documentation

Cline: official installation guide

AI & publishing 1/5

AI setup 2: know what each tool is

ChatGPT

A place to discuss an idea and prepare instructions. Planning here is optional.

Codex extension

Works inside VS Code beside your project files.

Codex desktop app

A separate application, rather than a VS Code extension. Open a local project there and review its changes. Official desktop naming and downloads can change.

AI & publishing 2/5

AI setup 2: know what each tool is

Kilo Code

An AI coding helper with model choices. Select a free model if that is what you intend to use.

Cline

Another coding agent. The correct spelling is Cline. Its extension and model-provider billing are separate things.

Claude Code

Anthropic’s coding agent. It can work through its VS Code extension; eligible account access is required.

AI & publishing 3/5

AI setup 2: know what each tool is

Qoder

A coding workspace for building and reviewing projects. Paul often gives it instructions directly.

Gemini Code Assist appears in Paul’s earlier workflow. Google ended its consumer IDE service on 18 June 2026 and directs those users to Antigravity. Do not follow an old free-account sign-in tutorial if it no longer works.

AI & publishing 4/5

AI setup 2: know what each tool is

Free does not mean unlimited. Codex access depends on your account. Kilo’s free models and background-model settings affect cost. Qoder uses plan allowances and credits. Check Usage before starting a long project.

Keep one agent in charge of the same files at a time. To change helper, save the files and ask the first helper to write a short progress note.

Official desktop app guide

AI & publishing 5/5

AI setup 2: know what each tool is

- Claude Code in VS Code
- Google’s consumer migration notice
- Kilo free-model settings
- Qoder plans and credits

AI & publishing

AI practical 1: let ChatGPT prepare your brief

A brief is a short description of what you want to build. Use this route when your idea is clear in your head but hard to explain.

1 Open ChatGPT and start a new conversation. Paste the message below. Replace the business details with your own idea.

AI & publishing

AI practical 1: let ChatGPT prepare your brief

CODE EXAMPLE

Copy online ↗

```
Help me plan a website for Beloved Training Centre.
Users should see courses and send an enquiry.
Use Laravel, Blade, CSS and SQLite for local practice.
I am a beginner using Windows and VS Code.
Ask up to five questions if important details are missin
g.
Then write a prompt for my coding agent.
Include pages, form fields, validation, mobile layout,
local setup, sample data and a clear completion checklis
t.
Build in small stages. Do not deploy or take payments.
```

What you are asking for

Business and users

Tell AI who the site helps.

Laravel and SQLite

Name the tools so AI does not choose a different stack.

AI & publishing 3/4

AI practical 1: let ChatGPT prepare your brief

Questions first

Give missing details before files are created.

Completion checklist

Define what must work so you can check the result.

2 Answer the questions. Read the resulting brief. Remove features you do not need.

AI & publishing

AI practical 1: let ChatGPT prepare your brief

3 Copy the final prompt. Paste it into Codex, Kilo or Qoder with your practice folder open.

RESULT

You should have a written plan with page names and a short build order. This is a target result, not a screenshot of an executed AI project.

TRY

Ask for only Home, Courses and Enquiry in the first version. A small working site is easier to understand.

AI & publishing 1/5

AI practical 2: build directly with your agent

You may skip ChatGPT and tell the coding agent your idea directly. This is also Paul’s usual approach in Qoder.

1. Open the project folder in VS Code or Qoder. Start a new agent conversation. Paste your approved brief.

AI & publishing

AI practical 2: build directly with your agent

CODE EXAMPLE

Copy online ↗

```
Read this project folder first. Tell me what is already
here.
Check the installed PHP, Composer and Node versions.
Explain any missing requirement before installing it.
Propose three small milestones.
First create the home page and explain how to run it.
Do not overwrite unrelated files or deploy the project.
After each milestone, list changed files and tests.
Keep secrets out of the code and use .env.example.
```

2 Read the plan. Let the agent create the first milestone. Review requested terminal commands before allowing them.

AI & publishing 3/5

AI practical 2: build directly with your agent

3 Open the local address it gives you. Try the page. Ask for one correction at a time.

4 Continue with the courses page, then the enquiry form and database. Do not add payments to this beginner project.

Read the instruction

Read the folder

Avoids building in the wrong place or replacing existing work.

AI & publishing 4/5

AI practical 2: build directly with your agent

Check requirements

AI still needs the correct language tools installed.

Milestones

Makes each piece small enough to inspect.

.env.example

Stores setting names and harmless examples, not real passwords.

AI & publishing

AI practical 2: build directly with your agent

RESULT

A working local website plus a README explaining how to start it. Ask AI to show which checks actually ran. A message saying “done” is not proof.

CHECK

Open the site on a narrow browser window. Send an empty form: it should show useful errors. Send a fictional valid enquiry: confirm it was saved.

AI & publishing

AI practical 3: correct a result and keep context

You do not need technical language to report a problem. Say what you clicked, what you expected and what happened.

CODE EXAMPLE

Copy online ↗

```
On the Enquiry page, I clicked Send with an empty name.
I expected a message asking for my name.
Instead, it saved an empty enquiry.
Find the cause and fix the server-side validation.
Keep the existing layout. Test empty and valid submissio
ns.
Explain the change in plain English.
```

“Server-side” means PHP checks the form too. A browser check alone can be bypassed.

AI & publishing 2/3

AI practical 3: correct a result and keep context

If you want ChatGPT’s help, copy the agent’s summary and the exact error. Remove passwords, keys and customer details first.

CODE EXAMPLE Copy online ↗

```
Here is my project goal and the agent’s latest result.
Explain the error simply. Give me one next instruction
to paste into the coding agent.
Do not guess which files exist; ask for missing evidence
.
```

AI & publishing 3/3

AI practical 3: correct a result and keep context

When returning tomorrow, ask the agent to read README.md and a short PROGRESS.md before changing anything. PROGRESS.md should list completed work, remaining work and the command used to start the project.

RESULT

The reported problem is fixed, the earlier working features still work, and the progress note matches the files.

AI & publishing 1/6

AI practical 4: run the correct local server

Local means running on your own computer. A local address does not publish your website to the internet.

In VS Code choose Terminal → New Terminal. Check that the terminal is in your project folder. Commands differ by project; follow its README.

HTML, CSS and JavaScript

For a simple page, open index.html in the browser. Use the earlier Live Server lesson when the project needs a local web server.

AI & publishing 2/6

AI practical 4: run the correct local server

Plain PHP

From the folder containing index.php, run the PHP command below. PHP must already be installed.

Laravel

PHP and Composer must be installed. Finish the Laravel setup lesson first, including .env, the application key and the practice database.

AI & publishing 3/6

AI practical 4: run the correct local server

Django

Activate the project’s virtual environment and install its requirements before using manage.py.

Flutter web

Install Flutter and a supported browser first. flutter doctor tells you what is missing.

CODE EXAMPLE Copy online ↗

```
php -S localhost:8000
```

AI & publishing

AI practical 4: run the correct local server

Plain PHP: open http://localhost:8000. The -S option starts PHP’s development server. Keep its terminal open. Ctrl + C stops it.

CODE EXAMPLE

Copy online ↗

```
php artisan serve
```

Laravel: open the address printed by Artisan. If the page uses Vite, open a second terminal and run the project’s npm run dev command too.

AI & publishing

AI practical 4: run the correct local server

CODE EXAMPLE Copy online ↗

```
python manage.py runserver
```

Django: run this beside manage.py, in the activated environment. Open the address it prints.

CODE EXAMPLE Copy online ↗

```
flutter run -d chrome
```

Flutter: run this inside the Flutter project. It launches the web version in Chrome. Flutter uses Dart, not PHP.

AI & publishing 6/6

AI practical 4: run the correct local server

CHECK

If a command is not recognised, the tool is missing or not on PATH. Return to that stack’s installation lesson. Do not paste random fix commands from a chat.

AI & publishing 1/6

GitHub practical 1: save your first project online

Git records versions on your computer. GitHub stores a copy online. A repository is one project’s version history. A commit is a named checkpoint. Push sends commits to GitHub; pull brings them down.

1 Create your account at github.com. Then download GitHub Desktop from desktop.github.com. Install it and sign in through its browser sign-in screen.

AI & publishing 2/6

GitHub practical 1: save your first project online

2 In GitHub Desktop choose File → New repository. Name it beloved-ai-practice. Choose a location you can find. Select the Laravel Git ignore template for a Laravel project. Click Create repository.

3 Choose Repository → Show in Explorer. Copy your finished practice project into this new folder. Copy its files, not a second enclosing folder, and do not copy any old .git folder.

AI & publishing 3/6

GitHub practical 1: save your first project online

4 Open that new repository folder in VS Code. Keep the framework’s original .gitignore rules. Add the examples below if they are missing.

CODE EXAMPLE Copy online ↗

```
.env
.env.*
!.env.example
node_modules/
vendor/
*.sqlite
*.sqlite3
*.pem
*.key
```

AI & publishing 4/6

GitHub practical 1: save your first project online

Each line names files Git should leave out. The ! line allows the safe example settings file. Keep real database files, credentials and private keys out. A .gitignore rule does not remove secrets already committed.

5 Return to GitHub Desktop. In Changes, click each file to review it. Confirm that no .env, real database, password or key is listed. Enter “First working training website” in Summary. Click Commit to main.

AI & publishing 5/6

GitHub practical 1: save your first project online

6 Click Publish repository. Keep “Keep this code private” selected for your first practice project. Click Publish repository again.

7 Choose Repository → View on GitHub. Check that your code appears and private files do not. GitHub is now holding your checkpoint; this alone does not make a PHP website live.

Download GitHub Desktop

AI & publishing 6/6

GitHub practical 1: save your first project online

GitHub Desktop documentation

AI & publishing 1/5

GitHub practical 2: let AI prepare an update

Work in the repository folder you published. First make one small change, such as correcting the course contact text.

CODE EXAMPLE Copy online ↗

```
Change the contact heading to Ask about a course.
Keep the rest of the page unchanged.
Run the relevant checks and show the diff.
Check that no secrets or local database files are staged
.
Prepare a clear commit message. Do not push yet.
```

1 Run the site locally. Check the new heading and try the enquiry form again.

AI & publishing 2/5

GitHub practical 2: let AI prepare an update

2 In GitHub Desktop, review Changes. Commit the tested files, then click Push origin. On GitHub, refresh the repository and find your new commit.

An agent can also commit and push when you explicitly ask it to, it has Git installed, and your account is signed in. Do not give it your account password in a prompt.

AI & publishing

GitHub practical 2: let AI prepare an update

CODE EXAMPLE

Copy online ↗

```
The changes are reviewed and tested.
Confirm the repository URL and current branch.
Commit only this task’s files with a clear message.
Push to origin on this branch. Never force-push.
Report the commit ID and whether the push succeeded.
```

A branch is a separate line of work. For a shared or live project, ask AI to use a feature branch and open a pull request. Review that proposal before merging it into main.

AI & publishing 4/5

GitHub practical 2: let AI prepare an update

Useful checks in the terminal

CODE EXAMPLE Copy online ↗

```
git status
git remote -v
git log -1 --oneline
```

git status

Shows changed files and the current branch.

git remote -v

Shows which online repository receives a push. Do not share a URL containing a token.

AI & publishing 5/5

GitHub practical 2: let AI prepare an update

git log -1 --oneline

Shows the latest checkpoint ID and its message.

CHECK

The newest commit on GitHub must match the one you intended to publish. Stop if the repository or branch is wrong.

AI & publishing 1/5

Server practical 1: prepare a safe deployment

A server is a computer that stays available to visitors. Deploy means putting a tested version there and preparing it to run.

Use a training server or staging site first. These instructions assume a Linux host with SSH access and an existing working Laravel installation. First-time server setup is a separate job; do not assume that cloning code creates a working server.

AI & publishing 2/5

Server practical 1: prepare a safe deployment

Before continuing, confirm: compatible PHP and Composer; the needed PHP extensions; a database; a web server; HTTPS; a domain; and a tested database/files backup. The web root must be Laravel’s public folder, not the whole project.

Your host must provide the SSH user, address, port and project directory. Use its documented values. HOST, USER, OWNER and REPO below are placeholders, not literal account names.

AI & publishing 3/5

Server practical 1: prepare a safe deployment

CODE EXAMPLE Copy online ↗

```
ssh USER@HOST
```

ssh opens the server terminal. Check the host fingerprint against your hosting panel on the first connection. Enter your own credentials through the secure prompt, not in AI chat.

For a private repository, use a read-only deploy key tied to that repository. Create a dedicated key on the server; keep the private file there.

AI & publishing 4/5

Server practical 1: prepare a safe deployment

CODE EXAMPLE Copy online ↗

```
ssh-keygen -t ed25519 -f ~/.ssh/course_deploy
cat ~/.ssh/course_deploy.pub
```

ssh-keygen creates the key pair. Choose a new filename; never overwrite an existing key. cat displays only the .pub public key, which is the part you copy to GitHub.

In GitHub open the repository → Settings → Deploy keys → Add deploy key. Give it a name, paste the public key and leave Allow write access unticked. Save it.

AI & publishing 5/5

Server practical 1: prepare a safe deployment

Tell your host to configure SSH to use this dedicated key. A public key on GitHub is not enough if SSH is selecting a different private key.

GitHub: deploy-key setup

Laravel: production deployment requirements

AI & publishing 1/8

Server practical 2: first copy and production settings

This is a checklist for a configured training host. Have your hosting provider prepare the web-server and database settings first. Use the project’s documented Laravel version requirements.

1. In the server terminal, enter the directory assigned by the host. Clone into a new, empty project location using your configured deploy key.

AI & publishing 2/8

Server practical 2: first copy and production settings

CODE EXAMPLE Copy online ↗

```
git clone git@github.com:OWNER/REPO.git training-site
cd training-site
composer install --no-dev --optimize-autoloader
```

git clone downloads the repository. cd enters it. Composer installs the locked PHP dependencies without development-only packages. Stop if any command fails.

AI & publishing 3/8

Server practical 2: first copy and production settings

2 For this first installation only, copy .env.example to .env. Enter production settings using the host’s secure editor. Set APP_ENV=production, APP_DEBUG=false and APP_URL to the HTTPS address. Use the server database credentials.

CODE EXAMPLE Copy online ↗

```
cp .env.example .env
php artisan key:generate
```

Server practical 2: first copy and production settings

cp creates the private settings file. key:generate creates the application encryption key. Do this only for a new installation; do not replace the key during ordinary updates.

3 Confirm the correct database. Review the migrations and take a backup before applying them. If this is your empty training database, run the reviewed migrations.

AI & publishing 5/8

Server practical 2: first copy and production settings

CODE EXAMPLE Copy online ↗

```
php artisan migrate --force
```

This changes the database schema. --force permits production execution; it does not make the change safe. Never use migrate:fresh on a live database: it deletes tables.

AI & publishing 6/8

Server practical 2: first copy and production settings

4 If package-lock.json and the project’s build script exist, install the locked front-end dependencies and build the assets. Otherwise follow the project README.

CODE EXAMPLE Copy online ↗

```
npm ci
npm run build
```

Server practical 2: first copy and production settings

npm ci installs from the lockfile. npm run build creates production CSS/JavaScript assets. Node must be available on this build machine. A production server may instead receive assets built in CI.

5 Configure storage permissions through the host. If the project uses public file uploads, follow its storage-link instructions. Point the site to public/, enable HTTPS, then open the site and test it. Never use php artisan serve as the public production server.

AI & publishing 8/8

Server practical 2: first copy and production settings

Laravel database migrations

Laravel environment settings

AI & publishing

Server practical 3: pull a tested update

Now practise Paul’s update routine: local change → test → push → server pull → deploy → live check.

1. Push your tested commit to GitHub. Record its ID. Back up the server database and uploaded files. Confirm that you can restore the backup.

2. Connect by SSH. Enter the existing project directory. Run these checks one at a time.

AI & publishing

Server practical 3: pull a tested update

CODE EXAMPLE

Copy online ↗

```
cd /path/to/training-site
git status
git branch --show-current
git rev-parse HEAD
```

Replace the path with the real directory. status must show a clean working tree. Confirm the deployment branch, normally main, and record the old commit ID for recovery. If there are local changes, stop and investigate; do not erase them.

AI & publishing

Server practical 3: pull a tested update

3 For this small training site, enter maintenance mode. Then pull only a straightforward update. If any command fails, stop and resolve it before continuing.

CODE EXAMPLE

Copy online ↗

```
php artisan down
git pull --ff-only origin main
composer install --no-dev --optimize-autoloader
```

AI & publishing 4/7

Server practical 3: pull a tested update

down temporarily shows maintenance mode. pull downloads the main branch and refuses to invent a merge if histories differ. Composer installs the dependencies for the new code.

4 Build assets if this project requires them. Review new migrations. Apply them only after checking the backup and the intended database. Use the build and migration commands from the previous lesson.

AI & publishing

Server practical 3: pull a tested update

CODE EXAMPLE

Copy online ↗

```
php artisan optimize
php artisan up
```

optimize prepares Laravel’s caches. up ends maintenance mode. If the update uses long-running workers, restart them using the command documented for your Laravel version and host.

AI & publishing

Server practical 3: pull a tested update

5 Open the real HTTPS address in a fresh browser tab. Check the homepage, changed feature, form validation and one fictional test submission. Confirm the deployed commit ID with git rev-parse HEAD.

If deployment fails, visitors may still see maintenance mode. Restore a known-working release and compatible settings, then bring the site up after checks. Do not blindly reverse database migrations. A code rollback alone may not undo a schema change.

AI & publishing 7/7

Server practical 3: pull a tested update

RESULT

The live test shows your intended change, the commit matches GitHub, and existing features still work. These are guided deployment checks; no deployment to your server was performed for this handout.

AI & publishing 1/3

Choose the right home for your website

Plain HTML, CSS and browser JavaScript

Can use static hosting such as GitHub Pages.

PHP or Laravel

Needs PHP-capable hosting and the database/settings your project uses. GitHub Pages cannot run PHP.

Django

Needs Python application hosting, a production application server and the project database. The development runserver command is not production hosting.

AI & publishing 2/3

Choose the right home for your website

Flutter web

Build with flutter build web. Publish the generated build/web files to web hosting. A separate API still needs its own server.

A professional improvement is automated deployment after a reviewed change. First understand the manual routine. Later, let CI run tests, build assets and deploy an approved release with rollback support.

AI & publishing 3/3

Choose the right home for your website

Your finish line: you can explain where the files live, how to start the local site, which commit is online, and how to recover a failed update. That is control of your project—not just prompting.

Tool references in these added lessons were checked on 14 October 2026. Screens and account allowances change; follow the linked official guide when your screen differs.

AI & publishing 1/4

Your course completion record

Write the date beside each result you can produce without copying the answer.

Create folders and save a real HTML file

_____

Explain browser, server, DNS and hosting

_____

Finish all five HTML practicals

_____

AI & publishing 2/4

Your course completion record

Finish all five CSS practicals

_____

Finish all five JavaScript practicals

_____

Run all five PHP practicals

_____

Run all five SQL practicals

_____

AI & publishing 3/4

Your course completion record

Save and inspect a record in each complete website

_____

Review an AI change and test the result

_____

Publish a static site and check its public link

_____

AI & publishing 4/4

Your course completion record

CHECK

Choose the project you understand best. Change its content for a new purpose and explain your choices. That is stronger evidence of learning than a folder full of untouched answer files.

You can now use the contents as a reference. Return to a practical when you need to remember how something works.

AI & publishing 1/5

Optional tutor notes Teach the person in front of you

The sequence is deliberately slow at the beginning. A student who cannot find a saved file is not ready for a framework installer.

A simple classroom rhythm

Demonstrate one small action. Let everyone repeat it. Walk around and look at their screens. Ask them to change one detail by themselves. End with a visible result.

AI & publishing 2/5

Optional tutor notes Teach the person in front of you

Use short blocks: about 5 minutes to demonstrate, 15 minutes to practise and 5 minutes to check. These are suggested blocks, not deadlines. Give extra time for typing and mouse practice.

Computer basics

Open Notepad, select text and find a saved file

First webpage

Save index.html correctly and refresh the browser

AI & publishing 3/5

Optional tutor notes Teach the person in front of you

VS Code

Open the correct folder and edit two linked files

Design practice

Improve spacing and demonstrate a narrow layout

AI

Request one change and check the actual result

Platforms

Build a page and explain draft versus published

AI & publishing 4/5

Optional tutor notes Teach the person in front of you

Stack sampler

Run one local project and make a visible change

Final build

Demonstrate a tested site and explain its limits

Pair students for discussion, but let each person use the mouse and keyboard. Avoid having the confident partner do all the clicking.

AI & publishing 5/5

Optional tutor notes Teach the person in front of you

Give this feedback: “You saved the file correctly. Now let us check which folder it is in.” A broken page is a useful problem to investigate, not evidence that someone cannot learn.

Advanced does not mean rushing through more tool names. Revisit one chosen stack until students can understand the generated structure, check data handling and maintain a small working application.

AI & publishing 1/6

Optional tutor notes Prepare the room and accounts

Before the first class

Check each computer's keyboard, mouse, display, free space and Windows compatibility. Provide a teacher-approved cake.jpg image and a place to save work. Test Notepad's Save As behaviour on those exact computers.

Do not require an AI account for the first success. The opening computer and Notepad exercises can work offline. Download installers in advance where classroom internet is unreliable.

AI & publishing 2/6

Optional tutor notes Prepare the room and accounts

Before later stages

VS Code

Current supported release and Live Server extension

Codex

Official OpenAI extension, working sign-in and usage access

Platforms

Suitable accounts; WordPress Studio local installation

AI & publishing 3/6

Optional tutor notes Prepare the room and accounts

Laravel

Supported PHP, Composer, Herd and a clean Laravel 13 build

Django

Python 3.12, the py launcher and a Django 5.2 environment

Flutter

Git, SDK, Dart Code extension, Chrome and a working web run

React

Compatible Node.js LTS and the current Vite setup

AI & publishing 4/6

Optional tutor notes Prepare the room and accounts

Qoder

Editor login, selected plan and available Agent allowance

Publishing

GitHub access and a fictional static practice project

AI & publishing 5/6

Optional tutor notes Prepare the room and accounts

The framework walkthroughs were checked against official documentation; they still need a trial run on your classroom computers. Run every practical once on the classroom setup before teaching it. Downloads and account screens change. Record the versions that worked in a classroom setup sheet. Do not update the whole class during a lesson unless needed.

AI & publishing 6/6

Optional tutor notes Prepare the room and accounts

If an account is unavailable, demonstrate on the teacher's screen and let students continue their local project. Never share one password around the room. Check age and institutional account requirements before arranging student sign-ups.

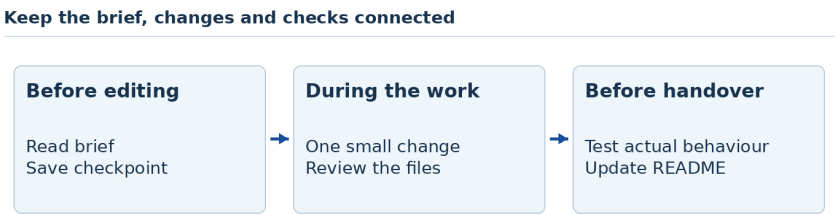
A tool asking for a subscription is a teaching decision for the teacher, not a problem a beginner should solve alone.

AI & publishing

Optional tutor notes Use AI like a professional colleague

Use clear requirements, small changes, saved versions and observed results. This routine works across AI tools.

VISUAL EXPLANATION



AI & publishing 2/5

Optional tutor notes Use AI like a professional colleague

Keep three short files

brief.txt says who the site serves and what it should do. README.txt says how to run it. A test checklist says how you know it works. Keep them with the project so another person can continue the work.

Add tools only when they solve a problem

Figma or Penpot

Agree on a layout before building it

AI & publishing 3/5

Optional tutor notes Use AI like a professional colleague

GitHub Desktop

Save and compare working versions

Browser developer tools

Investigate layout, errors and loading

Prettier for VS Code

Keep formatting consistent after students understand files

Playwright

Later, automate a repeated browser check with teacher support

AI & publishing 4/5

Optional tutor notes Use AI like a professional colleague

Documentation connections through MCP

Later, let an AI consult approved technical documentation

MCP is a way for an AI tool to connect to additional tools or information. It is optional here. An MCP connection is different from a WordPress plugin or a VS Code extension. Review its permissions before use.

AI & publishing 5/5

Optional tutor notes Use AI like a professional colleague

Use Codex for explanation and code work, and Qoder for the final build as planned. Avoid running two editing agents on the same files at the same time. Give the next tool the project, brief and tests instead of expecting it to remember the previous chat.

For advanced work, ask for evidence from tests. Try the feature yourself.

Sources: S39–S41.

AI & publishing 1/5

Optional tutor notes Important current tool details

Checked against official documentation on 8 October 2026. Recheck before a new class, especially prices, account access and installation screens.

Qoder

The current documentation describes a Free plan and paid plans. Free usage is limited; an eligible trial may temporarily add paid features and credits. Do not promise students free unlimited Agent work. A personal API key can involve separate provider charges.

AI & publishing 2/5

Optional tutor notes Important current tool details

Qoder's official notice says the Quest window inside Qoder IDE is due to retire on 15 October 2026 at 23:59 UTC+8. The Editor remains available. This handout therefore teaches Editor chat and Agent mode, not a Quest-dependent course. Recheck the current notice if teaching after that date.

AI & publishing 3/5

Optional tutor notes Important current tool details

Codex

Use the official OpenAI extension. Available usage depends on the account and plan. Check sign-in and permissions before the lesson. Do not require students to purchase API credits just to follow the manual HTML exercises.

Optional tutor notes Important current tool details

WordPress

WordPress.com is a hosted service. WordPress.org distributes the WordPress software. The software can be free while hosting and a domain still cost money. WordPress.com's current documentation allows plugin installation on paid plans; the free classroom visit does not depend on that feature.

AI & publishing 5/5

Optional tutor notes Important current tool details

Dreamweaver and Sites

Dreamweaver is an optional demonstration using an appropriately licensed installation. Google Sites and ChatGPT Sites are different products. ChatGPT Sites availability and sharing controls depend on the account and workspace.

Sources: S6, S8, S11, S19, S21, S31, S32, S42. These optional notes explain tool details without interrupting the first lessons.

AI & publishing 1/5

Optional tutor notes Publish the right kind of project

A domain is a name people type, such as example.com. Hosting is the service that makes a website available online. A local development server is not production hosting.

HTML, CSS and JavaScript

Static files; GitHub Pages suits the class exercise

React built with Vite

Run npm run build; deploy the dist output with the correct base path

AI & publishing 2/5

Optional tutor notes Publish the right kind of project

Flutter web

Run flutter build web; deploy build/web through a web host

Laravel

Supported PHP, web server, application settings and database

Django

Supported Python, production application server, settings and database

WordPress.org software

Supported WordPress hosting, database and managed updates

AI & publishing 3/5

Optional tutor notes Publish the right kind of project

A supervised production workflow

Choose a host that explicitly supports your stack. Read its current deployment guide. First deploy a test copy, often called staging. Set secret configuration through the host's settings, not public source files. Use a supported production server rather than artisan serve or Django runserver.

AI & publishing 4/5

Optional tutor notes Publish the right kind of project

Run the database setup safely, configure HTTPS, turn off debug output and test the real hosted address. Check file uploads and database storage survive restarts. Make a backup and practise restoring it before handling real customer information.

AI & publishing 5/5

Optional tutor notes Publish the right kind of project

For a public form, check validation, spam controls, restricted access and retention. For sign-in or payment, use well-maintained framework features and established providers with experienced review. A generated demo is not a security assessment.

Advanced assignment: ask AI to draft a deployment checklist for the chosen host and exact project. Verify every step against that host's official guide before acting.

Sources: S25, S27, S33, S43, S44. Hosting choices, free tiers and billing change; confirm them at the time of use.

AI & publishing 1/5

Quick help When something goes wrong

Start with what you can see. Change one thing, then test again.

Code appears instead of a page

Is it index.html rather than index.html.txt?

The old words remain

Did you save, refresh and open the same file?

CSS has no effect

Does the link say styles.css and match the file?

AI & publishing 2/5

Quick help When something goes wrong

A picture is missing

Check its folder, filename, extension and letter case

Live Server does not start

Open the folder and right-click index.html

A local address will not load

Is its server still running? Use the printed address

A terminal command fails

Read the first error; check tool installation and folder

AI & publishing 3/5

Quick help When something goes wrong

An AI change breaks something

Return to the saved version and request a smaller fix

A form says success

Check the record or delivery; do not trust text alone

A published page looks old

Wait for deployment, refresh and check the published files

Ask for help clearly

Write this when asking a person or AI for help:

AI & publishing 4/5

Quick help When something goes wrong

“I am using [tool]. I opened [file]. I did [action]. I expected [result]. Instead I saw [message or behaviour].”

For example: “In VS Code I changed styles.css and saved. I expected a blue heading. It stayed black. My HTML link says style.css.”

AI & publishing 5/5

Quick help When something goes wrong

Include a screenshot or the exact error when useful. Remove passwords, private names and account details first. Do not ask AI to rebuild the whole project for a small spelling mistake.

AI & publishing 1/5

Small glossary Words you have already used

File

One saved item, such as index.html

Folder

A container that keeps files together

Browser

The program that opens webpages, such as Edge

Editor

The program where you change code, such as VS Code

AI & publishing 2/5

Small glossary Words you have already used

HTML

The page's content and structure

CSS

Rules for its appearance and layout

JavaScript

Code that can respond to actions on the page

Extension

An extra feature installed in an editor

AI & publishing 3/5

Small glossary Words you have already used

Plugin

An add-on for a product such as WordPress

Responsive

A layout that adapts to available screen space

Terminal

A panel where you type commands

Local

Running or stored on your own computer

AI & publishing 4/5

Small glossary Words you have already used

Framework

A prepared structure and tools for building applications

Database

Organised saved information

Repository

A project tracked with version history

Commit

A recorded set of changes with a description

AI & publishing 5/5

Small glossary Words you have already used

Deploy

Put a working version on its hosting service

Debug

Find and fix the cause of a problem

You do not need to memorise this table. Return to a word when you meet it during a task. Understanding grows through use.

AI & publishing 1/5

Official sources 1

These links support the tool instructions. Checked 8 October 2026. Open a reference when you need more detail or a menu has changed.

S1 Windows file extensions

https://support.microsoft.com/en-us/windows/experience/storage-filemanagement/common-file-name-extensions-in-windows

S2 VS Code Windows setup

AI & publishing 2/5

Official sources 1

https://code.visualstudio.com/docs/setup/windows

S3 VS Code requirements

https://code.visualstudio.com/docs/supporting/requirements

S4 VS Code interface

https://code.visualstudio.com/docs/editing/getting-started/userinterface

S5 Live Server by Ritwick Dey

AI & publishing 3/5

Official sources 1

https://marketplace.visualstudio.com/items?itemName=ritwickdey.LiveServer

S6 Codex IDE extension

https://learn.chatgpt.com/docs/codex/ide

S7 GitHub Desktop first repository

https://docs.github.com/en/desktop/overview/creating-your-first-repository-using-github-desktop

AI & publishing 4/5

Official sources 1

S8 WordPress com and org compared

https://wordpress.com/support/com-vs-org/

S9 WordPress com getting started

https://wordpress.com/support/getting-started-with-wordpress-com/

S10 WordPress com pages

https://wordpress.com/support/pages/

AI & publishing 5/5

Official sources 1

S11 WordPress com plugins and plans

https://wordpress.com/support/plugins/

Interface reference: Visual Studio Code user interface

Interface reference: Qoder overview

AI & publishing 1/5

Official sources 2

S12 WordPress Studio setup

https://developer.wordpress.com/docs/developer-tools/studio/

S13 WordPress Studio sites

https://developer.wordpress.com/docs/developer-tools/studio/sites/

S14 WordPress Playground quick start

AI & publishing 2/5

Official sources 2

https://developer.wordpress.org/playground/handbook/quick-start-guide/

S15 Hello Dolly plugin publisher and behaviour

https://wordpress.org/plugins/hello-dolly/

S16 WordPress Studio import and export

AI & publishing 3/5

Official sources 2

https://developer.wordpress.com/docs/developer-tools/studio/import-export/

S17 Blogger create a blog

https://support.google.com/blogger/answer/1623800

S18 Google Sites create and publish

https://support.google.com/sites/answer/6372878

AI & publishing 4/5

Official sources 2

S19 Dreamweaver new documents

https://helpx.adobe.com/dreamweaver/desktop/manage-files/creating-opening-documents.html

S20 Penpot design platform

https://penpot.app/

S21 ChatGPT Sites and sharing

https://learn.chatgpt.com/docs/sites

AI & publishing 5/5

Official sources 2

S22 Laravel 13 installation

https://laravel.com/framework/docs/13.x/installation

AI & publishing 1/4

Official sources 3

S23 Django 5.2 first application

https://docs.djangoproject.com/en/5.2/intro/tutorial01/

S24 Flutter installation in VS Code

https://docs.flutter.dev/install/with-vs-code

S25 Flutter web build and deployment

https://docs.flutter.dev/platform-integration/web/building

AI & publishing 2/4

Official sources 3

S26 React learning guide

https://react.dev/learn

S27 Vite getting started

https://vite.dev/guide/

S28 Laravel Herd Windows installation

https://herd.laravel.com/docs/windows/getting-started/installation

AI & publishing 3/4

Official sources 3

S29 Laravel validation

https://laravel.com/docs/13.x/validation

S30 Django on Windows

https://docs.djangoproject.com/en/5.2/howto/windows/

S31 Qoder quick start

https://docs.qoder.com/quick-start

AI & publishing 4/4

Official sources 3

S32 Qoder plans and usage

https://docs.qoder.com/account/pricing

S33 GitHub Pages create a site

https://docs.github.com/en/pages/getting-started-with-github-pages/creating-a-github-pages-site

AI & publishing 1/5

Official sources 4

S34 MDN CSS box model

https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/Styling_basics/Box_model

S35 MDN CSS Grid

https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/CSS_layout/Grids

S36 W3C accessible page structure

AI & publishing 2/5

Official sources 4

https://www.w3.org/WAI/tutorials/page-structure/

S37 Google SEO starter guide

https://developers.google.com/search/docs/fundamentals/seo-starter-guide

S38 Chrome Lighthouse

https://developer.chrome.com/docs/lighthouse/overview/

S39 Prettier VS Code extension

AI & publishing 3/5

Official sources 4

https://marketplace.visualstudio.com/items?itemName=esbenp.prettier-vscode

S40 Playwright browser testing

https://playwright.dev/docs/intro

S41 OpenAI documentation MCP

https://developers.openai.com/learn/docs-mcp

S42 Qoder Quest retirement notice

AI & publishing 4/5

Official sources 4

https://docs.qoder.com/release-notes/quest-retirement-notice

S43 Laravel deployment

https://laravel.com/framework/docs/13.x/deployment

S44 Django deployment checklist

https://docs.djangoproject.com/en/5.2/howto/deployment/checklist/

AI & publishing 5/5

Official sources 4

Additional reference for choosing Flutter: https://docs.flutter.dev/platform-integration/web/faq

Django admin extension reference

https://docs.djangoproject.com/en/5.2/intro/tutorial02/

AI & publishing 1/4

Optional tutor notes Assess the new foundation practicals

Use a demonstration, not only a written answer. For each of the 25 separate practicals, check four things: the student created the files, obtained the expected output, explained the relevant code and completed the change task.

For each of the five websites, check navigation, the catalogue filter, server validation, a saved record, readable mobile layout and clear limits on what the site does.

AI & publishing 2/4

Optional tutor notes Assess the new foundation practicals

A student who has copied a page but cannot find the file needs more file practice. A student whose form says success but has no stored row needs help tracing the request. Use the error as the next lesson.

The separate SQL exercises run in DB Browser, not inside a webpage. Do not skip them because a generated application already has a database.

AI & publishing 3/4

Optional tutor notes Assess the new foundation practicals

The companion files are answer copies. Keep clean copies so learners can compare changes. A sensible class pace is one practical over as many sessions as needed; do not force all five into one lesson.

AI & publishing

Optional tutor notes Assess the new foundation practicals

Before public use of a complete project, review production hosting, session settings, secrets, backups, rate limits, privacy and accessibility. The guesthouse and event examples accept requests; they do not allocate stock, dates or seats. Add those features only with explicit requirements and appropriate tests.

AI & publishing 1/4

Foundation sources 1

Official references for the added lessons. Checked during this revision on 12–13 October 2026.

N1 Internet infrastructure

https://developer.mozilla.org/en-US/docs/Learn_web_development/Howto/Web_mechanics/How_does_the_Internet_work

N2 How the web works

AI & publishing 2/4

Foundation sources 1

https://developer.mozilla.org/en-US/docs/Learn_web_development/Getting_started/Web_standards/How_the_web_works

N3 URL components

https://developer.mozilla.org/en-US/docs/Learn_web_development/Howto/Web_mechanics/What_is_a_URL

N4 HTTP overview

AI & publishing 3/4

Foundation sources 1

https://developer.mozilla.org/en-US/docs/Web/HTTP/Overview

N5 Browser network panel

https://developer.chrome.com/docs/devtools/network/

N6 HTML elements

https://developer.mozilla.org/en-US/docs/Web/HTML/Reference/Elements

N7 JavaScript guide

AI & publishing 4/4

Foundation sources 1

https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide

N8 PHP tutorial

https://www.php.net/manual/en/tutorial.php

AI & publishing 1/3

Foundation sources 2

N9 DB Browser installation

https://sqlitebrowser.org/dl/

N10 SQLite SQL reference

https://www.sqlite.org/lang.html

N11 CSS reference

https://developer.mozilla.org/en-US/docs/Web/CSS/Reference

N12 Prepared SQL with PDO

AI & publishing 2/3

Foundation sources 2

https://www.php.net/manual/en/pdo.prepare.php

N13 PHP sessions

https://www.php.net/manual/en/book.session.php

N14 PHP development server

https://www.php.net/manual/en/features.commandline.webserver.php

N15 PHP validation filters

AI & publishing 3/3

Foundation sources 2

https://www.php.net/manual/en/filter.constants.php

N16 PHP output escaping

https://www.php.net/manual/en/function.htmlspecialchars.php

YOUR COURSE ROADMAP

Choose a subject

00 **Welcome**
5 lessons

01 **Computer & internet**
12 lessons

02 **Notepad & VS Code**
9 lessons

03 **HTML**
9 lessons

04 **CSS**
8 lessons

YOUR COURSE ROADMAP

Choose a subject

05 **JavaScript**
8 lessons

06 **PHP**
10 lessons

07 **SQL**
10 lessons

08 **Complete websites**
31 lessons

09 **AI & publishing**
43 lessons

Welcome 1/2

Welcome

OPEN LESSON
Your phone is the guide

OPEN LESSON
Welcome to your first website

OPEN LESSON
Meet your course author

OPEN LESSON
Download your course files

Welcome 2/2

Welcome

OPEN LESSON

Find your way around the course

Computer & internet 1/3

Computer & internet

OPEN LESSON

Part 01 Computer and internet

OPEN LESSON

Meet the mouse and keyboard

OPEN LESSON

Select copy paste and undo

OPEN LESSON

Make a folder and save your work

Computer & internet 2/3

Computer & internet

OPEN LESSON

What the internet actually is

OPEN LESSON

Open a browser and visit a website

OPEN LESSON

Names addresses and places to store a site

OPEN LESSON

Follow a webpage from request to screen

Computer & internet 3/3

Computer & internet

OPEN LESSON

Watch real requests in the browser

OPEN LESSON

Data travels in pieces

OPEN LESSON

Online local upload and download

OPEN LESSON

Meet the five subjects we will practise

Notepad & VS Code

OPEN LESSON

Part 02 Your first webpage

OPEN LESSON

Make your first webpage in Notepad

OPEN LESSON

Open your page and change it

OPEN LESSON

Understand the few tags you used

Notepad & VS Code

OPEN LESSON

Make a safe copy before changing more

OPEN LESSON

Install Visual Studio Code

OPEN LESSON

Open your folder in VS Code

OPEN LESSON

Add Live Server to VS Code

Notepad & VS Code

OPEN LESSON

Get the practice files ready

HTML 1/3

HTML

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Part 03 HTML gives content a structure

OPEN LESSON
Save and reopen the right file

OPEN LESSON
Build the document shell

OPEN LESSON
Read tags and attributes

HTML

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A personal introduction page

OPEN LESSON

A restaurant menu page

OPEN LESSON

An event information page

OPEN LESSON

A workshop registration form

HTML

OPEN LESSON

A school club home page

CSS 1/2

CSS

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Part 04 CSS gives the page its appearance

OPEN LESSON

See which rule changes the page

OPEN LESSON

Select and style the page

OPEN LESSON

A clear business introduction

CSS 2/2

CSS

OPEN LESSON

A product card with breathing room

OPEN LESSON

A flexible navigation bar

OPEN LESSON

A responsive course card grid

OPEN LESSON

A readable enquiry form

JavaScript 1/2

JavaScript

OPEN LESSON

Part 05 JavaScript makes a page respond

OPEN LESSON

Find the first error

OPEN LESSON

Values decisions and actions

OPEN LESSON

A visitor greeting on a page

JavaScript 2/2

JavaScript

OPEN LESSON
A printing cost calculator

OPEN LESSON
A small service catalogue

OPEN LESSON
A frequently asked questions control

OPEN LESSON
A catalogue search page

PHP 1/3

PHP

OPEN LESSON

Part 06 PHP works on the server

OPEN LESSON

Meet the terminal slowly

OPEN LESSON

Install and check PHP

OPEN LESSON

Start a server and fix setup errors

PHP 2/3

PHP

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Where the code runs

OPEN LESSON

A server generated welcome page

OPEN LESSON

A server calculated quotation page

OPEN LESSON

A dynamic price list page

PHP 3/3

PHP

OPEN LESSON

A validated estimate form

OPEN LESSON

A session based visit counter page

SQL 1/3

SQL

OPEN LESSON

Part 07 SQL works with stored data

OPEN LESSON

Understand records and relationships

OPEN LESSON

Open DB Browser and run a query

OPEN LESSON

Run one statement and read the result

SQL

OPEN LESSON

Create a product table

OPEN LESSON

Find the products a visitor needs

OPEN LESSON

Correct and remove specific records

OPEN LESSON

Join customer and order information

SQL

OPEN LESSON

Build a report and undo a trial change

OPEN LESSON

Foundation checkpoint Before combining the languages

Complete websites 1/8

Complete websites

OPEN LESSON

Part 08 Put the five subjects together

OPEN LESSON

Five complete websites How the parts work together

OPEN LESSON

Website 1 Build a bakery website

OPEN LESSON

Website 1 · config php

Complete websites 2/8

Complete websites

OPEN LESSON

Website 1 · schema sql

OPEN LESSON

Website 1 · setup php

OPEN LESSON

Website 1 · app php

OPEN LESSON

Website 1 · header php

Complete websites

OPEN LESSON

Website 1 · footer php

OPEN LESSON

Website 1 · public index php

OPEN LESSON

Website 1 · public catalog php

OPEN LESSON

Website 1 · public request php

Complete websites 4/8

Complete websites

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Website 1 · public thanks php

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Website 1 · public style css

OPEN LESSON

Website 1 · public app js

OPEN LESSON

Website 1 Inspect the saved record

Complete websites

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Website 1 Explain the HTML CSS and JavaScript

OPEN LESSON

Website 1 Explain PHP SQL and the request flow

OPEN LESSON

Website 1 See the result & test it

OPEN LESSON

Website 2 A tailoring and branding enquiry website

Complete websites

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Website 2 Configuration code

OPEN LESSON

Website 2 See the result & test it

OPEN LESSON

Website 3 A training centre course website

OPEN LESSON

Website 3 Configuration code

Complete websites

OPEN LESSON

Website 3 See the result & test it

OPEN LESSON

Website 4 A guesthouse room enquiry website

OPEN LESSON

Website 4 Configuration code

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Website 4 See the result & test it

Complete websites 8/8

Complete websites

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Website 5 A community event registration website

OPEN LESSON

Website 5 Configuration code

OPEN LESSON

Website 5 See the result & test it

AI & publishing 1/11

AI & publishing

OPEN LESSON
Part 09 Build with AI and explore more tools

OPEN LESSON
Prepare the next AI practice project

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Ask ChatGPT for help in ordinary English

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Install Codex in VS Code

AI & publishing 2/11

AI & publishing

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Let Codex make one small improvement

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Save versions with GitHub Desktop

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Work like a careful website designer

OPEN LESSON

Choose the right kind of website tool

AI & publishing 3/11

AI & publishing

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Make a page on WordPress com

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Try WordPress software on your computer

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Change a WordPress theme and try a plugin

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Publish a short article with Blogger

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AI & publishing

OPEN LESSON
Build a simple site with Google Sites

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Briefly try Dreamweaver and a design tool

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Try building with ChatGPT Sites

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What is a stack

AI & publishing 5/11

AI & publishing

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Meet the terminal slowly

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Start a Laravel project

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Give Laravel your own page

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Start a Django project

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Start a Flutter web project

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Give AI a clear finished test

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AI & publishing

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Put the simple website online

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Show what you can do

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Paul’s workflow: from an idea to a live website

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AI setup 1: open a folder and choose a helper

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AI setup 2: know what each tool is

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AI practical 1: let ChatGPT prepare your brief

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AI practical 2: build directly with your agent

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AI practical 3: correct a result and keep context

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AI practical 4: run the correct local server

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GitHub practical 1: save your first project online

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GitHub practical 2: let AI prepare an update

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Server practical 1: prepare a safe deployment

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Server practical 2: first copy and production settings

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Server practical 3: pull a tested update

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Choose the right home for your website

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Website 5 Configuration code

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Optional tutor notes Teach the person in front of you

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Optional tutor notes Prepare the room and accounts

YOUR LEARNING TOOLS 2/4

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Optional tutor notes Use AI like a professional colleague

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Optional tutor notes Important current tool details

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Optional tutor notes Publish the right kind of project

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Quick help When something goes wrong

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